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Abstract:	We study the fractional p-Laplacian equation $(-\Delta)_s^p u + V u ^{p-2}u = \ln(u) u ^{\theta-2}u + \beta u ^{p_{s^*}-2}u, \quad \text{in } \mathbb{R}^N$ where $\beta \in (0, 1)$, $s \in (0, 1)$, $N \in \mathbb{N}$, $p \in (2, \beta\sqrt{N/s} + (1-\beta)N)$, $\theta \in (p, p_{s^*})$, $u \in W^{s,p}(\mathbb{R}^N)$, and $V: \mathbb{R}^N \rightarrow \mathbb{R}$ is a bounded potential. We establish existence of ground state solutions for both values of β when V is either periodic or asymptotically periodic. To avoid introducing an adjustable parameter multiplying some term of the equation, we instead exploit the solitons of $\mathcal{D}^{s,p}(\mathbb{R}^N)$ to show that the ground state energy lies below the vanishing threshold of the concentration-compactness argument in the critical case. This yields the condition $p < \sqrt{N/s}$, which is more readily verifiable than the (usually) nonconstructive threshold for the aforementioned parameter. We also devise a general equality between the Nehari and mountain pass levels under verifiable conditions on the fiber maps. Finally, we prove an abstract multiplicity theorem for functionals with discrete Palais-Smale attractors and apply it to obtain infinitely many geometrically distinct solutions when $\beta=0$ and V is periodic.

Ground state and infinite multiplicity for fractional p -Laplacian equations on $\mathbb{R}^N$ with logarithmic nonlinearity and (possible) critical growth; periodic and asymptotically periodic potentials

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Abstract

We study the fractional p -Laplacian equation

$$(-\Delta)_s^p u + V|u|^{p-2}u = \ln(|u|)|u|^{\theta-2}u + \beta|u|^{p_s^*-2}u,$$

where $\beta \in \{0, 1\}$, $s \in (0, 1)$, $N \in \mathbb{N}$, $p \in (2, \beta\sqrt{N/s} + (1-\beta)N/s)$, $\theta \in (p, p_s^*)$, $u \in W^{s,p}(\mathbb{R}^N)$, and $V: \mathbb{R}^N \rightarrow \mathbb{R}$ is a bounded potential.

We establish existence of ground state solutions for both values of β when V is either periodic or asymptotically periodic. To avoid introducing an adjustable parameter multiplying some term of the equation, we instead exploit the solitons of $\mathcal{D}^{s,p}(\mathbb{R}^N)$ to show that the ground state energy lies below the vanishing threshold of the concentration-compactness argument in the critical case. This yields the condition $p < \sqrt{N/s}$, which is more readily verifiable than the (usually) nonconstructive threshold for the aforementioned parameter.

We also devise a general equality between the Nehari and mountain pass levels under verifiable conditions on the fiber maps. Finally, we prove an abstract multiplicity theorem for functionals with discrete Palais-Smale attractors and apply it to obtain infinitely many geometrically distinct solutions when $\beta = 0$ and V is periodic.

Keywords— Fractional p -Laplacian operator, logarithmic nonlinearity, critical exponents in context of PDEs, periodic potential, unbounded domain, ground state solution, infinite multiplicity of solutions

1 Introduction and main results

In this work, we are interested in the fractional p -Laplacian (denoted here by $(-\Delta)_p^s$; a definition can be found, for example, in [32]) class of equations with the form

$$(-\Delta)_p^s u + V|u|^{p-2}u = \ln(|u|)|u|^{\theta-2}u + \beta|u|^{p_s^*-2}u, \quad (\text{P})$$

where $\beta \in \{0, 1\}$, $s \in (0, 1)$, $N \in \mathbb{N}$, $p \in (2, \beta\sqrt{N/s} + (1-\beta)N/s)$, $\theta \in (p, p_s^*)$, $u \in W^{s,p}(\mathbb{R}^N)$, and $V: \mathbb{R}^N \rightarrow \mathbb{R}$ is some potential satisfying

(V₁) There exists $m_V, M_V \in (0, \infty)$ such that $V: \mathbb{R}^N \rightarrow \mathbb{R}$ satisfies $0 < m_V \leq V(x) \leq M_V = \|V\|_\infty$ for all $x \in \mathbb{R}^N$.

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Additionally, our main concrete results will always explicitly state that V satisfies one of the following:

(V₂) $V(x+z)=V(x)$ for all $x \in \mathbb{R}^N$ and $z \in \mathbb{Z}$. ($\mathbb{Z}^N$ -periodic)

(V₃) There exist functions $V_{\text{per}}: \mathbb{R}^N \rightarrow \mathbb{R}$ and $W \in L^{\frac{N}{N-p}}(\mathbb{R}^N)$ such that $V(x) = V_{\text{per}}(x) - W(x)$ for all $x \in \mathbb{R}^N$. Moreover, W is strictly positive and V satisfies (V₁) and (V₂). (asymptotically $\mathbb{Z}^N$ -periodic)

For simplicity, we consider periodicity along the $\mathbb{Z}^N$ lattice, although very minor adjustments would be necessary to make our results apply to any regular lattice over $\mathbb{R}^N$.

In each of the four combinations of either $\beta=0$ or $\beta=1$ with V satisfying either (V₂) or (V₃), we obtain a nontrivial ground state weak solution to [problem \(P\)](#).

Theorem 1.1. *Let V satisfy (V₁) and (V₂) and*

$p \in \left(2, \beta\sqrt{\frac{N}{s}} + (1-\beta)\frac{N}{s}\right)$. *Then there exists a nontrivial ground state weak solution to [problem \(P\)](#).*

Theorem 1.2. *Let V satisfy (V₁) and (V₃) and $p \in \left(2, \beta\sqrt{\frac{N}{s}} + (1-\beta)\frac{N}{s}\right)$. Then there exists a nontrivial ground state weak solution to [problem \(P\)](#).*

Our proofs of the results above rely on the equality of the Nehari [26] and mountain pass levels, defined, for a continuously differentiable energy functional $\mathcal{J}: \mathbb{B} \rightarrow \mathbb{R}$ from a Banach space $\mathbb{B}$, as, respectively,

$$c_{\mathcal{N}} := \inf_{u \in \mathcal{N}} \mathcal{J}(u), \text{ where } \mathcal{N} := \{u \in \mathbb{B} - \{0\} \mid \mathcal{J}'(u)(u) = 0\}, \text{ and} \quad (1.1)$$

$$c := \inf_{\gamma \in \Gamma} \sup_{t \in [0,1]} \mathcal{J}(\gamma(t)), \text{ where } \Gamma := \{\gamma \in \mathcal{C}([0,1], \mathbb{B}) \mid \gamma(0) = 0, \mathcal{J}(\gamma(1)) < \mathcal{J}(0)\}. \quad (1.2)$$

However, some hypotheses of the standard result presented in [29] do not apply to [problem \(P\)](#). Thus, we prove the equality $c = c_{\mathcal{N}}$ ourselves, in an abstract setting with hypotheses stated mainly in terms of the fibering maps [26]

$$\varphi_u: (0, \infty) \rightarrow \mathbb{R}; \quad \varphi_u(t) := \mathcal{J}(tu), \text{ where } u \in \mathbb{B} - \{0\}. \quad (1.3)$$

We are not aware of any existing result in the literature that yields the aforementioned equality under hypotheses compatible with the present problem.

Theorem 1.3 (Equality of c and $c_{\mathcal{N}}$). *Let $\mathbb{B}$ be a Banach space and $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$. Additionally, suppose that*

(MP₁) *There exists $\mathbf{r} \in (0, \infty)$, $\alpha \in (\mathcal{J}(0), \infty)$ such that $\mathcal{J}(u) \geq \alpha$ for every $u \in \partial\mathcal{B}(0, \mathbf{r})$;*

(N₁) *For each $u \in \mathbb{B} - \{0\}$, there exists $t_u \in (0, \infty)$ such that $\varphi'_u(t) > 0$ when $0 < t < t_u$ and $\varphi'_u(t) < 0$ when $t > t_u$;*

(N₂) *For every $u \in \mathbb{B} - \{0\}$, there exists $s_u \in (0, \infty)$ such that $\varphi_u(s_u u) < \mathcal{J}(0)$.*

Then $c = c_{\mathcal{N}}$.

We also prove existence of infinitely many geometrically distinct (i.e., different modulo $\mathbb{Z}^N$ -translations) weak solutions to [problem \(P\)](#) in the periodic noncritical case. In fact, this is a corollary of an abstract multiplicity result we formulated in terms of a countable group action on a separable, reflexive Banach space.

Definition 1. *Let $\mathbb{B}$ be a Banach space, $\mathbb{B}'$ its dual space and $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$. A sequence $(u_n) \subset \mathbb{B}$ is called a (PS) (Palais-Smale) sequence for the functional $\mathcal{J}$ if*

$$\sup_{n \in \mathbb{N}} \mathcal{J}(u_n) < \infty \quad \text{and} \quad \|\mathcal{J}'(u_n)\|_{\mathbb{B}'} \rightarrow 0. \quad (1.4)$$

Moreover, if there exists $\varrho \in \mathbb{R}$ such that $\mathcal{J}(u_n) \rightarrow \varrho$, we say that (u_n) is a (PS) sequence at the ϱ level (or, in short, a (PS) _{ϱ} sequence).

Theorem 1.4 (Infinitely many geometrically distinct solutions in an abstract setting). *Let $\mathbb{B}$ be a separable, reflexive Banach space over $\mathbb{R}$, $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$, $\mathcal{G}$ be a group whose elements form a countable set and $h: \mathcal{G} \times \mathbb{B} \rightarrow \mathbb{B}$ be a group action. Assume also that $\mathcal{J}$ satisfies [\(MP₁\)](#) and the following conditions:*

- ($\mathcal{J}_1$) $\mathcal{J}$ is even; ($\mathcal{J}_2$) For all $g \in \mathcal{G}$ and $u \in \mathbb{B}$, $\mathcal{J}(g(u)) = \mathcal{J}(u)$;
- ($\mathcal{J}_3$) If $u \in \mathbb{B}$ and (u_n) be a (PS) sequence for $\mathcal{J}$ such that $u_n \rightharpoonup u$ in $\mathbb{B}$, then $\mathcal{J}'(u) = 0$;
- ($\mathcal{J}_4$) For every finite-dimensional subspace $\mathbb{Y} \subset \mathbb{B}$, there exists $R = R(\mathbb{Y}) \in (0, \infty)$ such that $\mathcal{J}(u) < \mathcal{J}(0)$ for all $u \in \mathcal{B}(0, R)^c \cap \mathbb{Y}$;
- ($\mathcal{G}_1$) For all $g \in \mathcal{G}$, the map $h(g, \cdot)$ is linear; ($\mathcal{G}_2$) For all $g \in \mathcal{G}$ and $u \in \mathbb{B}$, $\|h(g, u)\|_{\mathbb{B}} = \|u\|_{\mathbb{B}}$;
- ($\mathcal{G}_3$) For all $u \in \mathbb{B} - \{0\}$ and $v \in \mathbb{B}$, $\inf_{g \in \mathcal{G}, h(g, v) \neq u} \|u - h(g, v)\|_{\mathbb{B}} > 0$;
- ($\mathcal{G}_4$) Let (u_n) and (v_n) be (PS) sequences for $\mathcal{J}$ such that $\|u_n - v_n\|_{\mathbb{B}} \not\rightarrow 0$. Then there exists $u, v \in \mathbb{B}$ with $u \neq v$ and $(g_n) \in \mathcal{G}$ such that, up to subsequence, $h(g_n, u_n) \rightharpoonup u$ and $h(g_n, v_n) \rightharpoonup v$. Moreover, $(h(g_n, u_n)), (h(g_n, v_n))$ are (PS) sequences for $\mathcal{J}$.

Then $\mathcal{J}$ has infinitely many geometrically distinct critical points.

Corollary 1.5. *Let V satisfy [\(V₁\)](#) and [\(V₂\)](#) and $\beta = 0$. Then there exists infinitely many geometrically distinct weak solutions to [problem \(P\)](#).*

Over the last few decades, the analysis of non-local Partial Differential Equations (PDEs) involving the fractional Laplacian operator (see [\[8\]](#) for a definition) has undergone substantial development, moving beyond the initial framework of purely theoretical investigations to establish itself as a tool in several applied fields. These include finance [\[6\]](#) and ultra-relativistic limits of quantum mechanics [\[11\]](#) — a more comprehensive list can be found in the introduction of [\[8\]](#). This dynamic has led to a growing demand for analytical methods capable of handling the complexity inherent to these operators.

More recently — in the last five years or so — many existence and multiplicity results have been established for the fractional p -Laplacian Schrödinger equation

$$(-\Delta)_p^s u + V|u|^{p-2}u = f(u),$$

where $u: \mathbb{R}^N \rightarrow \mathbb{R}$ is the unknown, $V: \mathbb{R}^N \rightarrow \mathbb{R}$ is a potential, and $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function which may be explicitly given or supposed to satisfy some hypotheses, depending on the work [\[16, 19, 27\]](#). However, we have found no publications treating a fractional p -Laplacian equation for $p > 2$ with logarithmic nonlinearity and a potential which is either periodic or asymptotically periodic. Many works dealing with similar problems have either constant potential term [\[7, 31\]](#) or have it satisfy a “well-type” condition [\[1, 10, 14, 15, 28, 30\]](#); several of them also include an adjustable positive parameter multiplying some term of their equation and have their results depend on this parameter being less (or greater) than some (usually non-explicit) threshold. We draw special attention to [\[10\]](#) and [\[15\]](#) — which have proved existence of ground state solution to the problems

$$(-\Delta)_p^s u + Vu = \mu u \ln(|u|^2) + K|u|^{2^*_s-2}u, \text{ where } K: \mathbb{R}^N \rightarrow \mathbb{R}, \text{ and}$$

$$(-\Delta)_p^s u + V|u|^{p-2}u = |u|^{p-2}u \ln(|u|) + |u|^{p^*-2}u(x),$$

respectively — as the equations they treat are the ones which most resemble [problem \(P\)](#) in the current literature, at least as far as these authors know.

Our proofs of existence of ground state (all four cases) are inspired by [\[5\]](#); there, they obtain existence of ground state solution for the equation

$$\Delta^2 u \pm \Delta_q u + Vu = f(u) + \beta|u|^{\frac{2N}{N-4}-2}u$$

for suitable $q \in (0, \infty)$ and $N \in \mathbb{N}$. Their results hold when the function $f: \mathbb{R}^N \rightarrow \mathbb{R}$ satisfies some hypotheses that make it incompatible with the logarithmic nonlinearity in [problem \(P\)](#); among them is that $f(t) \geq \tau t^{r-1}$ always, where $r \in (q, \infty)$ and $\tau \in (0, \infty)$ is an adjustable parameter — which needs to be above some threshold for their existence theorem to be valid in the critical case ($\beta=1$). We make several adjustments to their arguments in order to have them apply to [problem \(P\)](#), but two of them stand out due to their novelty and technical challenge. The first one is showing that the derivative of the Gagliardo seminorm is weakly compact; this is nontrivial because the fractional kernel appears to prevent a standard argument using Lebesgue’s dominated convergence theorem.

The second major difficulty is to show that, in the critical case ($\beta=1$), the mountain pass level lies below the vanishing (in the Lions’ sense [\[18\]](#)) threshold. Rather than introducing an auxiliary parameter — as is done in [\[5\]](#) and many other works cited above — we construct a suitable mountain pass path from rescaled extremals of the fractional Sobolev inequality (often called solitons, see [\[3\]](#) for more details) and prove directly that its maximal energy is below the vanishing threshold. This yields the explicit and verifiable condition $2 < p < \sqrt{\frac{N}{s}}$ and avoids the nonconstructive thresholds highlighted above.

Two further contributions of this work are the abstract criterion ensuring equality of the Nehari and mountain pass levels ([theorem 1.3](#)) and the abstract multiplicity result ([theorem 1.4](#)). Both theorems isolate the essential structure behind ideas that have previously appeared only in specialized forms and should be directly applicable to a broad class of variational problems. The proof of [theorem 1.4](#) is, in particular, heavily inspired by the arguments in [\[22\]](#) (and its corrigendum [\[23\]](#)). Some technical clarifications were drawn from [\[25\]](#), while the underlying idea of a discrete Palais-Smale attractor can be traced back to [\[2\]](#). We improve the result in [\[22\]](#) by recasting it in an abstract variational setting, and by identifying the minimal hypotheses under which it remains valid. We also provide complete proofs of several intermediate steps that were left implicit in the prior work. A noteworthy novelty in the proof of [corollary 1.5](#) when compared to [\[22\]](#) — due to abstracting away from the Hilbert space $H^s(\mathbb{R}^N)$ — is the proof that [\(E₄\)](#) holds for the respective mathematical objects associated with [problem \(P\)](#).

This work is structured as follows: we start [section 2](#) by recalling some important definitions and facts; then we define the functional associated with [problem \(P\)](#) and provide some auxiliary results and expressions that will be used throughout the text. Next, we prove [theorem 1.3](#) in [section 3](#) and, in [section 4](#), we derive some estimates that will be used to show that the solution for the periodic, critical case is ground state. Sections [5](#) and [6](#) are devoted to the proofs of [theorem 1.1](#) and [theorem 1.2](#), respectively. Finally, [theorem 1.4](#) and [corollary 1.5](#) are worked out in [section 7](#). For better organization of the text, we compile some auxiliary results in [appendix A](#).

2 The variational framework

We start by recalling some definitions about and embeddings of fractional Sobolev spaces, most of which can be found in [\[8\]](#).

Let $\Omega \subset \mathbb{R}^N$ be open. For each $u \in L^p(\Omega)$, the Gagliardo seminorm on Ω is given by

$$[u]_{s,p,\Omega}^p := \int_{\Omega} \int_{\Omega} \mathcal{K}(x,y) |u(x) - u(y)|^p dx dy, \text{ where } \mathcal{K}: \mathbb{R}^{2N} \rightarrow \mathbb{R}; \quad \mathcal{K}(x,y) := \frac{1}{|x-y|^{N+sp}}.$$

This seminorm can then be used to define the fractional Sobolev space

$$W^{s,p}(\Omega) := \left\{ u \in L^p(\Omega) \mid [u]_{s,p,\Omega} < \infty \right\}, \text{ with norm } \|u\|_{W^{s,p}(\Omega)} := \left([u]_{s,p,\Omega}^p + \|u\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}.$$

We will also briefly need, in [section 4](#), the space

$$\mathcal{D}^{s,p}(\mathbb{R}^N) := \left\{ u \in L^{p^*}(\mathbb{R}^N) \mid [u]_{s,p,\mathbb{R}^N} < \infty \right\}, \text{ with norm } [\cdot]_{s,p,\mathbb{R}^N}.$$

If Ω is also an extension domain for $W^{s,p}$, then it holds that $W^{s,p}(\Omega) \hookrightarrow L^q(\Omega)$ for $q \in [p, p_s^*]$, where $\hookrightarrow$ means “continuously embedded in”. The proof of these embeddings can be found in [\[8\]](#), as well as more information about extension domains for $W^{s,p}$. For the purposes of this work, the fact that

$\mathbb{R}^N$ itself, any open ball in $\mathbb{R}^N$, and the complement thereof are extension domains for $W^{s,p}$ is the only information needed on this topic. One can also find in the literature that $W^{s,p}(\mathbb{R}^N) \subset\subset L^q_{\text{loc}}(\mathbb{R}^N)$ for any $q \in [1, p_s^*)$, where $\subset\subset$ means “compactly embedded in”.

The space in which we will search for non-trivial weak solutions of [problem \(P\)](#) is

$$\mathbb{X}(\Omega) := \left\{ u \in W^{s,p}(\Omega) \mid \int_{\Omega} V(x) |u(x)|^p dx < \infty \right\},$$

equipped with the norm

$$\|u\|_{\mathbb{X}(\Omega)} = \left([u]_{s,p,\Omega}^p + \|u\|_{V,\Omega}^p \right)^{\frac{1}{p}}, \text{ where } \|u\|_{V,\Omega} := \left(\int_{\Omega} V(x) |u(x)|^p dx \right)^{\frac{1}{p}}.$$

Note that [\(V₁\)](#) guarantees that, as sets, $\mathbb{X}(\Omega) = W^{s,p}(\Omega)$ and that $\|\cdot\|_{\mathbb{X}(\Omega)}$ is equivalent to $\|\cdot\|_{W^{s,p}(\Omega)}$. Naturally, $\mathbb{X}(\Omega)$ is a reflexive Banach space.

For most of this text, the domain of the elements of the function spaces with which we will work — the set Ω — will be $\mathbb{R}^N$ itself. When no other underlying domains are being used, we will shorten the expressions $\mathbb{X}(\mathbb{R}^N)$, $\|\cdot\|_{L^q(\mathbb{R}^N)}$, $[\cdot]_{s,p,\mathbb{R}^N}$, $\|u\|_{V,\Omega}$ and $\|\cdot\|_{\mathbb{X}(\mathbb{R}^N)}$ to $\mathbb{X}$, $\|u\|_q$, $[u]$, $\|u\|_V$ and $\|\cdot\|_{\mathbb{X}}$, respectively.

We state, for future reference, some embeddings of $\mathbb{X}(\Omega)$ and consequences thereof.

Proposition 2.1. *Let $\Omega \in \mathbb{R}^N$ be open. Then $\mathbb{X}(\Omega) \hookrightarrow W^{s,p}(\Omega)$.*

Corollary 2.2. *Let $\Omega \in \mathbb{R}^N$ be an open extension domain for $W^{s,p}(\mathbb{R}^N)$ and $q \in [p, p_s^*]$. Then $\mathbb{X}(\Omega) \hookrightarrow L^q(\Omega)$.*

Corollary 2.3. *Let $q \in [1, p_s^*)$. Then $\mathbb{X} \subset\subset L^q_{\text{loc}}(\mathbb{R}^N)$.*

Corollary 2.4. *Let $(u_n) \subset \mathbb{X}$ and $u \in \mathbb{X}$ such that $u_n \rightharpoonup u$ in $\mathbb{X}$. Then, up to a subsequence,*

$$u_n \xrightarrow{a.e.} u \text{ on } \mathbb{R}^N.$$

From now on, there will be many expressions of the form $\ln(t)$, where t here is acting as a placeholder for several variables whose logarithm we shall take. One might be startled by the fact that, usually, there will be no safeguard to prevent t from vanishing. This, however, will not be a problem; these terms are always multiplied by some factor of the form $f(t)$, where f vanishes faster when $t \rightarrow 0$ than the argument of the logarithm, i.e., $f(t)\ln(t) \rightarrow 0$ as $t \rightarrow 0$. Therefore, we implicitly consider the products $f(t)\ln(t)$ as their own function that is continually extended to be valued 0 when $t=0$. In this way, we can avoid polluting the arguments that follow with worries about the singularity of the logarithm in the origin.

The functional associated with [problem \(P\)](#) is given by

$$\mathcal{I}: \mathbb{X} \rightarrow \mathbb{R}; \quad \mathcal{I}(u) := \frac{\|u\|_{\mathbb{X}}^p}{p} + \frac{\|u\|_{\theta}^{\theta}}{\theta^2} - \frac{\mathcal{L}(u)}{\theta} - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*}, \text{ where } \mathcal{L}: \mathbb{X} \rightarrow \mathbb{R}; \quad \mathcal{L}(u) := \int_{\mathbb{R}^N} |u(x)|^{\theta} \ln(|u(x)|) dx. \quad (2.1)$$

The embeddings of [corollary 2.2](#) immediately guarantee that all norms in the expression of $\mathcal{I}$ are finite. However, if $\mathcal{I}$ is to be well-defined, we must also guarantee that $\mathcal{L}(u)$ is bounded below and above for all $u \in \mathbb{X}$. This will be done in the next proposition.

Proposition 2.5. *Let $u \in \mathbb{X}$ such that $\|u\|_{\mathbb{X}} > 0$. Then there exists constants $C_1, C_2, C_3 \in (0, \infty)$ such that $\mathcal{L}(u) \leq C_1 \|u\|_{\mathbb{X}}^{\theta} + \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^{\theta}$, $\mathcal{L}(u) \geq -C_2 \|u\|_{\mathbb{X}}^{\theta} + \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^{\theta}$ and $|\mathcal{L}(u)| \leq C_3 \|u\|_{\mathbb{X}}^{\theta} + |\ln(\|u\|_{\mathbb{X}})| \|u\|_{\theta}^{\theta}$.*

Proof. First, we prove $\mathcal{L}(u) \geq -C_2 \|u\|_{\mathbb{X}}^{\theta} + \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^{\theta}$. Note that

$$\int_{\mathbb{R}^N} |u(x)|^{\theta} \ln\left(\frac{1}{|u(x)|}\right) dx - \ln\left(\frac{1}{\|u\|_{\mathbb{X}}}\right) \int_{\mathbb{R}^N} |u(x)|^{\theta} dx = \int_{\mathbb{R}^N} |u(x)|^{\theta} \ln\left(\frac{\|u\|_{\mathbb{X}}}{|u(x)|}\right) dx. \quad (2.2)$$

By using [prop. A.1](#) with $r=\theta-p>0$ and applying the embedding in [corollary 2.2](#), we get

$$\begin{aligned} \int_{\mathbb{R}^N} |u(x)|^\theta \ln\left(\frac{\|u\|_{\mathbb{X}}}{|u(x)|}\right) dx &\leq \frac{1}{\theta-p} \int_{\mathbb{R}^N} |u(x)|^\theta \left(\frac{\|u\|_{\mathbb{X}}}{|u(x)|}\right)^{\theta-p} dx \\ &= \frac{\|u\|_{\mathbb{X}}^{\theta-p}}{\theta-p} \int_{\mathbb{R}^N} |u(x)|^p dx = \frac{\|u\|_{\mathbb{X}}^{\theta-p} \|u\|_p^p}{\theta-p} \leq \frac{\|u\|_{\mathbb{X}}^{\theta-p} \tilde{C}_2^p \|u\|_{\mathbb{X}}^p}{\theta-p} = C_2 \|u\|_{\mathbb{X}}^\theta, \end{aligned}$$

where $\tilde{C}_2 \in (0, \infty)$ is the embedding constant and $C_2 := \frac{\tilde{C}_2^p}{\theta-p} > 0$. Substituting back into [ineq. \(2.2\)](#), rearranging, and using properties of the logarithm results in $\mathcal{I}(u) \geq -C_2 \|u\|_{\mathbb{X}}^\theta + \ln(\|u\|_{\mathbb{X}}) \|u\|_{\mathbb{X}}^\theta$. The proof of the first inequality is similar, but it uses $r=p_s^*-\theta$ and starts instead from

$$\int_{\mathbb{R}^N} |u(x)|^\theta \ln(|u(x)|) dx - \ln(\|u\|_{\mathbb{X}}) \int_{\mathbb{R}^N} |u(x)|^\theta dx = \int_{\mathbb{R}^N} |u(x)|^\theta \ln\left(\frac{|u(x)|}{\|u\|_{\mathbb{X}}}\right) dx. \quad (2.3)$$

Finally, the last inequality is a direct consequence of the previous two. $\square$

Naturally, $\mathcal{I} \in \mathcal{C}^1(\mathbb{X}, \mathbb{R})$. Furthermore, for any $u, v \in \mathbb{X}$, the derivative $\mathcal{I}'$ of $\mathcal{I}$ at u in the direction of v can be written as

$$\mathcal{I}'(u)(v) = \langle u, v \rangle_{\mathbb{X}} - \mathcal{L}'(u)(v) - \beta \langle u, v \rangle_{p_s^*}, \text{ where } \langle u, v \rangle_{\mathbb{X}} := \left([u, v]^p + \int_{\mathbb{R}^N} V(x) |u(x)|^{p-2} u(x) v(x) dx \right),$$

$$[u, v] := \left(\int \int_{\Omega^2} \mathcal{K}(x, y) |u(x) - u(y)|^{p-2} (u(x) - u(y)) (v(x) - v(y)) dx dy \right)^{\frac{1}{p}},$$

$$\mathcal{L}'(u)(v) = \int_{\mathbb{R}^N} |u(x)|^{\theta-2} u(x) v(x) \ln(|u(x)|) dx, \quad \langle u, v \rangle_{p_s^*} := \int_{\mathbb{R}^N} |u(x)|^{p_s^*-2} u(x) v(x) dx.$$

In particular, when $v=u$, we have

$$\mathcal{I}'(u)(u) = \|u\|_{\mathbb{X}}^p - \mathcal{L}(u) - \beta \|u\|_{p_s^*}^{p_s^*}. \quad (2.4)$$

3 Auxiliary results about mountain pass and Nehari levels

The goal of this section is to establish general conditions such that the famous Nehari and mountain pass levels are equal. Then we prove that $\mathcal{I}$ satisfies those conditions.

All the results regarding $\mathcal{I}$ in this section assume only that V satisfies [\(V₁\)](#) and are valid for either value of β . Furthermore, in order to state broadly applicable results, we consider $\mathbb{B}$ a Banach space and $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$ a continuously differentiable functional.

Because $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$, each φ_u (see [eq. \(1.3\)](#)) is a continuously differentiable function of t and $\lim_{t \rightarrow 0} \varphi_u(t) = \mathcal{J}(0)$ for every $u \in \mathbb{B} - \{0\}$.

The first result of this section tells us when it is possible to project a nontrivial element of $\mathbb{B}$ into the Nehari manifold.

Proposition 3.1. *Let $u \in \mathbb{B} - \{0\}$ and $t \in (0, \infty)$. Then φ_u has a critical point at t if, and only if, $tu \in \mathcal{N}$.*

Proof. Using the identification $\mathbb{R} \cong \mathcal{L}(\mathbb{R}, \mathbb{R})$ (with this latter set being the vector space of all linear maps from $\mathbb{R}$ to $\mathbb{R}$), we have that $\varphi'_u(t) = \mathcal{J}'(tu)(u)$. Thus, by linearity,

$$tu \in \mathcal{N} \iff 0 = \mathcal{J}'(tu)(tu) = t \mathcal{J}'(tu)(u) = t \varphi'_u(t) \iff \varphi'_u(t) = 0. \quad \square$$

There is not much of use that can be said about the Nehari manifold without adding more assumptions. However, [\(N₁\)](#) — which is admittedly a strong hypothesis — goes a long way in allowing us to extract useful information about the Nehari manifold (and about $\mathcal{J}$ itself). For instance, if $\mathcal{J}$ satisfies [\(N₁\)](#) and $u \in \mathbb{B} - \{0\}$, then the critical point t_u of φ_u is unique (and is the maximum of φ_u); this means that the function $\mathfrak{t}: \mathbb{B} - \{0\} \rightarrow (0, \infty)$; $\mathfrak{t}(u) := t_u$ is well-defined. Henceforth, we shall denote t_u by $\mathfrak{t}(u)$.

Additionally, [\(N₁\)](#) is already enough to relate the minimax level

$$c_F := \inf \left\{ \sup_{t \geq 0} \varphi_u(t) \mid u \in \mathbb{B} - \{0\} \right\} \quad (3.1)$$

and the Nehari level $c_{\mathcal{N}}$. Such result is stated and proved below.

Proposition 3.2. *Assume that $\mathcal{J}$ satisfies (N_1) . Then $c_{\mathcal{N}} = c_F$.*

Proof. On one hand,

$$\begin{aligned} c_F &= \inf \left\{ \sup_{t \geq 0} \mathcal{J}(tu) \mid u \in \mathbb{B} - \{0\} \right\} \leq \inf \left\{ \sup_{t \geq 0} \mathcal{J}(tu) \mid u \in \mathcal{N} \right\} \stackrel{(N_1)}{=} \inf \left\{ \sup_{t \geq 0} \varphi_u(t) \mid u \in \mathcal{N} \right\} \\ &= \inf \left\{ \varphi_u(\mathfrak{t}(u)) \mid u \in \mathcal{N} \right\} = \inf \left\{ \mathcal{J}(\mathfrak{t}(u)u) \mid u \in \mathcal{N} \right\} = \inf \left\{ \mathcal{J}(u) \mid u \in \mathcal{N} \right\} = \inf_{u \in \mathcal{N}} \mathcal{J}(u) = c_{\mathcal{N}}, \end{aligned}$$

where the inequality comes from the fact that $\mathcal{N} \subset \mathbb{B} - \{0\}$ and the third to last equality holds because of the uniqueness of $\mathfrak{t}(u)$.

On the other hand, we have that $\mathfrak{t}(u)u \in \mathcal{N}$ for any $u \in \mathbb{B} - \{0\}$ by [prop. 3.1](#). Thus, $c_{\mathcal{N}} = \inf_{v \in \mathcal{N}} \mathcal{J}(v) \leq \mathcal{J}(\mathfrak{t}(u)u) = \sup_{t \geq 0} \mathcal{J}(tu)$. Taking the infimum over $u \in \mathbb{B} - \{0\}$ on both sides then gives

$$c_{\mathcal{N}} = \inf \left\{ c_{\mathcal{N}} \mid u \in \mathbb{B} - \{0\} \right\} \leq \inf \left\{ \sup_{t \geq 0} \mathcal{J}(tu) \mid u \in \mathbb{B} - \{0\} \right\} = c_F. \quad \square$$

In order to accomplish the first goal of this section, i.e., show conditions under which the Nehari and mountain pass levels coincide, we need to better understand how the Nehari manifold splits the Banach space. For this purpose, consider the function $\phi : \mathbb{B} \rightarrow \mathbb{R}$; $\phi(u) = \mathcal{J}'(u)(u)$ — which is continuous because $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$ — and the sets $\mathcal{N}_{>} := \phi^{-1}((0, \infty))$, $\mathcal{N}_{<} := \phi^{-1}((-\infty, 0))$. It is easy to see that $\mathcal{N} = \phi^{-1}(\{0\}) - \{0\}$ and then, by construction, that $\mathbb{B} - \{0\} = \mathcal{N}_{>} \sqcup \mathcal{N} \sqcup \mathcal{N}_{<}$. By characterizing $\partial \mathcal{N}_{>}$, the following result shows how the Nehari manifold partitions the space when (N_1) is holds.

Proposition 3.3. *Suppose that $\mathcal{J}$ satisfies (N_1) . Then $\partial \mathcal{N}_{>} - \{0\} = \mathcal{N}$.*

Proof. Because any continuous function f satisfies $\overline{f(A)} \subset f(\overline{A})$, we have

$$\begin{aligned} \mathcal{N}_{>} \cup (\partial \mathcal{N}_{>} - \{0\}) &\subset (\mathcal{N}_{>} \cup \partial \mathcal{N}_{>}) - \{0\} = \overline{\mathcal{N}_{>}} - \{0\} = \overline{\phi^{-1}((0, \infty))} - \{0\} \\ &\subset \phi^{-1}(\overline{(0, \infty)}) - \{0\} = \phi^{-1}([0, \infty)) - \{0\} = \phi^{-1}(\{0\} \cup (0, \infty)) - \{0\} \\ &= (\phi^{-1}(\{0\}) \cup \phi^{-1}((0, \infty))) - \{0\} = ((\mathcal{N} \cup \{0\}) \cup \mathcal{N}_{>}) - \{0\} = \mathcal{N}_{>} \cup \mathcal{N}, \end{aligned} \quad (3.2)$$

where the last line comes from the fact that $0 \notin \mathcal{N}_{>}$, which means that $\mathcal{N}_{>} \cap \partial \mathcal{N}_{>} = \emptyset$. This, along with [expression \(3.2\)](#), allows us to conclude that $\partial \mathcal{N}_{>} - \{0\} \subset \mathcal{N}$.

We will now show that the reverse inclusion holds, i.e., $\mathcal{N} \subset \partial \mathcal{N}_{>} - \{0\}$. For this, let $u \in \mathcal{N}$ and take an arbitrary $r \in (0, \infty)$. Since the definition of $\mathcal{N}$ implies that $u \neq 0$, it suffices to show that u is a boundary point of $\mathcal{N}_{>}$. Note that we have $\mathcal{N} \subset \mathcal{N}_{>}^c$ because we defined $\mathcal{N}_{>}$ in such a way that $\mathcal{N} \cap \mathcal{N}_{>} = \emptyset$. This means that u itself is a point in $\mathcal{B}(u, r) \cap \mathcal{N}_{>}^c$. On the other hand, (N_1) implies that

$$0 < \varphi'_u(t) = \mathcal{J}'(tu)(u), \quad \therefore \quad 0 < t\varphi'_u(t) = t\mathcal{J}'(tu)(u) = \mathcal{J}'(tu)(tu) = \phi(tu) \quad (3.3)$$

for every $t \in (0, \mathfrak{t}(u))$. Moreover, $\mathfrak{t}(u) = 1$ because $u \in \mathcal{N}$. Now take $r' := \frac{1}{2} \min\{r, \|u\|_{\mathbb{B}}\}$ and consider the mapping $\gamma_u : [0, 1] \rightarrow \mathbb{B}$; $\gamma_u(t) = tu$. By construction, γ_u is trivially continuous, $\gamma_u(0) = 0 \in \mathcal{B}(u, r')^c$, and $\gamma_u(1) = u \in \mathcal{B}(u, r')$. Since $[0, 1]$ is connected and $\mathcal{B}(u, r')$ is open, [prop. A.6](#) implies that there exists $v \in \gamma_u([0, 1])$ such that $v \in \partial \mathcal{B}(u, r')$.

As $v \in \gamma_u([0, 1])$, there exists $t_0 \in [0, 1]$ such that $v = \gamma_u(t_0) = t_0 u$. By our choice of r' , we have that $t_0 \neq 0$ and, because $v \in \partial \mathcal{B}(u, r')$, we also have $t_0 \neq 1$. This, along with the fact that $r > r'$, implies that $v \in \mathcal{B}(u, r)$. Moreover, since $t_0 \in (0, 1) = (0, \mathfrak{t}(u))$, [ineq. \(3.3\)](#) allows us to conclude that $v \in \mathcal{N}_{>}$. As u itself is not in $\mathcal{N}_{>}$ (by the definitions of $\mathcal{N}$ and $\mathcal{N}_{>}$), we conclude that it is a boundary point of $\mathcal{N}_{>}$. $\square$

We now briefly recall some definitions and key facts about the mountain pass geometry before we proceed to prove a relation between the mountain pass level and the levels defined previously in eqs. [\(1.1\)](#) and [\(3.1\)](#). We say that $\mathcal{J}$ satisfies the mountain pass geometry if [\(MP₁\)](#) holds and

(MP₂) There exists $\mathbf{e} \in \overline{\mathcal{B}(0, \mathbf{r})}^c$ such that $\mathcal{J}(\mathbf{e}) < \mathcal{J}(0)$.

It is not hard to see that $(N_2) \implies (MP_2)$, which is why we will not mention it any further. It is also well documented in the literature that if $\mathcal{J}$ satisfies the mountain pass geometry, then the mountain pass level c is well-defined and strictly positive.

We now accomplish the first stated goal of this section by proving its most important abstract result.

Proof of theorem 1.3. Recall that $c_{\mathcal{N}} = c_F$, by prop. 3.2. We then begin by showing that $c_{\mathcal{N}} = c_F \geq c$.

Let $u \in \mathbb{B} - \{0\}$. By (N_2) , there exists $s_u \in (0, \infty)$ such that $\varphi_u(s_u) < \mathcal{J}(0)$. Thus, the function

$$\gamma_u : [0, 1] \rightarrow \mathbb{B}; \quad \gamma_u(t) = t s_u u \quad (3.4)$$

is in $\mathcal{C}([0, 1], \mathbb{B})$, satisfying $\gamma_u(0) = 0$ and $\mathcal{J}(\gamma_u(1)) < \mathcal{J}(0)$, i.e., $\Gamma' := \{\gamma_u \in \mathcal{C}([0, 1], \mathbb{B}) \mid u \in \mathbb{B} - \{0\}\} \subset \Gamma$.

By prop. 3.1 and (N_1) , there is a unique $t_{s_u u} \in (0, \infty)$ such that $(t_{s_u u}) s_u u \in \mathcal{N}$. It is then easy to see that

$$\sup_{t \geq 0} \mathcal{J}(tu) = \mathcal{J}(t(u)u) = \mathcal{J}((t_{s_u u}) s_u u) = \sup_{t \in [0, 1]} \mathcal{J}(\gamma_u(t)).$$

Taking the infimum of $u \in \mathbb{B} - \{0\}$ and using that $\Gamma' \subset \Gamma$, we conclude that

$$\begin{aligned} c_F &= \inf \left\{ \sup_{t \geq 0} \mathcal{J}(tu) \mid u \in \mathbb{B} - \{0\} \right\} = \inf \left\{ \sup_{t \in [0, 1]} \mathcal{J}(\gamma_u(t)) \mid u \in \mathbb{B} - \{0\} \right\} \\ &= \inf_{\gamma \in \Gamma'} \sup_{t \in [0, 1]} \mathcal{J}(\gamma(t)) \geq \inf_{\gamma \in \Gamma} \sup_{t \in [0, 1]} \mathcal{J}(\gamma(t)) = c. \end{aligned}$$

It now just remains to prove that $c_{\mathcal{N}} \leq c$. For this purpose, let $\gamma \in \Gamma$. Our objective is to show that there exists $u_{\gamma} \in \gamma([0, 1]) \cap \mathcal{N}$. First, suppose by contradiction that $\gamma([0, 1]) \subset \mathcal{N}_{>}^c = \{0\} \cup \mathcal{N} \cup \mathcal{N}_{<}^c$. If any point in $\gamma([0, 1])$ is in $\mathcal{N}$, we are already done; so assume, without loss of generality, that $\gamma([0, 1]) \subset \{0\} \cup \mathcal{N}_{<}^c$. Since, by the definition of Γ , $\mathcal{J}(\gamma(1)) < \mathcal{J}(0)$, we can infer that $\gamma(1) \neq 0$. By continuity of γ , we know that there is some $\delta \in (0, \infty)$ such that $\gamma([1 - \delta, 1]) \cap \{0\} = \emptyset$. We can then take $t_0 := \inf\{t \in [0, 1] \mid \gamma([t, 1]) \cap \{0\} = \emptyset\}$, and we one can easily see that $t_0 < 1 - \delta < 1$. Moreover, again by continuity, it must be the case that $\gamma(t_0) = 0$. Thus, $\gamma((t_0, 1]) \subset \mathcal{N}_{<}^c$. By rescaling the argument use for γ , we could construct a new path $\tilde{\gamma} \in \Gamma$ such that $\tilde{\gamma}((0, 1]) \subset \mathcal{N}_{<}^c$. Thus, without loss of generality, we can suppose that $\gamma((0, 1]) \subset \mathcal{N}_{<}^c$.

Let $\bar{\gamma} : (0, 1] \rightarrow \mathbb{B} - \{0\}$; $\bar{\gamma}(t) := \gamma(t) / \|\gamma(t)\|_{\mathbb{B}}$, so that $\gamma(t) = \|\gamma(t)\|_{\mathbb{B}} \bar{\gamma}(t)$ for all $t \in (0, 1]$. Note that, since $0 \notin \mathcal{N}_{<}^c$, we have that $\|\gamma(t)\|_{\mathbb{B}} > 0$ for all $t \in (0, 1]$, which means that $\bar{\gamma}$ is well-defined. As $\gamma((0, 1]) \subset \mathcal{N}_{<}^c$, we have

$\|\gamma(t)\|_{\mathbb{B}} \varphi'_{\bar{\gamma}(t)}(\|\gamma(t)\|_{\mathbb{B}}) = \|\gamma(t)\|_{\mathbb{B}} \mathcal{J}'(\|\gamma(t)\|_{\mathbb{B}} \bar{\gamma}(t))(\bar{\gamma}(t)) = \mathcal{J}'(\|\gamma(t)\|_{\mathbb{B}} \bar{\gamma}(t))(\|\gamma(t)\|_{\mathbb{B}} \bar{\gamma}(t)) = \mathcal{J}'(\gamma(t)) < 0$ for all $t \in (0, 1]$, i.e., $\varphi'_{\bar{\gamma}(t)}(\|\gamma(t)\|_{\mathbb{B}}) < 0$ always. This, (N_1) , and the definition of $\mathbf{t}$ imply that $\|\gamma(t)\|_{\mathbb{B}} > \mathbf{t}(\bar{\gamma}(t))$ for all $t \in (0, 1]$. Since γ is continuous and $\gamma(0) = 0$, we have

$$0 = \lim_{t \rightarrow 0} \|\gamma(t)\|_{\mathbb{B}} \geq \lim_{t \rightarrow 0} \mathbf{t}(\bar{\gamma}(t)) > 0,$$

where the last inequality comes from the definition of $\mathbf{t}$. Then the squeeze theorem gives us $\lim_{t \rightarrow 0} \mathbf{t}(\bar{\gamma}(t)) = 0$. We can now find a sequence $(t_n) \subset (0, 1]$ with $t_n \rightarrow 0$ such that $\mathbf{t}(\bar{\gamma}(t_n)) \rightarrow 0$. In particular, we can extract a subsequence that always satisfies $\mathbf{t}(\bar{\gamma}(t_n)) < \mathbf{r}$. For any $n \in \mathbb{N}$, (N_1) and the definition of $\mathbf{t}$ imply that $h'_{\bar{\gamma}(t_n)} < 0$ for all $t \in (\mathbf{t}(\bar{\gamma}(t_n)), \infty)$, i.e., $\varphi_{\bar{\gamma}(t_n)}$ is decreasing on $(\mathbf{t}(\bar{\gamma}(t_n)), \mathbf{r}]$. Thus,

$$\mathcal{J}(\mathbf{t}(\bar{\gamma}(t_n)) \bar{\gamma}(t_n)) > \mathcal{J}(\mathbf{r} \bar{\gamma}(t_n)) \geq \alpha > \mathcal{J}(0)$$

for all $n \in \mathbb{N}$, where the two last inequalities come from (MP_1) and the fact that $\|\mathbf{r} \bar{\gamma}(t_n)\|_{\mathbb{B}} = \|\mathbf{r}\| \|\bar{\gamma}(t_n)\|_{\mathbb{B}} = \mathbf{r}$. However, since

$$\|\mathbf{t}(\bar{\gamma}(t_n)) \bar{\gamma}(t_n)\|_{\mathbb{B}} = \mathbf{t}(\bar{\gamma}(t_n)) \|\bar{\gamma}(t_n)\|_{\mathbb{B}} = \mathbf{t}(\bar{\gamma}(t_n)) \rightarrow 0,$$

we can use the continuity of $\mathcal{J}$ to write the contradiction

$$\mathcal{J}(0) = \lim_{n \rightarrow \infty} \mathcal{J}(\mathbf{t}(\bar{\gamma}(t_n)) \bar{\gamma}(t_n)) \geq \alpha > \mathcal{J}(0).$$

Thus, we can always suppose that there exists some $t_1 \in [0, 1]$ for which $\gamma(t_1) \in \mathcal{N}_{>}^c$. Let $v_{\gamma} := \gamma(1)$ and note that $\lim_{t \rightarrow 0} \varphi_{v_{\gamma}}(t) = \mathcal{J}(0)$ and $\varphi_{v_{\gamma}}(1) = \mathcal{J}(v_{\gamma}) < \mathcal{J}(0)$. Then $h'_{v_{\gamma}}(t_2) < 0$ at some $t_2 \in [0, 1]$. As $t_2 \leq 1$, it then follows from (N_1) that $0 > h'_{v_{\gamma}}(t_2) \leq h'_{v_{\gamma}}(1) = \mathcal{J}'(v_{\gamma})(v_{\gamma})$, i.e., $v_{\gamma} \in \mathcal{N}_{<}^c \subset \mathcal{N}_{>}^c$. Since γ is continuous, $(0, 1]$ is connected and $\mathcal{N}_{>}^c$ is open, we can use prop. A.6 to find $t_2 \in [t_0, 1]$ such that $u_{\gamma} := \gamma(t_2) \in \partial \mathcal{N}_{>}^c - \{0\} = \mathcal{N}$ (the equality comes from prop. 3.3). This element may not be unique, but we will see that this does not matter.

If $u \in \mathcal{N}$, then it follows trivially from (N₁) that u_{γ_u} is unique and $u_{\gamma_u} = u$, where γ_u is the path defined in eq. (3.4). Now consider the sets $\Gamma'' := \{\gamma_u \in \Gamma' \mid u \in \mathcal{N}\}$ and $A := \{u_\gamma \in \mathbb{B} \mid \gamma \in \Gamma'\}$; one may choose any u_{γ_0} when it is not uniquely defined for a $\gamma_0 \in \Gamma$. We claim that $A = \mathcal{N}$. The inclusion $A \subset \mathcal{N}$ follows trivially from the construction of each u_γ . On the other hand, since $\Gamma'' \subset \Gamma' \subset \Gamma$, we have

$$\mathcal{N} = \{u \in \mathbb{B} \mid u \in \mathcal{N}\} = \{u_{\gamma_u} \in \mathbb{B} \mid u \in \mathcal{N}\} = \{u_{\gamma_u} \in \mathbb{B} \mid \gamma_u \in \Gamma''\} \subset \{u_\gamma \in \mathbb{B} \mid \gamma \in \Gamma'\} = A.$$

Therefore, we finally conclude that

$$c = \inf_{\gamma \in \Gamma} \sup_{t \geq 0} \mathcal{J}(\gamma(t)) \geq \inf_{\gamma \in \Gamma} \mathcal{J}(u_\gamma) = \inf_{u \in A} \mathcal{J}(u) = \inf_{u \in \mathcal{N}} \mathcal{J}(u) = c_{\mathcal{N}}. \quad \square$$

Remark. A simple counterexample that shows that (N₂) cannot be removed is the function

$$f: \mathbb{R} \rightarrow \mathbb{R}; \quad f(x) := \begin{cases} x^2(2x^2 - 2x + 1)^{-1}, & x \in [0, \infty); \\ (5x^3 - 10x^2)(-10x^3 + 26x - 20)^{-1}, & x \in (-\infty, 0). \end{cases}$$

We close this section with the proof that all the results herein apply to $\mathcal{I}$.

Lemma 3.4. *The functional $\mathcal{I}$ satisfies (MP₁), (N₁), and (N₂).*

Proof. Let $u \in \mathbb{X} - \{0\}$, $i \in \{0, 1\}$ and consider the continuous function

$$f_{u,i}: (0, \infty) \rightarrow \mathbb{R}; \quad f(t) = A_{u,i}t^{p-\theta} + B_{u,i} - C_{u,i}\ln(t) - D_{u,i}t^{p_s^*-\theta}, \text{ where} \quad (3.5)$$

$$A_{u,0} := \frac{\|u\|_{\mathbb{X}}^p}{p}, \quad B_{u,0} := \frac{\|u\|_{\theta}^{\theta}}{\theta} \left(\frac{1}{\theta} - \ln(t) - \frac{\mathcal{L}(u)}{\|u\|_{\theta}^{\theta}} \right), \quad C_{u,0} := \frac{\|u\|_{\theta}^{\theta}}{\theta}, \quad D_{u,0} := \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*}, \quad (3.6)$$

$$A_{u,1} := \|u\|_{\mathbb{X}}^p, \quad B_{u,1} := -\mathcal{L}(u), \quad C_{u,1} := \|u\|_{\theta}^{\theta} \quad \text{and} \quad D_{u,1} := \beta \|u\|_{p_s^*}^{p_s^*}.$$

It is easy to check that $A_{u,i}, C_{u,i} \in (0, \infty)$, $B_{u,i} \in \mathbb{R}$ and $D_{u,i} \in [0, \infty)$. Since $p < \theta < p_s^*$, we have

$$\lim_{t \rightarrow 0^+} f(t) = \infty \quad \text{and} \quad \lim_{t \rightarrow \infty} f(t) = -\infty. \quad (3.7)$$

This means that there must exist some $t_{u,i} \in (0, \infty)$ such that $f(t_{u,i}) = 0$ by the intermediate value theorem. This $t_{u,i}$ is unique, otherwise there would exist, without loss of generality, some $\tilde{t}_{u,i} \in (t_{u,i}, \infty)$ such that

$$At_{u,i}^{p-\theta} + B_{u,i} - C_{u,i}\ln(t_{u,i}) - D_{u,i}t_{u,i}^{p_s^*-\theta} = 0 = A\tilde{t}_{u,i}^{p-\theta} + B_{u,i} - C_{u,i}\ln(\tilde{t}_{u,i}) - D_{u,i}\tilde{t}_{u,i}^{p_s^*-\theta}, \quad \therefore$$

$$A_{u,i}(t_{u,i}^{p-\theta} - \tilde{t}_{u,i}^{p-\theta}) = C_{u,i}(\ln(t_{u,i}) - \ln(\tilde{t}_{u,i})) + D_{u,i}(t_{u,i}^{p_s^*-\theta} - \tilde{t}_{u,i}^{p_s^*-\theta}).$$

This would contradict $A_{u,i}, C_{u,i} > 0$, $D_{u,i} \geq 0$, $t_{u,i}^{p-\theta} - \tilde{t}_{u,i}^{p-\theta} > 0$, $\ln(t_{u,i}) - \ln(\tilde{t}_{u,i}) < 0$, and $t_{u,i}^{p_s^*-\theta} - \tilde{t}_{u,i}^{p_s^*-\theta} < 0$.

Let $t \in (0, \infty)$. By using that $\ln(|tu(x)|) = \ln(t) + \ln(|u(x)|)$ in eqs. (2.1) and (2.4), we get

$$\mathcal{I}(tu) = \frac{\|u\|_{\mathbb{X}}^p}{p} t^p + \frac{\|u\|_{\theta}^{\theta}}{\theta} \left(\frac{\mathcal{L}(u)}{\theta} - \ln(t) - \frac{1}{\|u\|_{\theta}^{\theta}} \right) t^{\theta} - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} t^{p_s^*}, \quad (3.8)$$

$$\mathcal{I}'(tu)(tu) = \|u\|_{\mathbb{X}}^p t^p - (\ln(t)\|u\|_{\theta}^{\theta} + \mathcal{L}(u))t^{\theta} - \beta \|u\|_{p_s^*}^{p_s^*} t^{p_s^*}, \quad (3.9)$$

$$\varphi_u(t) = \mathcal{I}(tu) = A_{u,0}t^p + B_{u,0}t^{\theta} - C_{u,0}t^{\theta}\ln(t) - D_{u,0}t^{p_s^*} = t^{\theta} f_{u,0}(t), \text{ and} \quad (3.10)$$

$$\varphi'_u(t) = \frac{\mathcal{I}'(tu)(tu)}{t} = A_{u,1}t^{p-1} + B_{u,1}t^{\theta-1} - C_{u,1}t^{\theta-1}\ln(t) - D_{u,1}t^{p_s^*-1} = t^{\theta-1} f_{u,1}(t). \quad (3.11)$$

The uniqueness of $t_{u,i}$ and the limits given in equations (3.7) applied to eq. (3.11) directly imply (N₁). The same results applied instead to eq. (3.10) imply (N₂) (recall that $\mathcal{I}(0) = 0$).

In order to prove (MP₁), we use prop. 2.5 and corollary 2.2 to write

$$\begin{aligned} \mathcal{I}(u) &= \frac{\|u\|_{\mathbb{X}}^p}{p} + \frac{\|u\|_{\theta}^{\theta}}{\theta^2} - \frac{1}{\theta} \int_{\mathbb{R}^N} |u(x)|^{\theta} \ln(|u(x)|) dx - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} \geq \frac{\|u\|_{\mathbb{X}}^p}{p} - \frac{1}{\theta} \int_{\mathbb{R}^N} |u(x)|^{\theta} \ln(|u(x)|) dx - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} \\ &\geq \frac{\|u\|_{\mathbb{X}}^p}{p} - \frac{C_1}{\theta} \|u\|_{\mathbb{X}}^{\theta} - \frac{1}{\theta} \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^{\theta} - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} \geq \frac{\|u\|_{\mathbb{X}}^p}{p} - \frac{C_1}{\theta} \|u\|_{\mathbb{X}}^{\theta} - \frac{1}{\theta} \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^{\theta} - \beta \frac{C_2}{p_s^*} \|u\|_{\mathbb{X}}^{p_s^*} \end{aligned}$$

for some $C_1, C_2 \in (0, \infty)$. Furthermore, if $\|u\|_{\mathbb{X}} \leq 1$, then

$$\mathcal{I}(u) \geq \frac{\|u\|_{\mathbb{X}}^p}{p} - \frac{C_1}{\theta} \|u\|_{\mathbb{X}}^\theta - \beta \frac{C_2}{p_s^*} \|u\|_{\mathbb{X}}^{p_s^*} = \|u\|_{\mathbb{X}}^p \left(\frac{1}{p} - \frac{C_1}{\theta} \|u\|_{\mathbb{X}}^{\theta-p} - \beta \frac{C_2}{p_s^*} \|u\|_{\mathbb{X}}^{p_s^*-p} \right) = \|u\|_{\mathbb{X}} g(\|u\|_{\mathbb{X}}), \quad (3.12)$$

where g is the continuous function given by $g: [0, \infty) \rightarrow \mathbb{R}$; $g(t) = \frac{1}{p} - \frac{C_1}{\theta} t^{\theta-p} - \beta \frac{C_2}{p_s^*} t^{p_s^*-p}$.

Since $p < \theta < p_s^*$, it is clear that $\lim_{t \rightarrow 0^+} g(t) = \frac{1}{p} > 0$. The continuity of g then tells us that there must exist $\mathbf{r} \in (0, 1)$ such that $g(t) > 0$ for all $t \in (0, \mathbf{r}]$. Then $\mathcal{I}(u) \geq \mathbf{r} f(\mathbf{r}) > \frac{\mathbf{r}}{2} f(\mathbf{r}) > 0$ whenever $\|u\|_{\mathbb{X}} = \mathbf{r}$, which is (MP₁) with $\alpha := \frac{\mathbf{r}}{2} f(\mathbf{r}) > 0 = \mathcal{I}(0)$. $\square$

4 Estimates for the critical case

Consider the inverse of the embedding constant of $\mathbb{X} \hookrightarrow L^{p_s^*}(\mathbb{R}^N)$, given by

$$\mathcal{S}_{s,p} := \inf \left\{ \frac{\|u\|_{\mathbb{X}}}{\|u\|_{p_s^*}} \mid u \in \mathbb{X} - \{0\} \right\}, \quad (4.1)$$

which is, therefore, strictly positive. The main purpose of this section is to prove the following lemma:

Lemma 4.1. *Let $\beta = 1$, V satisfy (V₁), and $\omega := \frac{pp_s^*}{p_s^*-p}$. Then there exists $u \in \mathbb{X}$ such that*

$$\sup_{t \in [0,1]} \mathcal{I}(tu) < \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \mathcal{S}_{s,p}^\omega. \quad (4.2)$$

This result will then be used in sections 5 and 6 to show that, when $\beta = 1$, the candidate we find for weak solution of problem (P) is nontrivial. Since lemma 4.1 is the goal of this section, we will assume of V only (V₁), and the constant β will always satisfy $\beta = 1$; therefore, it will not be explicitly written. We will prove lemma 4.1 by explicitly constructing an element in $\mathbb{X}$ satisfying ineq. (4.2). This construction depends on the existence of a minimizer, in $\mathbb{X}$, to an expression similar to that of eq. (4.1), which we will prove now.

Proposition 4.2. *There exists $U \in \mathbb{X}$ with $\|U\|_{p_s^*} = 1$ such that*

$$\frac{[U]}{\|U\|_{p_s^*}} = \inf \left\{ \frac{[u]}{\|u\|_{p_s^*}} \mid u \in W^{s,p}(\mathbb{R}^N) - \{0\} \right\}. \quad (4.3)$$

Proof. Existence of $U \in \mathcal{D}^{s,p}(\mathbb{R}^N)$ satisfying eq. (4.3) is given in [17], page 168 (section I.4, example iii).

We now use and justify the crucial hypothesis that $p \in \left(2, \sqrt{\frac{N}{s}}\right)$, which implies that $p > \frac{N(p-1)}{(N-sp)}$. This and proposition 3.3 of [4] gives us that $U \in L^p(\mathbb{R}^N)$. It then follows from (V₁) that $U \in \mathbb{X}$. The fact that $U \neq 0$ comes trivially from it satisfying eq. (4.3); therefore, it can be normalized by its p_s^* -norm. $\square$

From now on, U will always refer to the function described in prop. 4.2. The next step in constructing an element that satisfies ineq. (4.2) is to apply some rescaling to U . Given an scaling parameter $\varepsilon \in (0, \infty)$, we define $U_\varepsilon(x) := \varepsilon^{-\sigma} U\left(\frac{x}{\varepsilon}\right)$, where $\sigma := \frac{N}{p_s^*}$. This is a natural choice of rescaling; in fact, we did not find out until after all calculations were complete that it had already been used in the literature (e.g., [20, 21]). This is due to the fact that $[\cdot]$ and $\|\cdot\|_{p_s^*}$ are invariant by this scaling option. To facilitate referencing in the text that follows, we state these facts, as well as the transformation of an arbitrary q -norm by the rescaling, in the next proposition.

Proposition 4.3. *Let $q \in [p, p_s^*]$. Then $\|U_\varepsilon\|_q^q = \varepsilon^{N-\sigma q} \|U\|_q^q$ and $[U_\varepsilon] = [U]$. In particular, $\|U_\varepsilon\|_{p_s^*} = \|U\|_{p_s^*}$ and, if $q < p_s^*$, we have $N - \sigma q > 0$.*

The remaining of this section is dedicated to proving that, for ε small enough, U_ε is the element we are after, i.e., U_ε satisfies ineq. (4.2). The next result will give some intuition about the preceding claim.

Proposition 4.4. $\lim_{\varepsilon \rightarrow 0} \|U_\varepsilon\|_{\mathbb{X}} = [U] = \mathcal{S}_{s,p}$.

Proof. By [prop. 4.3](#) and [\(V₁\)](#), we have $[U]^p + m_V \|U\|_p^p \varepsilon^{N-\sigma p} \leq \|U_\varepsilon\|_{\mathbb{X}}^p \leq [U]^p + \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p}$. Thus, making $\varepsilon \rightarrow 0$ proves the first equality in [prop. 4.4](#). For the second equality, note that

$$\begin{aligned} [U] &= \frac{[U]}{\|U\|_{p_s^*}} = \inf \left\{ \frac{[u]}{\|u\|_{p_s^*}} \mid u \in W^{s,p}(\mathbb{R}^N) - \{0\} \right\} \\ &= \inf \left\{ \frac{[u]}{\|u\|_{p_s^*}} \mid u \in \mathbb{X} - \{0\} \right\} \leq \inf \left\{ \frac{\|u\|_{\mathbb{X}}}{\|u\|_{p_s^*}} \mid u \in \mathbb{X} - \{0\} \right\} = \mathcal{S}_{s,p} \end{aligned} \quad (4.4)$$

by [prop. 4.2](#) and [\(V₁\)](#). On the other hand,

$$\mathcal{S}_{s,p} = \inf \left\{ \frac{\|u\|_{\mathbb{X}}}{\|u\|_{p_s^*}} \mid u \in \mathbb{X} - \{0\} \right\} \leq \frac{\|U_\varepsilon\|_{\mathbb{X}}}{\|U_\varepsilon\|_{p_s^*}} = \|U_\varepsilon\|_{\mathbb{X}},$$

which, by making $\varepsilon \rightarrow 0$, implies $\mathcal{S}_{s,p} \leq [U]$. Combining this with [ineq. \(4.4\)](#) concludes the proof. $\square$

Before we move on to analyzing the behavior of the functional along tU_ε , we need another result regarding U_ε : for ε small enough, its norm in $\mathbb{X}$ remains bounded below by a positive constant and bounded above; the bounding constants do not depend on ε .

Proposition 4.5. *Let $\varepsilon \in (0, \infty)$. Then $[U] < \|U_\varepsilon\|_{\mathbb{X}}$ and, for all ε small enough, $\|U_\varepsilon\|_{\mathbb{X}} < \|U\|_{\mathbb{X}}$.*

Proof. The first inequality follows trivially from [\(V₁\)](#). On the other hand, by [prop. 4.3](#), we have

$$\begin{aligned} \|U_\varepsilon\|_{\mathbb{X}}^p &= [U_\varepsilon]^p + \int_{\mathbb{R}^N} V(x) |U_\varepsilon(x)|^p dx = [U]^p + \int_{\mathbb{R}^N} V(x) |U_\varepsilon(x)|^p dx \\ &\leq [U]^p + \|V\|_\infty \|U_\varepsilon\|_p^p = [U]^p + \|V\|_\infty \varepsilon^{N-\sigma p} \|U\|_p^p. \end{aligned} \quad (4.5)$$

Thus, the second inequality is valid whenever $\varepsilon < (m_V / \|V\|_\infty)^{\frac{1}{N-\sigma p}}$, where m_V is given by [\(V₁\)](#). $\square$

We are now ready to investigate $\mathcal{I}(tU_\varepsilon)$. In the next result, we find its general expression.

Proposition 4.6. *Let $t, \varepsilon \in (0, \infty)$. Then*

$$\mathcal{I}(tU_\varepsilon) = \frac{t^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} + \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) + \sigma \ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right).$$

Proof. By [eq. \(3.8\)](#), the identities of [prop. 4.3](#) and using that $\|U\|_{p_s^*} = 1$ (from [prop. 4.2](#)), we have

$$\begin{aligned} \mathcal{I}(tU_\varepsilon) &= \frac{t^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} \|U_\varepsilon\|_{p_s^*}^{p_s^*} + \frac{t^\theta}{\theta} \|U_\varepsilon\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) - \frac{\mathcal{L}(U_\varepsilon)}{\|U_\varepsilon\|_\theta^\theta} \right) \\ &= \frac{t^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} + \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) - \frac{\mathcal{L}(U_\varepsilon)}{\varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta} \right). \end{aligned} \quad (4.6)$$

By using the theorem of change of variables, we get

$$\begin{aligned} \mathcal{L}(U_\varepsilon) &= \int_{\mathbb{R}^N} \varepsilon^{-\sigma\theta} \left| U\left(\frac{\varepsilon y}{\varepsilon}\right) \right|^\theta \ln \left(\left| \varepsilon^{-\sigma} U\left(\frac{\varepsilon y}{\varepsilon}\right) \right| \right) \varepsilon^N dy = \varepsilon^{N-\sigma\theta} \int_{\mathbb{R}^N} |U(y)|^\theta \ln(|\varepsilon^{-\sigma} U(y)|) dy \\ &= \varepsilon^{N-\sigma\theta} \int_{\mathbb{R}^N} |U(y)|^\theta \left(\ln(|\varepsilon^{-\sigma}|) + \ln(U(y)) \right) dy = \varepsilon^{N-\sigma\theta} (-\sigma \ln(\varepsilon) \|U\|_\theta^\theta + \mathcal{L}(U)). \end{aligned}$$

Plugging back into [eq. \(4.6\)](#) and manipulating we conclude that

$$\mathcal{I}(tU_\varepsilon) = \frac{t^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} + \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) + \sigma \ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right). \quad \square$$

We will now treat $\mathcal{I}(tU_\varepsilon)$ as a function of t with parameter ε . Furthermore, we will split it into two parts. For $\varepsilon \in (0, \infty)$, we define the functions

$$\mathcal{H}_\varepsilon : (0, \infty) \rightarrow \mathbb{R}; \quad \mathcal{H}_\varepsilon(t) := \frac{t^p}{p} \|U_\varepsilon\|_\mathbb{X}^p - \frac{t^{p_s^*}}{p_s^*} \quad \text{and} \quad (4.7)$$

$$\mathcal{G}_\varepsilon : (0, \infty) \rightarrow \mathbb{R}; \quad \mathcal{G}_\varepsilon(t) := \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right). \quad (4.8)$$

[Proposition 4.6](#) makes it obvious that $\mathcal{I}(tU_\varepsilon) := \mathcal{H}_\varepsilon(t) + \mathcal{G}_\varepsilon(t)$. Some observations are in order: first, both functions are smooth, with

$$\begin{aligned} \mathcal{H}'_\varepsilon(t) &= t^{p-1} \|U_\varepsilon\|_\mathbb{X}^p - t^{p_s^*-1}, \quad \mathcal{G}'_\varepsilon(t) = t^{\theta-1} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(-\ln(t) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right), \text{ and} \\ \mathcal{H}''_\varepsilon(t) &= t^{p-2} ((p-1) \|U_\varepsilon\|_\mathbb{X}^p - (p_s^*-1) t^{p_s^*-p}). \end{aligned}$$

Second, $\mathcal{H}_\varepsilon$ has a unique critical point at $t_\varepsilon := \|U_\varepsilon\|_\mathbb{X}^{\frac{p}{p_s^*-p}}$ (which is its global maximum), and

$$\mathcal{H}_\varepsilon(t_\varepsilon) = \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \|U_\varepsilon\|_\mathbb{X}^\omega > 0, \quad (4.9)$$

which notably has the same right-hand side of [ineq. \(4.2\)](#) when $\varepsilon \rightarrow 0$. Therefore, the supremum of this function always slightly exceeds the desired amount.

The third observation is that $\mathcal{G}_\varepsilon(t_\varepsilon)$ becomes strictly negative when ε is small enough. This, hopefully, makes the reader optimist that $\mathcal{G}_\varepsilon$ can be made negative enough to counteract the slight surplus of $\mathcal{H}_\varepsilon$. We will not prove this claim at this moment; instead it will be proved later when needed. Nonetheless, we state below another fact regarding $\mathcal{G}_\varepsilon(t_\varepsilon)$ and prove immediately.

Proposition 4.7. $\lim_{\varepsilon \rightarrow 0} \mathcal{G}_\varepsilon(t_\varepsilon) = 0$.

Proof. Note that

$$\mathcal{G}_\varepsilon(t_\varepsilon) = \frac{\|U\|_\theta^\theta}{\theta} \left(\frac{1}{\theta} - \frac{p}{p_s^*-p} \ln(\|U_\varepsilon\|_\mathbb{X}) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right) \|U_\varepsilon\|_\mathbb{X}^{\frac{\theta p}{p_s^*-p}} \varepsilon^{N-\sigma\theta}. \quad (4.10)$$

Let us now introduce the shorthand $W(t) = \frac{1}{\theta} - \frac{p}{p_s^*-p} \ln(t) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta}$. Then, by [prop. 4.5](#), we have

$$W(\|U\|_\mathbb{X}) + \sigma \ln(\varepsilon) < \left(\frac{1}{\theta} - \frac{p}{p_s^*-p} \ln(\|U_\varepsilon\|_\mathbb{X}) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right) < W([U]) + \sigma \ln(\varepsilon), \quad (4.11)$$

for ε sufficiently small. Since $W(\|U\|_\mathbb{X})$ and $W([U])$ are constants, one can find sufficiently small ε such that both ends of [ineq. \(4.11\)](#) are negative. Therefore, applying [prop. 4.5](#) once more gets us to

$$\frac{\|U\|_\theta^\theta}{\theta} \|U\|_\mathbb{X}^{\frac{\theta p}{p_s^*-p}} (W(\|U\|_\mathbb{X}) + \sigma \ln(\varepsilon)) \varepsilon^{N-\sigma\theta} < \mathcal{G}_\varepsilon(t_\varepsilon) < \frac{\|U\|_\theta^\theta}{\theta} [U]^{\frac{\theta p}{p_s^*-p}} (W([U]) + \sigma \ln(\varepsilon)) \varepsilon^{N-\sigma\theta}.$$

Taking the limit as $\varepsilon \rightarrow 0$ concludes the proof. $\square$

We are now ready to discuss — in general terms — the strategy that will be used in the proof of [lemma 4.1](#). The overall idea is to show that

$$\sup_{t \geq 0} \mathcal{I}(tU_\varepsilon) \leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + \mathcal{O}(\varepsilon^{2(N-\sigma p)}). \quad (4.12)$$

We will then show that the surplus of $\mathcal{H}_\varepsilon(t_\varepsilon)$ is $\mathcal{O}(\varepsilon^{N-\sigma p})$, while $\mathcal{G}_\varepsilon(t_\varepsilon)$ is negative and shrinks as $\mathcal{O}(\varepsilon^{N-\sigma\theta} \ln(\varepsilon))$. Therefore, there will exist an ε small enough such that the positive contributions of $\mathcal{H}_\varepsilon(t_\varepsilon)$ are outweighed by a negative $\mathcal{G}_\varepsilon(t_\varepsilon)$.

However, showing [ineq. \(4.12\)](#) will not be trivial. Our plan for this task is to split it in three parts: we will show that there exists $r \in (0, \infty)$ such that [ineq. \(4.12\)](#) is valid when the supremum is taken over the set $\{|t - t_\varepsilon| < r \mid t \in [0, \infty)\}$. Then, we will prove that there exists $R \in (0, \infty)$ such that $\sup_{t > R} \mathcal{I}(tU_\varepsilon)$ is negative. Finally, we will obtain a bound for the supremum of $\mathcal{I}(tU_\varepsilon)$ on the remaining set, i.e.,

$$\sup_{t \in F_\varepsilon} \mathcal{I}(tU_\varepsilon) \leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon), \quad \text{where } F_\varepsilon := [0, t_\varepsilon - r] \cup [t_\varepsilon + r, R]. \quad (4.13)$$

A final note before we start showing [ineq. \(4.12\)](#): it is fundamental for our overarching strategy that we guarantee that the constants r and R remain independent of ε .

Proposition 4.8. *Let $\varepsilon \in (0, \infty)$. Then there exists $\delta \in (0, 1)$ and $C \in (0, \infty)$ such that, for $r := \frac{\delta[U]^{\frac{p}{p_s^*-p}}}{2}$,*

$$\sup_{|t-t_\varepsilon|<r} \mathcal{I}(tU_\varepsilon) \leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + C\varepsilon^{2(N-\sigma p)} \text{ for all } \varepsilon \text{ small enough.}$$

Proof. Since $\frac{p_s^*-1}{p-1} > 1$, there is some $\kappa \in \left(1, \frac{p_s^*-1}{p-1}\right)$. Let $\delta := 1 - \left(\frac{\kappa(p-1)}{p_s^*-1}\right)^{\frac{1}{p_s^*-p}}$.

By Taylor's theorem, there exists $\xi_1 = \xi_1(\tau, \varepsilon) \in (t_\varepsilon - |\tau|, t_\varepsilon + |\tau|)$ such that

$$\begin{aligned} \mathcal{H}_\varepsilon(t_\varepsilon + \tau) - \mathcal{H}_\varepsilon(t_\varepsilon) &= \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{H}'_\varepsilon(t_\varepsilon)\tau + \mathcal{H}''_\varepsilon(\xi_1(\tau, \varepsilon))\frac{\tau^2}{2} - \mathcal{H}_\varepsilon(t_\varepsilon) \\ &= \xi_1^{p-2}((p-1)\|U_\varepsilon\|_\infty^p - (p_s^*-1)\xi_1(\tau, \varepsilon)^{p_s^*-p})\frac{\tau^2}{2} \\ &\leq \xi_1^{p-2}((p-1)([U]^p + \|V\|_\infty\|U\|_p^p\varepsilon^{N-\sigma p}) - (p_s^*-1)\xi_1(\tau, \varepsilon)^{p_s^*-p})\frac{\tau^2}{2}, \end{aligned}$$

where we used [eq. \(4.5\)](#) and the fact that $\mathcal{H}'_\varepsilon(t_\varepsilon) = 0$ (since t_ε is its critical point). By [prop. 4.5](#), $[U]^{\frac{p}{p_s^*-p}} < t_\varepsilon$; thus,

$$r = \frac{\delta[U]^{\frac{p}{p_s^*-p}}}{2} < \delta[U]^{\frac{p}{p_s^*-p}} = [U]^{\frac{p}{p_s^*-p}} - (1-\delta)[U]^{\frac{p}{p_s^*-p}} < t_\varepsilon - (1-\delta)[U]^{\frac{p}{p_s^*-p}}, \quad \therefore$$

$$\xi_1 > t_\varepsilon - |\tau| > t_\varepsilon - r > t_\varepsilon - \left(t_\varepsilon - (1-\delta)[U]^{\frac{p}{p_s^*-p}}\right) = (1-\delta)[U]^{\frac{p}{p_s^*-p}} = \left(\frac{\kappa(p-1)}{p_s^*-1}[U]^p\right)^{\frac{1}{p_s^*-p}} \text{ if } |\tau| < r. \quad (4.14)$$

We draw the reader's attention to the fact that, by using κ and δ , we were able to make r and the bound on $\xi_1(\tau, \varepsilon)$ independent of ε and t_ε . Under the condition for r given in [ineq. \(4.14\)](#), we can write

$$\begin{aligned} \mathcal{H}_\varepsilon(t_\varepsilon + \tau) - \mathcal{H}_\varepsilon(t_\varepsilon) &\leq \xi_1^{p-2} \left((p-1)([U]^p + \|V\|_\infty\|U\|_p^p\varepsilon^{N-\sigma p}) - (p_s^*-1)\xi_1^{p_s^*-p} \right) \frac{\tau^2}{2} \\ &\leq \xi_1^{p-2} \left((p-1)([U]^p + \|V\|_\infty\|U\|_p^p\varepsilon^{N-\sigma p}) - \kappa(p-1)[U]^p \right) \frac{\tau^2}{2} \\ &= \xi_1^{p-2} \left((1-\kappa)(p-1)([U]^p + \|V\|_\infty\|U\|_p^p\varepsilon^{N-\sigma p}) \right) \frac{\tau^2}{2}. \end{aligned}$$

Now, if $A := -\frac{(1-\kappa)}{4} \left(\frac{\kappa}{p_s^*-1}\right)^{\frac{p-2}{p_s^*-p}} ((p-1)[U]^p)^{\frac{p_s^*-2}{p_s^*-p}} > 0$ and $\varepsilon < \left(\frac{(\kappa-1)[U]^p}{2\|V\|_\infty\|U\|_p^p}\right)^{\frac{1}{N-\sigma p}}$, then

$$\begin{aligned} \mathcal{H}_\varepsilon(t_\varepsilon + \tau) - \mathcal{H}_\varepsilon(t_\varepsilon) &\leq \xi_1^{p-2} \left((1-\kappa)(p-1)[U]^p + (p-1)\|V\|_\infty\|U\|_p^p\varepsilon^{N-\sigma p} \right) \frac{\tau^2}{2} \\ &\leq \xi_1^{p-2} \left((1-\kappa)(p-1)[U]^p + \frac{(\kappa-1)(p-1)}{2}[U]^p \right) \frac{\tau^2}{2} = \xi_1^{p-2} \frac{(1-\kappa)(p-1)}{2} [U]^p \frac{\tau^2}{2} = -A\tau^2. \quad (4.15) \end{aligned}$$

Applying Taylor's theorem about t_ε to $\mathcal{G}_\varepsilon$ and proceeding similarly, we get

$$\begin{aligned} \mathcal{G}_\varepsilon(t_\varepsilon + \tau) - \mathcal{G}_\varepsilon(t_\varepsilon) &= \mathcal{G}_\varepsilon(t_\varepsilon) + \mathcal{G}'_\varepsilon(\xi_2)\tau - \mathcal{G}_\varepsilon(t_\varepsilon) = \xi_2^{\theta-1}\varepsilon^{N-\sigma\theta}\|U\|_\theta^\theta \left(-\ln(\xi_2) + \sigma\ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) \tau \\ &\leq \xi_2^{\theta-1}\varepsilon^{N-\sigma\theta}\|U\|_\theta^\theta \left(-\ln\left((1-\delta)[U]^{\frac{p}{p_s^*-p}}\right) + \sigma\ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) \tau \end{aligned}$$

for some $\xi_2 = \xi_2(\varepsilon, \tau) \in (t_\varepsilon - |\tau|, t_\varepsilon + |\tau|)$ (which, for $|\tau| < r$, can easily be shown to obey [ineq. \(4.14\)](#)).

If $B := \left(\frac{\kappa(p-1)}{p_s^*-1} [U]^p \right)^{\frac{\theta-1}{p_s^*-p}} \|U\|_\theta^\theta$ and $\varepsilon < \exp \left(\frac{1}{\sigma} \left(-1 + \ln \left((1-\delta) [U]^{\frac{p}{p_s^*-p}} \right) + \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) \right)$, then

$$\begin{aligned} \mathcal{G}_\varepsilon(t_\varepsilon + \tau) - \mathcal{G}_\varepsilon(t_\varepsilon) &\leq \xi_2^{\theta-1} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(-\ln \left((1-\delta) [U]^{\frac{p}{p_s^*-p}} \right) + \sigma \ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) \tau \\ &\leq -\xi_2^{\theta-1} \|U\|_\theta^\theta \varepsilon^{N-\sigma\theta} \tau < - \left((1-\delta) [U]^{\frac{p}{p_s^*-p}} \right)^{\theta-1} \|U\|_\theta^\theta \varepsilon^{N-\sigma\theta} \tau = -B \varepsilon^{N-\sigma\theta} \tau. \end{aligned} \quad (4.16)$$

Combining [ineq. \(4.16\)](#) with [ineq. \(4.15\)](#) yields, for all ε sufficiently small,

$$\sup_{|t-t_\varepsilon|<r} \mathcal{I}(tU_\varepsilon) = \sup_{|\tau|<r} \mathcal{I}((t_\varepsilon + \tau)U_\varepsilon) = \sup_{|\tau|<r} (\mathcal{H}_\varepsilon(t_\varepsilon + \tau) + \mathcal{G}_\varepsilon(t_\varepsilon + \tau)) \leq \sup_{|\tau|<r} (\mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) - A\tau^2 - B\varepsilon^{N-\sigma p}\tau).$$

As $\tau \mapsto -A\tau^2 - B\varepsilon^{N-\sigma p}\tau$ has a global maximum at $\tau = -\frac{B\varepsilon^{N-\sigma p}}{2A}$, we conclude, for $C := \frac{B^2}{4A} > 0$, that

$$\begin{aligned} \sup_{|t-t_\varepsilon|<r} \mathcal{I}(tU_\varepsilon) &\leq \sup_{|\tau|<r} (\mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) - A\tau^2 - B\varepsilon^{N-\sigma p}\tau) \\ &\leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) - A \left(-\frac{B\varepsilon^{N-\sigma p}}{2A} \right)^2 - B\varepsilon^{N-\sigma p} \left(-\frac{B\varepsilon^{N-\sigma p}}{2A} \right) \\ &= \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + \frac{B^2 \varepsilon^{2(N-\sigma p)}}{4A} = \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + C\varepsilon^{2(N-\sigma p)}. \end{aligned} \quad \square$$

Proposition 4.9. *Let $\varepsilon \in (0, \infty)$. There exists $R \in \left(\|U\|_\chi^{\frac{p}{p_s^*-p}} + r, \infty \right)$ (where r is as in [prop. 4.8](#)) such that $\mathcal{H}_\varepsilon(R) < 0$ and, for all ε small enough, $\sup_{t>R} \mathcal{I}(tU_\varepsilon) < 0$.*

Proof. Let $R := \max \left\{ \|U\|_\chi^{\frac{p}{p_s^*-p}} + r + 1, \left(\frac{p_s^*}{p} \|U\|_\chi^p \right)^{\frac{1}{p_s^*-p}} \right\}$. If $t \in (R, \infty)$ and ε is as small as in [prop. 4.5](#), then

$$\mathcal{H}_\varepsilon(t) = \frac{t^p}{p} \|U_\varepsilon\|_\chi^p - \frac{t^{p_s^*}}{p_s^*} = t^p \left(\frac{\|U_\varepsilon\|_\chi^p}{p} - \frac{t^{p_s^*-p}}{p_s^*} \right) < t^p \left(\frac{\|U_\varepsilon\|_\chi^p}{p} - \frac{R^{p_s^*-p}}{p_s^*} \right) \leq t^p \left(\frac{\|U_\varepsilon\|_\chi^p}{p} - \frac{\|U\|_\chi^p}{p} \right) < 0,$$

i.e., $\mathcal{H}_\varepsilon(R) < 0$. Imposing $\varepsilon < \exp \left(\frac{1}{\sigma} \left(\frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} - \frac{1}{\theta} + \ln \left(\left(\frac{p_s^*}{p} \|U\|_\chi^p \right)^{\frac{1}{p_s^*-p}} \right) \right) \right)$, the proof is finished by

$$\begin{aligned} \mathcal{G}_\varepsilon(t) &= \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(t) + \sigma \ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) < \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \ln(R) + \sigma \ln(\varepsilon) - \frac{\mathcal{L}(U)}{\|U\|_\theta^\theta} \right) \\ &< \frac{t^\theta}{\theta} \varepsilon^{N-\sigma\theta} \|U\|_\theta^\theta \left(\ln \left(\left(\frac{p_s^*}{p} \|U\|_\chi^p \right)^{\frac{1}{p_s^*-p}} \right) - \ln(R) \right) < 0. \end{aligned} \quad \square$$

For [ineq. \(4.12\)](#) to be fully proven, it remains to show [ineq. \(4.13\)](#). However, this part of the argument is harder than the other two, and we will need to prove some auxiliary results before we attack it directly. In showing these results (and in the proof of [lemma 4.1](#)), we will extensively use the following inequality, which we state as a proposition for future reference.

Proposition 4.10. *Let $a, q, r \in (0, \infty)$ and $f: (0, \infty) \rightarrow \mathbb{R}$ be a function such that $\sup_{0 < x < r} |f(x)| < a$. Then there exists constants $K_1, K_2 \in \mathbb{R}$ such that*

$$a^q + K_1 f(\varepsilon) \leq (a + f(\varepsilon))^q \leq a^q + K_2 f(\varepsilon)$$

for every $\varepsilon \in (0, r)$. Furthermore, if $f(x) > 0$ in a neighborhood of 0, then $K_1, K_2 > 0$.

Proof. Consider the function $g: \mathbb{R} \rightarrow \mathbb{R}$; $g(x) = (a \pm x)^q$. For every $x \in (0, \infty)$, we can use Taylor's theorem centered at 0 applied to g to find $\xi \in (0, x)$ such that

$$g(x) = (a \pm x)^q = g(0) \pm g'(\xi)x = a^q \pm q(a \pm \xi)^{q-1}x.$$

Let $\varepsilon \in (0, r)$. By making $x = |f(\varepsilon)|$ in the previous equation, we get, for some new $\xi \in (0, |f(\varepsilon)|)$,

$$(a \pm |f(\varepsilon)|)^q = a^q \pm q(a \pm \xi)^{q-1} |f(\varepsilon)|. \quad (4.17)$$

The sign of $q-1$ is unknown. However, since $a > 0$ and $0 < \xi < |f(\varepsilon)| < a$, we can infer that $|a \pm \xi|$ remains bounded below by a positive constant and bounded above. Thus, we can find $K_1, K_2 \in \mathbb{R}$ satisfying the proposition regardless of the sign of $q-1$.

Furthermore, if there exists a neighborhood of 0 where f remains strictly positive, we can use $+$ instead of $\pm$ in [ineq. \(4.17\)](#), which, along with the positivity of a and q , yields that $K_1, K_2 > 0$. $\square$

Our proof of [ineq. \(4.13\)](#) will rely on an uniform bound of the type

$$\inf_{t \in F_\varepsilon} (\mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{H}_\varepsilon(t)) > \lambda, \quad (4.18)$$

where $\lambda \in (0, \infty)$ is independent of ε . Our strategy to show it uses the fact that the supremum of $\mathcal{H}_\varepsilon$ over the compact set F_ε must be attained at the boundary. Because $\mathcal{H}_\varepsilon$ at $t=0$ (resp. $t=R$) is zero (resp. negative) and $\mathcal{H}_\varepsilon(t_\varepsilon - r) > 0$, the maximum must occur at either $t = t_\varepsilon - r$ or $t_\varepsilon + r$. The next result shows that the former is the case.

Proposition 4.11. *Let $\varepsilon \in (0, \infty)$ and $\tau \in (0, t_\varepsilon)$. Then $\mathcal{H}_\varepsilon(t_\varepsilon - \tau) > \mathcal{H}_\varepsilon(t_\varepsilon + \tau)$.*

Proof. By the definition of t_ε we have, for any $t \in (0, \infty)$, that

$$\begin{aligned} \mathcal{H}_\varepsilon(tt_\varepsilon) &= \frac{(tt_\varepsilon)^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{(tt_\varepsilon)^{p_s^*}}{p_s^*} = \frac{t^p}{p} t_\varepsilon^p \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} t_\varepsilon^{p_s^*} = \frac{t^p}{p} \left(\|U_\varepsilon\|_{\mathbb{X}}^{\frac{p}{p_s^*-p}} \right)^p \|U_\varepsilon\|_{\mathbb{X}}^p - \frac{t^{p_s^*}}{p_s^*} \left(\|U_\varepsilon\|_{\mathbb{X}}^{\frac{p}{p_s^*-p}} \right)^{p_s^*} \\ &= \frac{t^p}{p} \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} - \frac{t^{p_s^*}}{p_s^*} \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} = \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} \left(\frac{t^p}{p} - \frac{t^{p_s^*}}{p_s^*} \right) = \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} \varphi(t), \end{aligned}$$

where $\varphi: [0, \infty) \rightarrow \mathbb{R}$; $\varphi(t) := \frac{t^p}{p} - \frac{t^{p_s^*}}{p_s^*}$. Noting that $\frac{\tau}{t_\varepsilon} =: c \in (0, 1)$, we conclude that

$$\begin{aligned} \mathcal{H}_\varepsilon(t_\varepsilon - \tau) - \mathcal{H}_\varepsilon(t_\varepsilon + \tau) &= \mathcal{H}_\varepsilon\left(\left(1 + \frac{\tau}{t_\varepsilon}\right)t_\varepsilon\right) - \mathcal{H}_\varepsilon\left(\left(1 - \frac{\tau}{t_\varepsilon}\right)t_\varepsilon\right) \\ &= \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} \left(\varphi\left(1 + \frac{\tau}{t_\varepsilon}\right) - \varphi\left(1 - \frac{\tau}{t_\varepsilon}\right) \right) = \|U_\varepsilon\|_{\mathbb{X}}^{\frac{pp_s^*}{p_s^*-p}} \left(\varphi(1-c) - \varphi(1+c) \right), \end{aligned}$$

where $c := \frac{\tau}{t_\varepsilon}$. Thus, it suffices to show that $\varphi(1-c) - \varphi(1+c) > 0$.

Let $h_c: (2, \infty) \rightarrow \mathbb{R}$; $h_c(r) := \frac{(1-c)^r - (1+c)^r}{r}$. Then

$$\begin{aligned} \varphi(1-c) - \varphi(1+c) &= \frac{(1-c)^p}{p} - \frac{(1-c)^{p_s^*}}{p_s^*} - \left(\frac{(1+c)^p}{p} - \frac{(1+c)^{p_s^*}}{p_s^*} \right) \\ &= \frac{(1-c)^p - (1+c)^p}{p} - \frac{(1-c)^{p_s^*} - (1+c)^{p_s^*}}{p_s^*} = h_c(p) - h_c(p_s^*). \end{aligned}$$

Therefore, the problem is reduced to showing that h_c is decreasing, i.e., that $h'(r) < 0$.

Motivated by the fact that

$$\begin{aligned} h'_c(r) &= \frac{1}{r^2} (\ln(1-c)(1-c)^r - \ln(1+c)(1+c)^r) - ((1-c)^r - (1+c)^r) \\ &= \frac{1}{r^2} ((1-c)^r (r \ln(1-c) - 1) - (1+c)^r (r \ln(1+c) - 1)), \end{aligned}$$

we now define, for any $r \in [p, p_s^*]$, the functions $f_r: (-1, 1)$; $f_r(c) = (1+c)^r (r \ln(1+c) - 1)$ and $g_r: (0, 1) \rightarrow \mathbb{R}$; $g_r(c) := f_r(-c) - f_r(c)$. It is easy to check that $h'_c(r) = \frac{g_r(c)}{r^2}$ and $f'_r(c) = r^2(1+c)^{r-1} \ln(1+c)$.

Thus, using [prop. A.3](#), we get

$$\begin{aligned}
g'_r(c) &= -r^2(1-c)^{r-1}\ln(1-c) - r^2(1+c)^{r-1}\ln(1+c) \\
&= -r^2 \left(-(1-c)^{r-1}c \int_0^1 \frac{1}{1-ct} dt + (1+c)^{r-1}c \int_0^1 \frac{1}{1+ct} dt \right) \\
&= -cr^2 \left((1+c)^{r-1} \int_0^1 \frac{1}{1+ct} dt - (1-c)^{r-1} \int_0^1 \frac{1}{1-ct} dt \right) \\
&= -cr^2 \left(\int_0^1 \frac{(1+c)^{r-1}}{1+ct} - \frac{(1-c)^{r-1}}{1-ct} dt \right) = -cr^2 \left(\int_0^1 \frac{(1+c)^{r-1}(1-ct) - (1-c)^{r-1}(1+ct)}{1-c^2t^2} dt \right).
\end{aligned}$$

Since $c \in (0,1)$, we have, for all $t < 1$,

$$\begin{aligned}
(1+c)^{r-1}(1-ct) - (1-c)^{r-1}(1+ct) &> (1+c)^{r-1}(1-c) - (1-c)^{r-1}(1+c) \\
&= (1-c^2)((1+c)^{r-2} - (1-c)^{r-2}) > 0,
\end{aligned}$$

meaning that $g'_r(c) < 0$, i.e., g_r is a strictly decreasing function. As $\lim_{c \rightarrow 0} g_r(c) = 0$, the proof is done $\square$

Proposition 4.12. *There exists $\lambda \in (0, \infty)$ such that [ineq. \(4.18\)](#) holds for all $\varepsilon \in (0, \infty)$ small enough.*

Proof. As discussed previously and now enlightened by [prop. 4.11](#), we know that $\sup_{t \in F_\varepsilon} \mathcal{H}_\varepsilon(t) = \mathcal{H}_\varepsilon(t_\varepsilon - r)$. Together with [eq. \(4.9\)](#) and properties of the infimum and supremum, this leads us to

$$\begin{aligned}
\inf_{t \in F_\varepsilon} (\mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{H}_\varepsilon(t)) &= \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \|U_\varepsilon\|_\mathbb{X}^\omega + \inf_{t \in F_\varepsilon} \left(-\mathcal{H}_\varepsilon(t) \right) \\
&= \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \|U_\varepsilon\|_\mathbb{X}^\omega - \sup_{t \in F_\varepsilon} \mathcal{H}_\varepsilon(t) = \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \|U_\varepsilon\|_\mathbb{X}^\omega - \mathcal{H}_\varepsilon(t_\varepsilon - r).
\end{aligned} \tag{4.19}$$

Note that [\(V₁\)](#) and [prop. 4.3](#) imply that

$$0 < [U]^p + m_V \|U\|_p^p \varepsilon^{N-\sigma p} < \|U_\varepsilon\|_\mathbb{X}^p < [u]^p + \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p}.$$

Thus, [prop. 4.10](#) and the definitions of t_ε and r (in [prop. 4.8](#)) imply that

$$\begin{aligned}
\left(1 - \frac{\delta}{2}\right) [U]^{\frac{p}{p_s^* - p}} + K_1 m_V \|U\|_p^p \varepsilon^{N-\sigma p} &\leq \left([U]^p + \int_{\mathbb{R}^N} V(x) |U_\varepsilon(x)|^p dx \right)^{\frac{1}{p_s^* - p}} - \frac{\delta}{2} [U]^{\frac{p}{p_s^* - p}} \\
&= \|U_\varepsilon\|_\mathbb{X}^{\frac{p}{p_s^* - p}} - \frac{\delta}{2} [U]^{\frac{p}{p_s^* - p}} = (t_\varepsilon - r) \leq \left(1 - \frac{\delta}{2}\right) [U]^{\frac{p}{p_s^* - p}} + K_2 \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p} \varepsilon^{N-\sigma p}.
\end{aligned} \tag{4.20}$$

Henceforth, we will use the shorthand $\delta' := 1 - \frac{\delta}{2} \in (0,1)$ and will absorb the positive constants $m_V, \|V\|_\infty$ and $\|U\|_p^p$ into K_1 and K_2 . Using [ineq. \(4.20\)](#) and [prop. 4.10](#) in the expression $\frac{\|U_\varepsilon\|_\mathbb{X}^p}{p} - \frac{1}{p_s^*}$, we obtain

$$\begin{aligned}
\frac{\|U_\varepsilon\|_\mathbb{X}^p}{p} - \frac{1}{p_s^*} (t_\varepsilon - r)^{p_s^* - p} &\leq \frac{1}{p} \left([U]^p + \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p} \right) - \frac{1}{p_s^*} \left(\delta' [U]^{\frac{p}{p_s^* - p}} + K_1 \varepsilon^{N-\sigma p} \right)^{p_s^* - p} \\
&\leq \frac{[U]^p}{p} + K_3 \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p} - \frac{\delta'^{p_s^* - p}}{p_s^*} [U]^p - K_4 \varepsilon^{N-\sigma p} = \left(\frac{1}{p} - \frac{\delta'^{p_s^* - p}}{p_s^*} \right) [U]^p + K_5 \varepsilon^{N-\sigma p},
\end{aligned} \tag{4.21}$$

where $K_3, K_4 \in (0, \infty)$ and $K_5 \in \mathbb{R}$. On the other hand, [prop. 4.10](#) and [ineq. \(4.20\)](#) also imply that

$$(t_\varepsilon - r)^p \leq \left(\delta' [U]^{\frac{p}{p_s^* - p}} + K_2 \varepsilon^{N-\sigma p} \right)^p \leq \left(\left(\delta' [U]^{\frac{p}{p_s^* - p}} \right)^p + K_6 \varepsilon^{N-\sigma p} \right), \tag{4.22}$$

where $K_6 \in (0, \infty)$. For some $K_7 \in \mathbb{R}$ and all $\varepsilon < 1$, the inequation above along with [ineq. \(4.21\)](#) leads us to

$$\begin{aligned}
\mathcal{H}_\varepsilon(t_\varepsilon - r) &= (t_\varepsilon - r)^p \left(\frac{\|U_\varepsilon\|_\mathbb{X}^p}{p} - \frac{1}{p_s^*} (t_\varepsilon - r)^{p_s^* - p} \right) \\
&\leq \left(\left(\delta' [U]^{\frac{p}{p_s^* - p}} \right)^p + K_6 \varepsilon^{N-\sigma p} \right) \left(\left(\frac{1}{p} - \frac{\delta'^{p_s^* - p}}{p_s^*} \right) [U]^p + K_5 \varepsilon^{N-\sigma p} \right) \\
&\leq \delta'^p \left(\frac{1}{p} - \frac{\delta'^{p_s^* - p}}{p_s^*} \right) [U]^{\frac{p^2}{p_s^* - p} + p} + K_7 \varepsilon^{N-\sigma p} = \left(\frac{\delta'^p}{p} - \frac{\delta'^{p_s^*}}{p_s^*} \right) [U]^\omega + K_7 \varepsilon^{N-\sigma p}.
\end{aligned} \tag{4.23}$$

By plugging this result back into eq. (4.19) and then using (V₁), props. 4.3 and 4.10, we get

$$\begin{aligned}
\inf_{t \in F_\varepsilon} (\mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{H}_\varepsilon(t)) &= \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \|U_\varepsilon\|_\infty^\omega - \mathcal{H}_\varepsilon(t_\varepsilon - r) \\
&\geq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) ([U]^p + m_V \|U\|_p^p \varepsilon^{N-\sigma p})^{\frac{p_s^*}{p_s^* - p}} - \left(\frac{\delta'^p}{p} - \frac{\delta'^{p_s^*}}{p_s^*} \right) [U]^\omega - K_7 \varepsilon^{N-\sigma p} \\
&\geq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega + K_8 \varepsilon^{N-\sigma p} - \left(\frac{\delta'^p}{p} - \frac{\delta'^{p_s^*}}{p_s^*} \right) [U]^\omega - K_7 \varepsilon^{N-\sigma p} \\
&\geq \left(\left(\frac{1}{p} - \frac{1}{p_s^*} \right) - \left(\frac{\delta'^p}{p} - \frac{\delta'^{p_s^*}}{p_s^*} \right) \right) [U]^\omega + K_9 \varepsilon^{N-\sigma p}.
\end{aligned}$$

Note that the function $\varphi: [0,1] \rightarrow \mathbb{R}$; $\varphi(t) := \frac{t^p}{p} - \frac{t^{p_s^*}}{p_s^*}$ has a global maximum at $t=1$. Since $\delta' \in (0,1)$, it follows that $\varphi(1) - \varphi(\delta') > 0$. Set $\lambda := \frac{1}{2}(\varphi(1) - \varphi(\delta'))[U]^\omega > 0$. If $K_9 \geq 0$, we are done. Otherwise, if ε is also smaller than $\left(\frac{\lambda}{|K_9|} \right)^{\frac{1}{N-\sigma p}}$, we have

$$\begin{aligned}
\inf_{t \in F_\varepsilon} (\mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{H}_\varepsilon(t)) &> \left(\left(\frac{1}{p} - \frac{1}{p_s^*} \right) - \left(\frac{\delta'^p}{p} - \frac{\delta'^{p_s^*}}{p_s^*} \right) \right) [U]^\omega + K_9 \varepsilon^{N-\sigma p} = 2\lambda - |K_9| \varepsilon^{N-\sigma p} \\
&> 2\lambda - |K_9| \left(\left(\frac{\lambda}{|K_9|} \right)^{\frac{1}{N-\sigma p}} \right)^{N-\sigma p} = 2\lambda - |K_9| \left(\frac{\lambda}{|K_9|} \right) = 2\lambda - \lambda = \lambda. \quad \square
\end{aligned}$$

Armed with props. 4.11 and 4.12, we are now going to bound the supremum of $\mathcal{I}(tU_\varepsilon)$ over F_ε .

Proposition 4.13. *There exists $\lambda \in (0, \infty)$ such that [ineq. \(4.13\)](#) holds for all $\varepsilon \in (0, \infty)$ small enough.*

Proof. By [prop. 4.12](#) and properties of the supremum and infimum, we know that, for ε sufficiently small,

$$\begin{aligned}
\sup_{t \in F_\varepsilon} \mathcal{I}(tU_\varepsilon) - \mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{G}_\varepsilon(t_\varepsilon) &\leq \left(\sup_{t \in F_\varepsilon} \mathcal{H}_\varepsilon(t) \right) + \left(\sup_{t \in F_\varepsilon} \mathcal{G}_\varepsilon(t) \right) - \mathcal{H}_\varepsilon(t_\varepsilon) - \mathcal{G}_\varepsilon(t_\varepsilon) \\
&= \left(\sup_{t \in F_\varepsilon} \left(\mathcal{H}_\varepsilon(t) - \mathcal{H}_\varepsilon(t_\varepsilon) \right) \right) + \sup_{t \in F_\varepsilon} \mathcal{G}_\varepsilon(t) - \mathcal{G}_\varepsilon(t_\varepsilon) < -\lambda + \sup_{t \in F_\varepsilon} \mathcal{G}_\varepsilon(t) - \mathcal{G}_\varepsilon(t_\varepsilon).
\end{aligned}$$

Since, for any $\varepsilon \in (0, \infty)$, $\mathcal{G}_\varepsilon$ has a global maximum at $\tilde{t}_\varepsilon := \exp\left(\sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta}\right)$, we can write

$$\sup_{t \in F_\varepsilon} \mathcal{G}_\varepsilon(t) \leq \sup_{t \in [0, \infty)} \mathcal{G}_\varepsilon(t) = \mathcal{G}_\varepsilon(\tilde{t}_\varepsilon) = \frac{1}{\theta^2} \exp\left(\sigma \theta \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta}\right) \varepsilon^{N-\sigma \theta} \|U\|_\theta^\theta. \quad (4.24)$$

Note that the right-hand side of [ineq. \(4.24\)](#) goes to zero as $\varepsilon \rightarrow 0$. Thus, it holds that $\sup\{\mathcal{G}_\varepsilon(t) \mid t \in F_\varepsilon\} \leq \frac{\lambda}{2}$ for all ε small enough. If necessary, we can further reduce ε to get, by [prop. 4.7](#), that $\mathcal{G}_\varepsilon(t_\varepsilon) \geq -\frac{\lambda}{2}$. $\square$

We are now ready to prove [lemma 4.1](#).

Proof of lemma 4.1. By [eq. \(4.9\)](#) and [prop. 4.7](#), we know that $\mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) > 0$ for ε sufficiently small. This means that props. 4.8, 4.9, and 4.13 jointly imply that

$$\sup_{t \in [0,1]} \mathcal{I}(tU_\varepsilon) \leq \sup_{t \geq 0} \mathcal{I}(tU_\varepsilon) \leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + C \varepsilon^{2(N-\sigma p)} \quad (4.25)$$

for some $C \in (0, \infty)$ and ε small enough.

By props. 4.3, 4.10, and [eq. \(4.9\)](#), we have, for some $K_1 \in (0, \infty)$,

$$\begin{aligned}
\mathcal{H}_\varepsilon(t_\varepsilon) &= \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \left([U]^\omega + \int_{\mathbb{R}^N} V(x) |U_\varepsilon(x)| dx \right)^{\frac{p_s^*}{p_s^* - p}} \\
&\leq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) ([U]^p + \|V\|_\infty \|U\|_p^p \varepsilon^{N-\sigma p})^{\frac{p_s^*}{p_s^* - p}} \leq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega + K_1 \varepsilon^{N-\sigma p}.
\end{aligned} \quad (4.26)$$

We now impose that ε is below the threshold of [prop. 4.5](#) and also satisfies $\varepsilon < \exp\left(\frac{1}{\sigma}\left(\frac{p}{p_s^* - p} \ln([U]) + \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} - \frac{1}{\theta}\right)\right)$; this guarantees that we can use the aforementioned proposition, and that the term in parentheses on the right-hand side of [eq. \(4.10\)](#) is strictly negative. Under these conditions, we get

$$\begin{aligned} \mathcal{G}_\varepsilon(t_\varepsilon) &= \frac{\|U_\varepsilon\|_{\mathbb{X}}^{\frac{\theta p}{(p_s^* - p)}}}{\theta} \varepsilon^{N - \sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \frac{p}{p_s^* - p} \ln(\|U_\varepsilon\|_{\mathbb{X}}) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right) \\ &\leq \frac{[U]^{\frac{\theta p}{(p_s^* - p)}}}{\theta} \varepsilon^{N - \sigma\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \frac{p}{p_s^* - p} \ln([U]) + \sigma \ln(\varepsilon) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right) = K_2 \varepsilon^{N - \sigma\theta} + K_3 \varepsilon^{N - \sigma\theta} \ln(\varepsilon), \end{aligned} \quad (4.27)$$

where $\frac{[U]^{\frac{\theta p}{(p_s^* - p)}}}{\theta} \|U\|_\theta^\theta \left(\frac{1}{\theta} - \frac{p}{p_s^* - p} \ln([U]) - \frac{\mathcal{Z}(U)}{\|U\|_\theta^\theta} \right) =: K_2 \in \mathbb{R}$ and $\frac{\sigma[U]^{\frac{\theta p}{(p_s^* - p)}}}{\theta} \|U\|_\theta^\theta =: K_3 \in (0, \infty)$.

Now we can put ineqs [\(4.25\)](#), [\(4.26\)](#), and [ineq. \(4.27\)](#) together to find that

$$\begin{aligned} \sup_{t \in [0, 1]} \mathcal{I}(tU_\varepsilon) &\leq \mathcal{H}_\varepsilon(t_\varepsilon) + \mathcal{G}_\varepsilon(t_\varepsilon) + C\varepsilon^{2(N - \sigma p)} \\ &\leq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega + K_1 \varepsilon^{N - \sigma p} + K_2 \varepsilon^{N - \sigma\theta} + K_3 \varepsilon^{N - \sigma\theta} \ln(\varepsilon) + C\varepsilon^{2(N - \sigma p)}, \end{aligned} \quad (4.28)$$

for ε sufficiently small. Note that $N - \sigma\theta < N - \sigma p < 2(N - \sigma p)$, and so, as long as $\varepsilon < 1$, we can write

$$\sup_{t \in [0, 1]} \mathcal{I}(tU_\varepsilon) \leq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega + (K_1 + K_2 + K_3 \ln(\varepsilon) + C) \varepsilon^{N - \sigma\theta}.$$

Finally, if ε^* is small enough for all previous assumptions and propositions to hold and also satisfies $\varepsilon^* < \exp\left(\frac{-K_1 - K_2 - C - 1}{K_3}\right)$, then, since $K_3 > 0$, we conclude that

$$\begin{aligned} \sup_{t \in [0, 1]} \mathcal{I}(tU_{\varepsilon^*}) &\leq \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega + (K_1 + K_2 + K_3 \ln(\varepsilon^*) + C) \varepsilon_*^{N - \sigma\theta} \\ &< \left(\frac{1}{p} - \frac{1}{p_s^*} \right) [U]^\omega - \varepsilon_*^{N - \sigma\theta} = \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \mathcal{S}_{s, p}^\omega - \varepsilon_*^{N - \sigma\theta} < \left(\frac{1}{p} - \frac{1}{p_s^*} \right) \mathcal{S}_{s, p}^\omega. \end{aligned} \quad \square$$

5 The periodic case

Since our goal in this section is to show [theorem 1.1](#), we always assume here — most often implicitly — that V satisfies [\(V₁\)](#) and [\(V₂\)](#). Our strategy for such a proof relies on the existence of a $(PS)_c$ sequence (see [equations \(1.4\)](#)), where c is the mountain pass level (see [eq. \(1.2\)](#)). The existence of such a sequence from [theorem 5.1](#) of [\[13\]](#); the fact that $\mathcal{I}$ satisfies the mountain pass geometry (see [lemma 3.4](#)) imply the hypotheses needed to use the aforementioned theorem. Accordingly, our first step now is to establish some properties of (PS) sequences for $\mathcal{I}$.

Proposition 5.1. *Let $C \in (0, \infty)$, (u_n) be a (PS) sequence and $(v_n) \subset \mathbb{X}$ such that $\sup_{n \in \mathbb{N}} \|v_n\|_{\mathbb{X}} < C$. Then $\mathcal{I}'(u_n)(v_n) \rightarrow 0$.*

Proof. Follows from using the definition of the norm of $\mathbb{X}'$ on $|\mathcal{I}'(u_n)(v_n)|$, applying the hypothesis that (v_n) is uniformly bounded, then taking the limit as $n \rightarrow \infty$ and finally using that (u_n) is (PS) . $\square$

Corollary 5.2. *Let (u_n) be a (PS) sequence. Then $\inf_{n \in \mathbb{N}} \mathcal{I}(u_n) \geq 0$.*

Proof. Since $p < \theta < p_s^*$, eqs. [\(2.1\)](#) and [\(2.4\)](#) imply that

$$\mathcal{I}(u) - \frac{1}{\theta} \mathcal{I}'(u)(u) = \frac{\theta - p}{\theta p} \|u\|_{\mathbb{X}}^p + \frac{\|u\|_\theta^\theta}{\theta^2} + \beta \frac{p_s^* - \theta}{p_s^* \theta} \|u\|_{p_s^*}^{p_s^*} \quad (5.1)$$

for any $u \in \mathbb{X}$. Swap u with u_n in the equation above, and note that the right-hand side of that equation is always non-negative. Thus, taking the limit inferior on both sides and using [prop. 5.1](#) (which we can because of [prop. 5.5](#)), we get $\liminf \mathcal{I}(u_n) \geq 0$, as desired. $\square$

The next result requires the following definition: for any $z \in \mathbb{Z}^N$, we define the translation-by- z function as $\tau_z: \mathbb{R}^N \rightarrow \mathbb{R}^N$; $\tau_z(x) = x + z$.

Proposition 5.3. *Let $\varepsilon, R \in (0, \infty)$, $(z_n) \subset \mathbb{Z}^N$ and $(u_n) \subset \mathbb{X}$ such that $\sup_{n \in \mathbb{N}} \|u_n\|_{\mathbb{X}} < \infty$ and $\lim \|u_n\|_{L^p(\mathcal{B}(z_n, R))} > 0$. Then there exists $u \in \mathbb{X} - \{0\}$ and subsequences such that $u_n \circ \tau_{z_n} \rightharpoonup u$.*

Proof. Consider the sequence $(v_n) := (u_n \circ \tau_{z_n})$. Two changes of variables and an application of (V₂) shows us that $\|\cdot\|_{\mathbb{X}}$ is $\mathbb{Z}^N$ -translation invariant, which means that $\|v_n\|_{\mathbb{X}} = \|u_n\|_{\mathbb{X}}$, and so (v_n) is also bounded in $\mathbb{X}$. This and the fact that $\mathbb{X}$ is reflexive means that there exists $u \in \mathbb{X}$ such that, up to a subsequence, $v_n \rightharpoonup u$ in $\mathbb{X}$. Passing again to a subsequence if necessary, corollary 2.3 implies that $v_n \rightarrow v$ in $L^p(\mathcal{B}(0, R))$. A change of variables and our second hypothesis leads us to

$$\|v\|_{L^p(\mathcal{B}(0, R))} = \lim \|v_n\|_{L^p(\mathcal{B}(0, R))} = \lim \|u_n\|_{L^p(\mathcal{B}(z_n, R))} > \varepsilon > 0. \quad \square$$

Proposition 5.4. *Let $\varrho \in (0, \infty)$, (u_n) a $(PS)_{\varrho}$ sequence, $(z_n) \subset \mathbb{Z}^N$. Then $(u_n \circ \tau_{z_n})$ is a $(PS)_{\varrho}$ sequence.*

Proof. A change of variables along with (V₂) shows that $\mathcal{I}$ is invariant by $\mathbb{Z}^N$ -translations. Since $z_n \in \mathbb{Z}^N$, we have that $\lim \mathcal{I}(u_n \circ \tau_{z_n}) = \lim \mathcal{I}(u_n) = \varrho$.

Now take an arbitrary $\phi \in \mathbb{X}$. By $\mathbb{Z}^N$ -translation invariance, we have that $\|\phi \circ \tau_{z_n}\|_{\mathbb{X}} = \|\phi\|_{\mathbb{X}}$, which is trivially bounded. By making some more changes of variables and using prop. 5.1, we then get

$$\mathcal{I}'(u_n \circ \tau_{z_n})(\phi) = \mathcal{I}'(u_n)(\phi \circ \tau_{z_n}) \rightarrow 0.$$

Since ϕ was arbitrary in $\mathbb{X}$, it follows that $\|\mathcal{I}'(u_n \circ \tau_{z_n})\|_{\mathbb{X}'} \rightarrow 0$, i.e., (v_n) is a $(PS)_{\varrho}$ sequence. $\square$

Proposition 5.5. *Let (u_n) be a (PS) sequence. Then $\sup_{n \in \mathbb{N}} \|u_n\|_{\mathbb{X}} < \infty$.*

Proof. Since (u_n) is (PS) , there exists $C \in (0, \infty)$ such that $\mathcal{I}(u_n) < C$ for all $n \in \mathbb{N}$. Then, for $n \in \mathbb{N}$, we can make $u = u_n$ in eq. (5.1) to get

$$C - \frac{1}{\theta} \mathcal{I}'(u_n)(u_n) > \left(\sup_{n \in \mathbb{N}} \mathcal{I}(u_n) \right) - \frac{1}{\theta} \mathcal{I}'(u_n)(u_n) \geq \mathcal{I}(u_n) - \frac{1}{\theta} \mathcal{I}'(u_n)(u_n) \geq \frac{\theta - p}{\theta p} \|u_n\|_{\mathbb{X}}^p.$$

Taking the absolute value using, using this last inequality and the definition of the norm of $\mathbb{X}'$, we get

$$C + \frac{1}{\theta} \|\mathcal{I}'(u_n)\|_{\mathbb{X}'} \|u_n\|_{\mathbb{X}} \geq C + \frac{1}{\theta} |\mathcal{I}'(u_n)(u_n)| \geq C - \frac{1}{\theta} \mathcal{I}'(u_n)(u_n) \geq \frac{\theta - p}{\theta p} \|u_n\|_{\mathbb{X}}^p.$$

Using again that (u_n) is (PS) , we find $n_0 \in \mathbb{N}$ such that $\|\mathcal{I}'(u_n)\|_{\mathbb{X}'} < \theta$ for all $n \in \mathbb{N} \cap (n_0, \infty)$. Thus,

$$C + \|u_n\|_{\mathbb{X}} = C + \frac{1}{\theta} \theta \|u_n\|_{\mathbb{X}} > C + \frac{1}{\theta} \|\mathcal{I}'(u_n)\|_{\mathbb{X}'} \|u_n\|_{\mathbb{X}} \geq \frac{\theta - p}{\theta p} \|u_n\|_{\mathbb{X}}^p \text{ for } n \text{ big enough.}$$

Let $A := \frac{C\theta p}{\theta - p}$ and $B := \frac{\theta p}{\theta - p}$, and note that they are both strictly positive. This and the last equation imply, for sufficiently large $n \in \mathbb{N}$, that $A + B \|u_n\|_{\mathbb{X}} \geq \|u_n\|_{\mathbb{X}}^p$, i.e., $\frac{A}{\|u_n\|_{\mathbb{X}}^p} + \frac{B}{\|u_n\|_{\mathbb{X}}^{p-1}} \geq 1$. Now if $\|u_n\|_{\mathbb{X}} \rightarrow \infty$, taking the limit as $n \rightarrow \infty$ on both sides of the inequation displayed above would yield $0 > 1$, a contradiction. This completes the proof. $\square$

This last result, together the fact that $\mathbb{X}$ is reflexive, will allow us to extract a weakly convergent subsequence of our (PS) sequence. However, there is no guarantee that this weak limit — which would be a candidate for a weak solution of problem (P), — is nontrivial.

Strongly inspired by the concentration-compactness principle [17], we set out to find a sequence of translations of our (PS) sequence with non-trivial weak limit.

Proposition 5.6. *Let $C, R \in (0, \infty)$, $(u_n) \subset \mathbb{X}$ such that $\limsup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^p = 0$ and $\|u_n\|_{\mathbb{X}} < C$ for all $n \in \mathbb{N}$. Then $u_n \rightarrow 0$ in $L^q(\mathbb{R}^N)$ for all $q \in (p, p_s^*)$*

Proof. Let $q \in (p, p_s^*)$. By Littlewood's inequality, there exists $\lambda \in (0, 1)$ such that, for any $y \in \mathbb{R}^N$,

$$\|u_n\|_{L^q(\mathcal{B}(y, R))} \leq \|u_n\|_{L^p(\mathcal{B}(y, R))}^{1-\lambda} \|u_n\|_{L^{p_s^*}(\mathcal{B}(y, R))}^\lambda \leq K \|u_n\|_{L^p(\mathcal{B}(y, R))}^{1-\lambda} \|u_n\|_{\mathcal{X}(\mathcal{B}(y, R))}^\lambda,$$

where we used [corollary 2.2](#) in the last inequality. Raising both sides to the power of q , we get

$$\int_{\mathcal{B}(y, R)} |u_n(x)|^q dx \leq K^q \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)} \|u_n\|_{\mathcal{X}(\mathcal{B}(y, R))}^p. \quad (5.2)$$

Now let $\mathcal{C}$ be a countable cover of $\mathbb{R}^N$ by open balls of radius R for which there exists $N_0 \in \mathbb{N}$ such that each point in $\mathbb{R}^N$ is contained in at most N_0 elements of $\mathcal{C}$. If y_i is the center of the i -th ball of $\mathcal{C}$, then by [eq. \(5.2\)](#),

$$\begin{aligned} \|u_n\|_q^q &= \int_{\mathbb{R}^N} |u_n(x)|^q dx \leq \sum_{i=1}^{\infty} \int_{\mathcal{B}(y_i, R)} |u_n(x)|^q dx \leq \sum_{i=1}^{\infty} K^q \|u_n\|_{L^p(\mathcal{B}(y_i, R))}^{q(1-\lambda)} \|u_n\|_{\mathcal{X}(\mathcal{B}(y_i, R))}^p \\ &\leq K^q \sum_{i=1}^{\infty} \|u_n\|_{\mathcal{X}(\mathcal{B}(y_i, R))}^p \sup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)} = K^q \sup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)} \sum_{i=1}^{\infty} \|u_n\|_{\mathcal{X}(\mathcal{B}(y_i, R))}^p \\ &\leq K^q \sup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)} N_0 \|u_n\|_{\mathcal{X}}^p = N_0 K^q \|u_n\|_{\mathcal{X}}^p \sup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)} \leq N_0 K^q C^p \sup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^{q(1-\lambda)}, \end{aligned}$$

where we used the hypothesis that $\|u_n\|_{\mathcal{X}} < C$ in the last inequality. Taking the limit on both sides and using the hypothesis that $\limsup_{y \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(y, R))}^p = 0$, we conclude that

$$0 \leq \lim \|u_n\|_q^q \leq N_0 K^q C^p \left(\limsup_{y \in \mathbb{R}^N} \int_{\mathcal{B}(y, R)} |u_n(x)|^p dx \right)^{\frac{q(1-\lambda)}{p}} = 0. \quad \square$$

We now rule out the possibility of $(PS)_\varrho$ sequences vanishing in the Lions' sense [\[18\]](#); in the critical case, we only rule out this possibility when ϱ is below the right-hand side of [ineq. \(4.2\)](#).

Proposition 5.7. *Let $\varrho \in (0, \infty)$ and (u_n) be a $(PS)_\varrho$ sequence. If either $\beta = 0$ or $\beta = 1$ and $\varrho < \left(\frac{1}{p} - \frac{1}{p_s^*}\right) \mathcal{S}_{s, p}^\omega$, then there exists $\varepsilon, R \in (0, \infty)$, and a sequence $(z_n) \subset \mathbb{Z}^N$ such that $\lim \|u_n\|_{L^p(\mathcal{B}(z_n, R))} > \varepsilon > 0$.*

Proof. We first prove this result while allowing that $(z_n) \subset \mathbb{R}^N$. Suppose, by contradiction, that the result is false in that case; then $\limsup_{z \in \mathbb{R}^N} \|u_n\|_{L^p(\mathcal{B}(z, R))} = 0$. Since (u_n) is $(PS)_\varrho$, it follows from [prop. 5.5](#) that it is bounded, and so by [prop. 5.6](#), we conclude that

$$\lim \int_{\mathbb{R}^N} |u_n(x)|^\theta dx = 0. \quad (5.3)$$

Let $\delta \in (0, \infty)$ with $p < \theta - \delta < \theta + \delta < p_s^*$. [Corollary A.2](#) then gives us $K \in (0, \infty)$ such that, for all $n \in \mathbb{N}$,

$$|\ln(|u_n(x)|)| < K(|u_n(x)|^\delta + |u_n(x)|^{-\delta}), \quad \therefore \quad (5.4)$$

$$0 \leq |\mathcal{Z}(u_n)| \leq \int_{\mathbb{R}^N} |u_n(x)|^\theta K(|u_n(x)|^\delta + |u_n(x)|^{-\delta}) dx = K(\|u_n\|_{\theta+\delta}^{\theta+\delta} + \|u_n\|_{\theta-\delta}^{\theta-\delta}), \quad \therefore \quad \lim \mathcal{Z}(u_n) = 0. \quad (5.5)$$

We will now separately consider the two cases: either $\beta = 0$ or $\beta = 1$ and $\varrho < (p^{-1} - p_s^{*-1}) \mathcal{S}_{s, p}^\omega$.

$\beta = 0$: Since (u_n) is a $(PS)_\varrho$ sequence, [props. 5.1](#) and [5.5](#) imply that $\lim \mathcal{I}'(u_n)(u_n) = 0$. This, along with [eqs. \(2.4\)](#), [\(5.3\)](#), and [\(5.5\)](#), gives us $\lim \|u_n\|_{\mathcal{X}} = 0$. Then we can combine the equation above with [eqs. \(2.1\)](#), [\(5.3\)](#), and [\(5.5\)](#) to get

$$\lim \mathcal{I}(u_n) = \lim \frac{\|u_n\|_{\mathcal{X}}^p}{p} + \lim \frac{\|u_n\|_\theta^\theta}{\theta^2} - \lim \frac{1}{\theta} \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx = 0.$$

Since (u_n) is $(PS)_\varrho$, we also have that $\mathcal{I}(u_n) \rightarrow \varrho$. This contradicts the fact that $\varrho \in (0, \infty)$.

$\beta=1$ and $\varrho < \left(\frac{1}{p} - \frac{1}{p_s^*}\right) \mathcal{S}_{s,p}^\omega$: Since (u_n) is a $(PS)_\varrho$ sequence, props. 5.1 and 5.5 imply that $\lim \mathcal{I}(u_n)(u_n) = 0$. This, along with eqs. (2.4), (5.3), and (5.5), now yields $\lim \|u_n\|_{\mathbb{X}}^p = \lim \|u_n\|_{p_s^*}^{p_s^*} =: L$. We can then see that

$$\lim \|u_n\|_{p_s^*}^p = \lim \|u_n\|_{p_s^*}^{\frac{p_s^*}{p}} = L^{\frac{p}{p_s^*}}, \therefore L^{\frac{p_s^*-p}{p_s^*}} = \frac{L}{L^{\frac{p}{p_s^*}}} = \lim \frac{\|u_n\|_{\mathbb{X}}^p}{\|u_n\|_{p_s^*}^p} \geq \mathcal{S}_{s,p}^p, \therefore L \geq \mathcal{S}_{s,p}^{\frac{p_s^*}{p_s^*-p}} = \mathcal{S}_{s,p}^\omega.$$

Applying eqs. (5.3) and (5.5) to the expression of $\lim \mathcal{I}(u_n)$, and then using the previous calculations gives

$$\begin{aligned} \lim \mathcal{I}(u_n) &= \lim \frac{\|u_n\|_{\mathbb{X}}^p}{p} + \lim \frac{\|u_n\|_\theta^\theta}{\theta^2} - \lim \frac{1}{\theta} \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx - \lim \frac{\|u_n\|_{p_s^*}^{p_s^*}}{p_s^*} \\ &= \lim \frac{\|u_n\|_{\mathbb{X}}^p}{p} - \lim \frac{\|u_n\|_{p_s^*}^{p_s^*}}{p_s^*} = \frac{L}{p} - \frac{L}{p_s^*} = \left(\frac{1}{p} - \frac{1}{p_s^*}\right) L \geq \left(\frac{1}{p} - \frac{1}{p_s^*}\right) \mathcal{S}_{s,p}^\omega. \end{aligned}$$

Since (u_n) is $(PS)_\varrho$, we also have that $\lim \mathcal{I}(u_n) = \varrho < \left(\frac{1}{p} - \frac{1}{p_s^*}\right) \mathcal{S}_{s,p}^\omega$, which contradicts the inequation above.

We now provide a sketch of the proof that (z_n) can be taken in $\mathbb{Z}^N$. Fix $\tilde{R} \in (0, \infty)$ and define the shorthand $g_n(z) := \|u_n\|_{L^p(\mathcal{B}(z, \tilde{R}))}$. For each $n \in \mathbb{N}$, one can find a sequence (z_m^n) satisfying $\lim_{m \rightarrow \infty} g_n(z_m^n) = \sup_{z \in \mathbb{R}^N} g_n(z)$ and, therefore, a $M_n \in \mathbb{N}$ such that $|g_n(z_m^n) - \sup_{z \in \mathbb{R}^N} g_n(z)| < \frac{1}{n}$ for all $m \geq M_n$. Since $n \in \mathbb{N}$ is arbitrary, the sequence $(z_{M_n}^n)$ is well-defined, and it is easy to show that it satisfies $\lim g_n(z_{M_n}^n) = \lim \sup_{z \in \mathbb{R}^N} g_n(z)$.

Let $[z_{M_n}^n] =: z_n \in \mathbb{Z}^N$, where the operator $[\cdot]$ is meant component-wise. For each $N \in \mathbb{N}$, z_n is a vertex of a unit N -dimensional hypercube, and $z_{M_n}^n$ lies in that same hypercube; it then follows that $\|z_n - z_{M_n}^n\| < \sqrt{N}$. Thus, for $R := \tilde{R} + \sqrt{N}$, it can be easily verified that (z_n) satisfies prop. 5.7. $\square$

We call the reader's attention to the fact that this previous result is also true when V satisfies (V₃) instead of (V₂), because its only reliance on V is that it satisfies (V₁).

The only piece missing for the proof of theorem 1.1 is the fact that weak limits of (PS) sequences are critical points of $\mathcal{I}$. For convenience, we split its proof in two statements: first, a type of Lebesgue's convergence theorem for the Gagliardo seminorm, and then the convergence of the remaining terms of $\mathcal{I}$.

Lemma 5.8. *Let $u \in \mathbb{X}$, $(u_n) \subset \mathbb{X}$, $u \in \mathbb{X}$ with $u_n \rightharpoonup u$ in $\mathbb{X}$, $K \in (0, \infty)$ such that $\|u_n\|_{\mathbb{X}} \leq K$ for all $n \in \mathbb{N}$, and $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$. Then, up to a subsequence,*

$$\lim \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} f_{u_n}(x, y) dx dy = \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} f_u(x, y) dx dy, \quad (5.6)$$

where, for each $v \in \mathbb{X}$, we use the shorthand

$$f_v(x, y) := \mathcal{K}(x, y) |v(x) - v(y)|^{p-2} (v(x) - v(y)) (\varphi(x) - \varphi(y)).$$

Proof. This result follows from a single application of the infinite measure version of Vitali's convergence theorem. Therefore, its proof consists only of showing that the appropriate hypotheses are satisfied by our problem. Those are: $(f_{u_n}) \subset L^1(\mathbb{R}^{2N})$, $f_{u_n} \xrightarrow{\text{a.e.}} f_u$ on $\mathbb{R}^{2N}$, (f_{u_n}) is uniformly integrable and tight over $\mathbb{R}^{2N}$. "Uniformly integrable" here means that for every $\varepsilon \in (0, \infty)$ there exists $\delta \in (0, \infty)$ such that for every measurable $E \subset \mathbb{R}^{2N}$ with $\mu(E) < \delta$, we have $\sup_{n \in \mathbb{N}} \int_E |f_{u_n}| d\mu < \varepsilon$; and by "tight over $\mathbb{R}^{2N}$ ", we mean that for every $\varepsilon \in (0, \infty)$ there exists $E_\varepsilon \subset \mathbb{R}^{2N}$ with $\mu(E_\varepsilon) < \infty$ such that $\sup_{n \in \mathbb{N}} \int_{\mathbb{R}^{2N} - E_\varepsilon} |f_{u_n}| d\mu < \varepsilon$.

The fact that $(f_{u_n}) \subset L^1(\mathbb{R}^{2N})$ is a consequence of the existence of the Fréchet derivative of $\mathcal{I}$. Furthermore, corollary 2.4 implies that, up to subsequence, $u_n \xrightarrow{\text{a.e.}} u$ on $\mathbb{R}^N$. Thus, $f_{u_n}(x, y) \xrightarrow{\text{a.e.}} f_u(x, y)$ on $\mathbb{R}^{2N}$.

Let $E \subset \mathbb{R}^{2N}$ be an arbitrary measurable set and consider the non-negative functions

$$g_n : \mathbb{R}^{2N} - D \rightarrow \mathbb{R}; \quad g_n(x, y) := \frac{|u_n(x) - u_n(y)|^{p-1}}{|x-y|^{\frac{N+sp}{p'}}} \text{ and } h : \mathbb{R}^{2N} - D \rightarrow \mathbb{R}; \quad h(x, y) := \frac{|\varphi(x) - \varphi(y)|^p}{|x-y|^{N+sp}},$$

where $n \in \mathbb{N}$, $D := \{(x, x) \in \mathbb{R}^{2N} \mid x \in \mathbb{R}^N\}$ and $p' = \frac{p}{p-1}$. By Hölder's inequality, we get

$$\begin{aligned} \int_E |f_{u_n}| d\mu &= \int_{\mathbb{R}^{2N}} |f_{u_n}| \mathbb{1}_E d\mu = \int_{\mathbb{R}^{2N}} |g_n h^{\frac{1}{p}} \mathbb{1}_E| d\mu \leq \|g_n\|_{L^{p'}(\mathbb{R}^{2N})} \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} \\ &\leq \left(\int_{\mathbb{R}^{2N}} |g_n|^{p'} d\mu \right)^{\frac{1}{p'}} \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} = \left(\iint_{\mathbb{R}^{2N}} \frac{|u_n(x) - u_n(y)|^p}{|x-y|^{N+ps}} dx dy \right)^{\frac{1}{p'}} \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} \\ &= \left([u_n]^p \right)^{\frac{1}{p'}} \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} \leq \|u_n\|_{\mathbb{X}}^{\frac{p}{p'}} \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} < K \|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})}. \end{aligned} \quad (5.7)$$

As $\mathcal{C}_c^\infty(\mathbb{R}^N) \subset W^{s,q}(\mathbb{R}^N)$ for any $q \in [1, \infty)$ [8], it holds that $h \in L^p(\mathbb{R}^N)$. Applying Hölder's inequality once more gives us

$$\|\mathbb{1}_E h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} = \left(\int_{\mathbb{R}^{2N}} \mathbb{1}_E h d\mu \right)^{\frac{1}{p}} \leq \|\mathbb{1}_E\|_{L^{p'}(\mathbb{R}^{2N})}^{\frac{1}{p}} \|h\|_{L^p(\mathbb{R}^{2N})}^{\frac{1}{p}} = \|h\|_{L^p(\mathbb{R}^{2N})}^{\frac{1}{p}} \mu(E)^{\frac{1}{pp'}}. \quad (5.8)$$

We now prove the uniform integrability of (f_{u_n}) . Given $\varepsilon \in (0, \infty)$, choose $\delta := \varepsilon^{pp'} / (K^p \|h\|_{L^p(\mathbb{R}^{2N})})^{-p'}$. If $\mu(E) < \delta$, then ineqs (5.7) and (5.8) give us that $\sup_{n \in \mathbb{N}} \int_E |f_{u_n}| d\mu < K \|h\|_{L^p(\mathbb{R}^{2N})}^{\frac{1}{p}} \mu(E)^{\frac{1}{pp'}} < \varepsilon$.

It now remains to show the tightness of (f_{u_n}) over $\mathbb{R}^{2N}$. Let $\Omega := \text{supp}(\varphi)$, $\Xi := \Omega^2 \cup (\Omega \times \Omega^c) \cup (\Omega^c \times \Omega)$, $A_M := \{(x, y) \in \Xi \mid |x - y| \leq M\}$, and $B_M := \{(x, y) \in \Xi \mid |x - y| > M\}$ for an arbitrary $M \in \mathbb{N}$; we claim that A_M is bounded. Suppose by contradiction that this was not the case, i.e., there exists an unbounded sequence $((x_n, y_n)) \in A_M$ unbounded. As $A_M \subset \Xi$ and Ω^2 is compact, we can find a subsequence entirely contained only in $(\Omega \times \Omega^c) \cup (\Omega^c \times \Omega)$. Without loss of generality, we can use the pigeonhole principle to extract a further subsequence contained in $(\Omega \times \Omega^c)$. Thus, $(x_n) \subset \Omega$ (a compact set), while $((x_n, y_n))$ is unbounded; these facts leads us to conclude that $|y_n| \rightarrow \infty$. However, the reverse triangle inequality, together with the definition of A_M , allows us to arrive at the contradiction $M \geq |x_n - y_n| \geq |x_n| - |y_n| \rightarrow \infty$. Therefore, A_M is bounded, which implies that $\mu(A_M) < \infty$.

Using ineq. (5.7) (with $E = B_M$) and the fact that $\mathcal{C}_c^\infty(\mathbb{R}^N) \subset W^{\frac{s}{2}, p}(\mathbb{R}^N)$ [8], we get

$$\begin{aligned} \int_{B_M} |f_{u_n}| d\mu &< K \|\mathbb{1}_{B_M} h^{\frac{1}{p}}\|_{L^p(\mathbb{R}^{2N})} = K \left(\iint_{B_M} \frac{|\varphi(x) - \varphi(y)|^p}{|x-y|^{N+sp}} dx dy \right)^{\frac{1}{p}} \\ &= K \left(\iint_{B_M} \frac{|\varphi(x) - \varphi(y)|^p}{|x-y|^{N+\frac{sp}{2}}} \frac{1}{|x-y|^{\frac{sp}{2}}} dx dy \right)^{\frac{1}{p}} \leq \frac{K}{M^{\frac{sp}{2}}} \left(\iint_{B_M} \frac{|\varphi(x) - \varphi(y)|^p}{|x-y|^{N+\frac{sp}{2}}} dx dy \right)^{\frac{1}{p}} \leq \frac{K[\varphi]_{\frac{s}{2}, p, \mathbb{R}^N}}{M^{\frac{sp}{2}}}. \end{aligned}$$

Now let $\varepsilon \in (0, \infty)$, $M_\varepsilon := \left\lceil \left(K[\varphi]_{\frac{s}{2}, p, \mathbb{R}^N} / \varepsilon \right)^{\frac{2}{sp}} \right\rceil + 1$, and choose $E_\varepsilon := A_{M_\varepsilon}$. Since $\Xi = A_{M_\varepsilon} \sqcup B_{M_\varepsilon}$ and, for any $v \in \mathbb{X}$, f_v vanishes outside Ξ , we conclude the tightness of f_{u_n} over $\mathbb{R}^{2N}$ as

$$\int_{\mathbb{R}^{2N} - A_{M_\varepsilon}} |f_{u_n}| d\mu = \int_{\Xi - A_{M_\varepsilon}} |f_{u_n}| d\mu = \int_{B_{M_\varepsilon}} |f_{u_n}| d\mu < \frac{K[\varphi]_{\frac{s}{2}, p, \mathbb{R}^N}}{M_\varepsilon^{\frac{sp}{2}}} < \varepsilon. \quad \square$$

Proposition 5.9. *Let $u \in \mathbb{X}$ and (u_n) a (PS) sequence for $\mathcal{I}$ with $u_n \rightharpoonup u$ in $\mathbb{X}$. Then $\mathcal{I}'(u) = 0$.*

Proof. Let $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$ and $\Omega := \text{supp}(\varphi)$. Since $u_n \rightharpoonup u$ in $\mathbb{X}$, corollary 2.4 gives us a subsequence satisfying $u_n \xrightarrow{\text{a.e.}} u$ in $\mathbb{R}^N$, and corollary 2.3 gives us that, up to a subsequence, $u_n \rightarrow u$ in $L^q(\Omega)$ whenever $q \in [1, p_s^*)$. As $p-1 \in [1, p_s^*)$, the reverse of the Lebesgue's dominated convergence theorem

implies that there exists a further subsequence of (u_n) and some $h_1 \in L^{p-1}(\Omega)$ such that $|u_n(x)| \leq h_1(x)$ for all $n \in \mathbb{N}$ and almost all $x \in \Omega$. Therefore,

$$\begin{aligned} V(x)|u_n(x)|^{p-2}u_n(x)\varphi(x) &\xrightarrow{\text{a.e.}} V(x)|u(x)|^{p-2}u(x)\varphi(x) \text{ on } \Omega \text{ and} \\ \left| V(x)|u_n(x)|^{p-2}u_n(x)\varphi(x) \right| &= |V(x)||u_n(x)|^{p-1}|\varphi(x)| \leq K_1\|V\|_\infty h_1(x)^{p-1}, \end{aligned} \quad (5.9)$$

where K_1 is the maximum of the continuous function with compact support φ . Therefore, by Lebesgue's dominated convergence theorem, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} V(x)|u_n(x)|^{p-2}u_n(x)\varphi(x)dx &= \lim_{n \rightarrow \infty} \int_{\Omega} V(x)|u_n(x)|^{p-2}u_n(x)\varphi(x)dx \\ &= \int_{\Omega} V(x)|u(x)|^{p-2}u(x)\varphi(x)dx = \int_{\mathbb{R}^N} V(x)|u(x)|^{p-2}u(x)\varphi(x)dx. \end{aligned} \quad (5.10)$$

Let $\delta \in (0, \infty)$ be such as in [ineq. \(5.4\)](#). It holds that $1 < \theta - 1 - \delta < \theta - 1 + \delta < p_s^*$ because $p > 2$. Thus, we can again apply the reverse of the Lebesgue's dominated convergence theorem twice in succession to find some $h_2 \in L^{\theta-1+\delta}(\Omega)$ and $h_3 \in L^{\theta-1-\delta}(\Omega)$ such that, passing to subsequences when necessary, $|u_n(x)| \leq h_2(x)$ and $|u_n(x)| \leq h_3(x)$ for all $n \in \mathbb{N}$ and almost all $x \in \Omega$. Therefore,

$$\begin{aligned} |u_n(x)|^{\theta-2}u_n(x)\ln(|u_n(x)|)\varphi(x) &\xrightarrow{\text{a.e.}} |u(x)|^{\theta-2}u(x)\ln(|u(x)|)\varphi(x) \\ \text{on } \Omega, \text{ and by } \text{corollary A.2, there exists } K_2 \in (0, \infty) \text{ such that, for all } n \in \mathbb{N}, \\ |u_n(x)|^{\theta-2}u_n(x)\ln(|u_n(x)|)\varphi(x) &= |u_n(x)|^{\theta-1}|\ln(|u_n(x)|)|\varphi(x)| \\ &\leq K_1K_2|u_n(x)|^{\theta-1}(|u_n(x)|^\delta + |u_n(x)|^{-\delta}) \leq K_1K_2(h_2(x)^{\theta-1+\delta} + h_3(x)^{\theta-1-\delta}), \end{aligned} \quad (5.11)$$

where K_1 is as in [eq. \(5.9\)](#) and we used that $\theta - 1 - \delta > 1$. It is easy to see that $h_2^{\theta-1+\delta} + h_3^{\theta-1-\delta} \in L^1(\Omega)$. Thus, by Lebesgue's dominated convergence theorem, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} |u_n(x)|^{\theta-2}u_n(x)\ln(|u_n(x)|)\varphi(x)dx &= \lim_{n \rightarrow \infty} \int_{\Omega} |u_n(x)|^{\theta-2}u_n(x)\ln(|u_n(x)|)\varphi(x)dx \\ &= \int_{\Omega} |u(x)|^{\theta-2}u(x)\ln(|u(x)|)\varphi(x)dx = \int_{\mathbb{R}^N} |u(x)|^{\theta-2}u(x)\ln(|u(x)|)\varphi(x)dx. \end{aligned} \quad (5.12)$$

Proceeding similarly, we can find $h_4 \in L^{p_s^*-1}(\Omega)$ such that, up to a subsequence, $|u_n(x)| \leq h_4(x)$ for all $n \in \mathbb{N}$ and almost all $x \in \Omega$. Therefore,

$$\beta|u_n(x)|^{p_s^*-2}u_n(x)\varphi(x) \xrightarrow{\text{a.e.}} \beta|u(x)|^{p_s^*-2}u(x)\varphi(x) \text{ on } \Omega \text{ and} \quad (5.13)$$

$$|\beta|u_n(x)|^{p_s^*-2}u_n(x)\varphi(x)| = \beta|u_n(x)|^{p_s^*-1}|\varphi(x)| \leq K_1h_4(x)^{p_s^*-1}. \quad (5.14)$$

Using Lebesgue's dominated convergence theorem again, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_{\mathbb{R}^N} \beta|u_n(x)|^{p_s^*-2}u_n(x)\varphi(x)dx &= \lim_{n \rightarrow \infty} \int_{\Omega} \beta|u_n(x)|^{p_s^*-2}u_n(x)\varphi(x)dx \\ &= \int_{\Omega} \beta|u(x)|^{p_s^*-2}u(x)\varphi(x)dx = \int_{\mathbb{R}^N} \beta|u(x)|^{p_s^*-2}u(x)\varphi(x)dx. \end{aligned} \quad (5.15)$$

Thus, [lemma 5.8](#) and eqs. (5.10), (5.12), and (5.15) let us assert that $\lim \mathcal{I}'(u_n)(\varphi) = \mathcal{I}'(u)(\varphi)$. On the other hand, as (u_n) is (PS), we can use [prop. 5.1](#) to conclude that $\mathcal{I}'(u_n)(\varphi) \rightarrow 0$. Since $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$ was arbitrary, it follows that $\mathcal{I}'(u)$ vanishes on $\mathcal{C}_c^\infty(\mathbb{R}^N)$.

It follows trivially from $\mathcal{C}_c^\infty(\mathbb{R}^N)$ being dense in $W^{s,p}(\mathbb{R}^N)$ [\[8\]](#) that $\mathcal{C}_c^\infty(\mathbb{R}^N)$ is dense in $\mathbb{X}$; therefore, for each $w \in \mathbb{X}$, there exists $(\varphi_n) \subset \mathcal{C}_c^\infty(\mathbb{R}^N)$ such that $\varphi_n \rightarrow w$ in $\mathbb{X}$. As discussed previously, $\mathcal{I}'(u) \in \mathbb{X}'$, which means that $\mathcal{I}'(u)$ is continuous. Thus, we finally conclude that $\mathcal{I}'(u)(w) = \lim \mathcal{I}'(u)(\varphi_n) = 0$. $\square$

We are now ready to prove [theorem 1.1](#). The strategy is relatively straightforward: we take a $(\text{PS})_c$ sequence and use shifts by the sequence (z_n) given in [prop. 5.7](#) to construct (v_n) , which has a nontrivial weak limit $v \in \mathbb{X}$ and also is a $(\text{PS})_c$ sequence. We then use the result above to show that $\mathcal{I}'(v_n)(\varphi)$ converges to $\mathcal{I}'(v)(\varphi)$ for all $\varphi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$. Finally, we use the density of $\mathcal{C}_c^\infty(\mathbb{R}^N)$ in $\mathbb{X}$ to conclude that v is a critical point of $\mathcal{I}$ and then [theorem 1.3](#) to show that it is at the ground state level.

Proof of theorem 1.1. It suffices to show that there exists $v \in \mathbb{X}$ such that $\mathcal{I}(v) = c_{\mathcal{N}}$ and $\mathcal{I}'(v)(\varphi) = 0$ for all $\varphi \in \mathbb{X}$.

As discussed in the beginning of this section, lemma 3.4 guarantees that one can find a $(PS)_c$ sequence, which we will denote by (u_n) . Because $c < \left(\frac{1}{p} - \frac{1}{p_s^*}\right) \mathcal{S}_{s,p}^\omega$, by lemma 4.1, we can use prop. 5.7 even in the case $\beta = 1$. Thus, let $\varepsilon, R \in (0, \infty)$, and $(z_n) \subset \mathbb{Z}^N$ be as in prop. 5.7. Since (u_n) is $(PS)_c$, prop. 5.5 gives us the hypotheses necessary to use prop. 5.3, i.e., there is some $v \in \mathbb{X} - \{0\}$ and subsequences such that $v_n := u_n \circ \tau_{z_n} \rightharpoonup v$ in $\mathbb{X}$.

By prop. 5.4, we know that (v_n) also is a $(PS)_c$ sequence. Moreover, by prop. 5.9, we have that $\|\mathcal{I}'(v)\|_{\mathbb{X}'} = 0$. Thus, it only remains to show that the solution v is ground state, i.e., that $\mathcal{I}(v) = c_{\mathcal{N}}$. Note that, by eq. (5.1), the fact that (v_n) is $(PS)_c$, and by the superadditivity of the liminf, we have

$$\begin{aligned} c = \liminf \left(\mathcal{I}(v_n) - \frac{1}{\theta} \mathcal{I}'(v_n)(v_n) \right) &\geq \frac{\theta - p}{\theta p} \left(\liminf \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \frac{|v_n(x) - v_n(y)|^p}{|x - y|^{N+sp}} dx dy \right. \\ &\quad \left. + \liminf \int_{\mathbb{R}^N} V |v_n|^p d\mu \right) + \frac{1}{\theta^2} \liminf \int_{\mathbb{R}^N} |v_n|^\theta d\mu + \beta \frac{p_s^* - \theta}{p_s^* \theta} \liminf \int_{\mathbb{R}^N} |v_n|^{p_s^*} d\mu. \end{aligned}$$

Since all the integrands are non-negative and $v_n \xrightarrow{\text{a.e.}} v$ on $\mathbb{R}^N$, Fatou's lemma gives us

$$\begin{aligned} c &\geq \frac{\theta - p}{\theta p} \left(\int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \frac{|v(x) - v(y)|^p}{|x - y|^{N+sp}} dx dy + \int_{\mathbb{R}^N} V |v|^p d\mu \right) + \frac{1}{\theta^2} \int_{\mathbb{R}^N} |v|^\theta d\mu \\ &\quad + \beta \frac{p_s^* - \theta}{p_s^* \theta} \int_{\mathbb{R}^N} |v|^{p_s^*} d\mu = \frac{\theta - p}{\theta p} \|v\|_{\mathbb{X}}^p + \frac{1}{\theta^2} \|v\|_\theta^\theta + \beta \frac{p_s^* - \theta}{p_s^* \theta} \|v\|_{p_s^*}^{p_s^*} = \mathcal{I}(v) - \frac{1}{\theta} \mathcal{I}'(v)(v) = \mathcal{I}(v), \end{aligned} \tag{5.16}$$

where the second to last equality comes from eq. (5.1) and the last equality comes from prop. 5.9.

Now, since $v \neq 0$ and $\mathcal{I}'(v)(v) = 0$, we have that $v \in \mathcal{N}$ by the definition of $\mathcal{N}$. Thus, $\mathcal{I}(v) \geq \inf_{u \in \mathcal{N}} \mathcal{I}(u) = c_{\mathcal{N}} = c$, where the last equality comes from prop. 3.1. But this last inequality combined with ineq. (5.16) imply that $\mathcal{I}(v) = c_{\mathcal{N}}$, which concludes the proof. $\square$

6 The asymptotically periodic case

In this section, $\mathcal{I}_{\text{asympt}}$ refers to a functional defined as in eq. (2.1) but with a potential V_{asympt} in the place of V . The potential V_{asympt} is always implicitly assumed to satisfy (V_1) and (V_3) . In particular, (V_3) implies that V_{per} also satisfies (V_1) .

Proposition 6.1. *Let $(u_n) \subset \mathbb{X}$ such that $u_n \rightharpoonup 0$ in $\mathbb{X}$. Then, up to a subsequence,*

$$\lim \int_{\mathbb{R}^N} W(x) |u_n(x)|^p dx = 0. \tag{6.1}$$

Proof. Passing to a subsequence if necessary, we have that $u_n \xrightarrow{\text{a.e.}} 0$ on $\mathbb{R}^N$ by corollary 2.4. In particular, this implies that $|u_n|^p \xrightarrow{\text{a.e.}} 0$.

Furthermore, by corollary 2.2, there exists $C \in (0, \infty)$ such that $\|u_n\|_{p_s^*} \leq C \|u_n\|_{\mathbb{X}}$. Since (u_n) is weakly convergent in $\mathbb{X}$, it is bounded there. This, together with the last inequality, implies that there exists $M \in (0, \infty)$ such that $\|u_n\|_{p_s^*} \leq M$ for all $n \in \mathbb{N}$. Thus,

$$\| |u_n|^p \|_{\frac{p_s^*}{p}} = \left(\int_{\mathbb{R}^N} |u_n(x)|^{p_s^*} dx \right)^{\frac{p}{p_s^*}} = \|u_n\|_{p_s^*}^p \leq M^p$$

for all $n \in \mathbb{N}$, i.e., $(|u_n|)^p$ is bounded in $L^{\frac{p_s^*}{p}}(\mathbb{R}^N)$. This, along with the pointwise convergence of $|u_n|^p$ to 0, implies that $|u_n|^p \rightharpoonup 0$ in $L^{\frac{p_s^*}{p}}(\mathbb{R}^N)$. Thus, for all v in the dual of $L^{\frac{p_s^*}{p}}(\mathbb{R}^N)$, it follows that

$$\lim \int_{\mathbb{R}^N} |u_n(x)|^p v(x) dx = 0.$$

As $p_s^*/p = N/(N-sp)$ by the definition of p_s^* , it follows that the Hölder conjugate of p_s^*/p is $N/(sp)$, i.e., the dual of $L^{\frac{p_s^*}{p}}(\mathbb{R}^N)$ is $L^{\frac{N}{sp}}(\mathbb{R}^N)$. As $W \in L^{\frac{N}{sp}}(\mathbb{R}^N)$, by (V₃), we conclude eq. (6.1). $\square$

By Hypothesis (V₃), V_{asympt} can be broken into the difference of two terms: V_{per} and W . One can then define a functional $\mathcal{I}_{\text{per}}$ as in eq. (2.1) but with a potential V_{per} in the place of V . It is immediate to check that, for any $u \in \mathbb{X} - \{0\}$,

$$\mathcal{I}_{\text{asympt}}(u) - \mathcal{I}_{\text{per}}(u) = -\frac{1}{p} \int_{\mathbb{R}^N} W(x) |u(x)|^p dx, \quad \mathcal{I}'_{\text{asympt}}(u) - \mathcal{I}'_{\text{per}}(u) = - \int_{\mathbb{R}^N} W(x) |u(x)|^{p-2} u dx. \quad (6.2)$$

We also define $\mathcal{N}_{\text{asympt}}$ (resp. $\mathcal{N}_{\text{per}}$) in the same way as $\mathcal{N}$ was defined in eq. (1.1) but using the functional $\mathcal{I}_{\text{asympt}}$ (resp. $\mathcal{I}_{\text{per}}$) instead of $\mathcal{I}$. Since $\mathcal{I}_{\text{asympt}}$ and $\mathcal{I}_{\text{per}}$ satisfy the mountain pass geometry (see lemma 3.4), there exist some strictly positive $c_{\text{asympt}}, c_{\text{per}} \in (0, \infty)$ defined as in eq. (1.2) for the functionals $\mathcal{I}_{\text{asympt}}$ and $\mathcal{I}_{\text{per}}$, respectively. Our next result establishes a strict relation between c_{asympt} and c_{per} .

Proposition 6.2. *The levels c_{asympt} and c_{per} satisfy the strict inequality $c_{\text{asympt}} < c_{\text{per}}$.*

Proof. As V_{per} itself satisfies (V₂), we can use theorem 1.1 to find $v \in \mathbb{X}$ such that $\mathcal{I}_{\text{per}}(v) = c_{\text{per}}$ and $\mathcal{I}'_{\text{per}}(v) = 0$. Since lemma 3.4 asserts that (N₁) holds, $\mathfrak{t}(v)$ is the unique point of maximum of φ_v . On the flip side, prop. 3.1 implies that $\mathfrak{t}(v)v \in \mathcal{N}_{\text{asympt}}$. Since $\mathcal{I}'_{\text{per}}(v) = 0$, we have that $v \in \mathcal{N}_{\text{per}}$. By uniqueness, it follows that $\mathfrak{t}(v) = 1$, which allows us to conclude that

$$\sup_{t \geq 0} \mathcal{I}_{\text{per}}(tv) = \mathcal{I}_{\text{per}}(v) = c_{\text{per}}. \quad (6.3)$$

Furthermore, using prop. 3.1 and that $\mathfrak{t}(v)v \in \mathcal{N}_{\text{asympt}}$, we get

$$d = \inf_{u \in \mathcal{N}} \mathcal{I}_{\text{asympt}}(u) \leq \mathcal{I}_{\text{asympt}}(\mathfrak{t}(v)v). \quad (6.4)$$

Note that, since (V₃) assures us that W is strictly positive, eq. (6.2) implies that $\mathcal{I}_{\text{asympt}}(\mathfrak{t}(v)v) < \mathcal{I}_{\text{per}}(\mathfrak{t}(v)v)$. We can then use this result, along with ineq. (6.4) and eq. (6.3), to conclude that

$$c_{\text{asympt}} \leq \mathcal{I}_{\text{asympt}}(\mathfrak{t}(v)v) < \mathcal{I}_{\text{per}}(\mathfrak{t}(v)v) \leq \sup_{t \geq 0} \mathcal{I}_{\text{per}}(tv) = \mathcal{I}_{\text{per}}(v) = c_{\text{per}}. \quad \square$$

Mind that this previous result works for either value of β , as long as the correct restriction on the value of p is valid.

We will denote by $\|\cdot\|_{\mathbb{X}_{\text{per}}}$ the usual $\|\cdot\|_{\mathbb{X}}$ with V swapped by V_{per} . Since both V_{asympt} and V_{per} satisfy (V₁) (although not necessarily for the same constants), it is obvious that both norms are equivalent. We do not distinguish between $\mathbb{X}_{\text{per}}$ and $\mathbb{X}$ because, similarly to what was discussed in section 2, they coincide as sets.

We now prove that a $(PS)_{c_{\text{asympt}}}$ sequence for the functional $\mathcal{I}_{\text{asympt}}$ with null weak limit must converge strongly to zero and then follow this proof by showing theorem 1.2.

Lemma 6.3. *Let (u_n) be a $(PS)_{c_{\text{asympt}}}$ sequence for the functional $\mathcal{I}_{\text{asympt}}$ and suppose that $u_n \rightharpoonup 0$ in $\mathbb{X}$. Then, up to a subsequence, $u_n \rightarrow 0$ in $\mathbb{X}$.*

Proof. The proof will proceed by contradiction. For this purpose, suppose that every subsequence of (u_n) satisfies $u_n \not\rightarrow 0$ in $\mathbb{X}$.

Since lemma 3.4 implies (N₁), we can consider the sequence $(u_n) := (\mathfrak{t}(u_n)) \subset (0, \infty)$. We will prove, also by contradiction, that there exists a subsequence of (t_n) such that $\lim t_n \leq 1$. If this was not the case, then there would exist $\delta \in (0, \infty)$ and a subsequence of (t_n) such that $t_n > 1 + \delta$ for every $n \in \mathbb{N}$. An appropriate form of eq. (3.9) for $\mathcal{I}'_{\text{per}}$ (which consists only of swapping $\|\cdot\|_{\mathbb{X}}$ by $\|\cdot\|_{\mathbb{X}_{\text{per}}}$), the definition of $\mathcal{N}_{\text{per}}$, and the fact that $t_n \neq 0$ allow us to string the following sequence of deductions:

$$\begin{aligned} t_n^p \|u_n\|_{\mathbb{X}_{\text{per}}}^p - \int_{\mathbb{R}^N} |t_n u_n(x)|^\theta \ln(|t_n u_n(x)|) dx - \beta t_n^{p_s^*} \|u_n\|_{p_s^*}^{p_s^*} &= \mathcal{I}'_{\text{per}}(t_n u_n)(t_n u_n) = 0, \quad \therefore \\ t_n^p \|u_n\|_{\mathbb{X}_{\text{per}}}^p - t_n^\theta \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx - t_n^\theta \ln(|t_n|) \|u_n\|_\theta^\theta - \beta t_n^{p_s^*} \|u_n\|_{p_s^*}^{p_s^*} &= 0, \quad \therefore \\ t_n^{p-\theta} \|u_n\|_{\mathbb{X}_{\text{per}}}^p - \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx - \ln(|t_n|) \|u_n\|_\theta^\theta - \beta t_n^{p_s^*-\theta} \|u_n\|_{p_s^*}^{p_s^*} &= 0. \end{aligned} \quad (6.5)$$

Subtracting $\mathcal{I}'_{\text{asympt}}(u_n)(u_n)$ (whose expression can be inferred from [eq. \(2.4\)](#)) from the last equation above leads us to

$$(t_n^{p-\theta}-1)[u_n]^p + \int_{\mathbb{R}^N} (t_n^{p-\theta}V_{\text{per}}(x)-V_{\text{asympt}}(x))|u_n(x)|^p dx - \ln(|t_n|)\|u_n\|_{\theta}^{\theta} - \beta(t_n^{p_s^*-\theta}-1)\|u_n\|_{p_s^*}^{p_s^*} \\ = -\mathcal{I}'_{\text{asympt}}(u_n)(u_n) \rightarrow 0, \quad (6.6)$$

where $\mathcal{I}'_{\text{asympt}}(u_n)(u_n) \rightarrow 0$ in the last step because (u_n) being a $(\text{PS})_{c_{\text{asympt}}}$ sequence allows us to apply [prop. 5.1](#).

First, since $p-\theta < 0$ and $t_n > 1+\delta$ always, we have

$$(t_n^{p-\theta}-1)[u_n]^p + \int_{\mathbb{R}^N} (t_n^{p-\theta}V_{\text{per}}(x)-V_{\text{asympt}}(x))|u_n(x)|^p dx \\ \leq \int_{\mathbb{R}^N} (V_{\text{per}}(x)-V_{\text{asympt}}(x))|u_n(x)|^p dx = \int_{\mathbb{R}^N} W(x)|u_n(x)|^p dx \quad (6.7)$$

for all $n \in \mathbb{N}$. Second, because $p_s^*-\theta > 0$, it holds, for all $n \in \mathbb{N}$, that $(t_n^{p_s^*-\theta}-1) \geq \delta$. Therefore, it is always true that

$$(t_n^{p_s^*-\theta}-1)\|u_n\|_{p_s^*}^{p_s^*} \geq \delta\|u_n\|_{p_s^*}^{p_s^*} \geq 0. \quad (6.8)$$

Third, since (u_n) is a $(\text{PS})_{c_{\text{asympt}}}$ sequence, [prop. 5.7](#) implies that there exists $R \in (0, \infty)$ and a sequence $(z_n) \subset \mathbb{Z}^N$ such that the sequence $(v_n) := (u_n \circ \tau_{z_n})$ satisfies $\lim \|v_n\|_{L^p(\mathcal{B}(0, R))} > 0$, up to a subsequence. Because $\mu(\mathcal{B}(0, R)) < \infty$, the standard theory gives us that $L^{\theta}(\mathcal{B}(0, R)) \hookrightarrow L^p(\mathcal{B}(0, R))$. Using that $\|\cdot\|_{\theta}$ is invariant by translations and that $t_n \geq 1+\delta$, we get

$$\lim(\ln(|t_n|)\|u_n\|_{\theta}^{\theta}) = \lim(\ln(|t_n|)\|v_n\|_{\theta}^{\theta}) \geq \lim(\ln(1+\delta)\|v_n\|_{\theta}^{\theta}) \geq \ln(1+\delta)\lim\|v_n\|_{L^{\theta}(\mathcal{B}(0, R))}^{\theta} > 0. \quad (6.9)$$

By combining ineqs (6.7) to (6.9) with [eq. \(6.6\)](#) and taking the limit, we get

$$0 = -\lim \mathcal{I}'_{\text{asympt}}(u_n)(u_n) \leq \lim \left(\int_{\mathbb{R}^N} W(x)|u_n(x)|^p dx - \ln(|t_n|)\|u_n\|_{\theta}^{\theta} \right) \\ = \lim \int_{\mathbb{R}^N} W(x)|u_n(x)|^p dx - \lim(\ln(|t_n|)\|u_n\|_{\theta}^{\theta}) = -\lim(\ln(|t_n|)\|u_n\|_{\theta}^{\theta}) < 0,$$

where we used [prop. 6.1](#), since it is assumed that $u_n \rightharpoonup 0$ in $\mathbb{X}$. This last equation is obviously a contradiction, showing us that there is, in fact, a subsequence of (t_n) with $\lim t_n \leq 1$. For this subsequence, there are now two exactly two possibilities we must address: $\lim t_n < 1$ or $\lim t_n = 1$.

$\lim t_n < 1$: We can then extract a subsequence of (t_n) that satisfies $t_n < 1$ for all $n \in \mathbb{N}$. On one hand, we can borrow the calculations done in [eq. \(5.1\)](#) and find that, for every $n \in \mathbb{N}$,

$$\mathcal{I}_{\text{per}}(t_n u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(t_n u_n)(t_n u_n) = \frac{\theta-p}{\theta p} t_n^p \|u_n\|_{\mathbb{X}_{\text{per}}}^p + \frac{t_n^{\theta}}{\theta^2} \|u_n\|_{\theta}^{\theta} + \frac{\beta(p_s^*-\theta)}{p_s^* \theta} t_n^{p_s^*} \|u_n\|_{p_s^*}^{p_s^*} \\ < \frac{\theta-p}{\theta p} \|u_n\|_{\mathbb{X}_{\text{per}}}^p + \frac{1}{\theta^2} \|u_n\|_{\theta}^{\theta} + \frac{\beta(p_s^*-\theta)}{p_s^* \theta} \|u_n\|_{p_s^*}^{p_s^*} = \mathcal{I}_{\text{per}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(u_n)(u_n). \quad (6.10)$$

On the other hand, because $t_n u_n \in \mathcal{N}_{\text{per}}$ for every $n \in \mathbb{N}$, we always have that

$$\mathcal{I}_{\text{per}}(t_n u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(t_n u_n)(t_n u_n) = \mathcal{I}_{\text{per}}(t_n u_n) \geq \inf_{u \in \mathcal{N}_{\text{per}}} \mathcal{I}_{\text{per}}(u) = c_{\text{per}}, \quad (6.11)$$

where the last equality comes from [prop. 3.1](#).

By combining ineqs (6.10) and (6.11), we get

$$c_{\text{per}} \leq \mathcal{I}_{\text{per}}(t_n u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(t_n u_n)(t_n u_n) < \mathcal{I}_{\text{per}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(u_n)(u_n). \quad (6.12)$$

We claim that $\mathcal{I}_{\text{per}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(u_n)(u_n) \rightarrow c_{\text{asympt}}$. Indeed, because $u_n \rightarrow 0$, we can use [prop. 6.1](#) along with [eq. \(6.2\)](#) and the fact that (u_n) is a $(\text{PS})_{c_{\text{asympt}}}$ sequence for $\mathcal{I}_{\text{asympt}}$, to get

$$\begin{aligned} \mathcal{I}_{\text{per}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(u_n)(u_n) &= \mathcal{I}_{\text{per}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{per}}(u_n)(u_n) + (\mathcal{I}_{\text{asympt}}(u_n) - \mathcal{I}_{\text{asympt}}(u_n)) \\ &+ \left(\frac{1}{\theta} \mathcal{I}'_{\text{asympt}}(u_n)(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{asympt}}(u_n)(u_n) \right) = \frac{1}{\theta} (\mathcal{I}'_{\text{asympt}}(u_n)(u_n) - \mathcal{I}'_{\text{per}}(u_n)(u_n)) \\ &(\mathcal{I}_{\text{per}}(u_n) - \mathcal{I}_{\text{asympt}}(u_n)) + \left(\mathcal{I}_{\text{asympt}}(u_n) - \frac{1}{\theta} \mathcal{I}'_{\text{asympt}}(u_n)(u_n) \right) \rightarrow c_{\text{asympt}}. \end{aligned}$$

We can conclude that $c_{\text{per}} \leq c_{\text{asympt}}$ by taking the limit of [ineq. \(6.12\)](#) and using the expression above. This begets a contradiction with [prop. 6.2](#); thus, up to subsequence, $u_n \rightarrow 0$ in $\mathbb{X}$.

$\lim t_n = 1$: By eqs. (2.1) and (3.8), we have

$$\begin{aligned} \mathcal{I}_{\text{per}}(t_n u_n) - \mathcal{I}_{\text{asympt}}(u_n) &= \frac{1}{p} \left((t_n^p - 1)[u_n] + \int_{\mathbb{R}^N} (t_n^p V_{\text{per}}(x) - V_{\text{asympt}}(x)) |u_n(x)|^p dx \right) \\ &+ \frac{t_n^\theta - 1}{\theta^2} \|u_n\|_\theta^\theta - \frac{1}{\theta} \left(\int_{\mathbb{R}^N} |t_n u_n(x)|^\theta \ln(|t_n u_n(x)|) dx \right. \\ &\left. + \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx \right) - \frac{\beta(t_n^{p_s^*} - 1)}{p_s^*} \|u_n\|_{p_s^*}^{p_s^*}. \end{aligned} \quad (6.13)$$

Since

$$t_n^p V_{\text{per}}(x) - V_{\text{asympt}}(x) = t_n^p V_{\text{per}}(x) - V_{\text{per}}(x) + W(x) = (t_n^p - 1) V_{\text{per}}(x) + W(x) \text{ and}$$

$$\int_{\mathbb{R}^N} |t_n u_n(x)|^\theta \ln(|t_n u_n(x)|) dx = t_n^\theta \left(\int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx + \ln(t_n) \|u_n\|_\theta^\theta \right),$$

[eq. \(6.13\)](#) can be simplified into

$$\begin{aligned} \mathcal{I}_{\text{per}}(t_n u_n) - \mathcal{I}_{\text{asympt}}(u_n) &= \frac{t_n^p - 1}{p} \left([u_n] + \int_{\mathbb{R}^N} V_{\text{per}}(x) |u_n(x)|^p dx \right) \\ &+ \frac{1}{p} \int_{\mathbb{R}^N} W(x) |u_n(x)|^p dx + \frac{t_n^\theta - 1}{\theta^2} \|u_n\|_\theta^\theta \\ &- \frac{t_n^\theta - 1}{\theta} \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx + \frac{t_n^\theta}{\theta} \ln(t_n) \|u_n\|_\theta^\theta - \frac{\beta(t_n^{p_s^*} - 1)}{p_s^*} \|u_n\|_{p_s^*}^{p_s^*}. \end{aligned} \quad (6.14)$$

Now note that there must exist $m_u, M_u \in (0, \infty)$ and a subsequence of (u_n) such that $m_u < \|u_n\|_{\mathbb{X}} < M_u$ for every $n \in \mathbb{N}$. The first inequality comes from the contradictory hypothesis that $u_n \not\rightarrow 0$ in $\mathbb{X}$ and the second from [prop. 5.5](#). We can then use [prop. 2.5](#) and the continuous embedding given in [corollary 2.2](#) to find $C_1 \in (0, \infty)$ such that

$$\left| \int_{\mathbb{R}^N} |u_n(x)|^\theta \ln(|u_n(x)|) dx \right| \leq C_1$$

for every $n \in \mathbb{N}$. Furthermore, these same bounds on $\|u_n\|_{\mathbb{X}}$, alongside the continuous embeddings given in [corollary 2.2](#) and [\(V₁\)](#), allow us to refactor [eq. \(6.14\)](#) into

$$\begin{aligned} |\mathcal{I}_{\text{per}}(t_n u_n) - \mathcal{I}_{\text{asympt}}(u_n)| &\leq C_2 |t_n^p - 1| + C_3 |t_n^\theta - 1| + C_4 t_n^\theta |\ln(t_n)| + \beta C_5 |t_n^{p_s^*} - 1| \\ &+ \frac{1}{p} \left| \int_{\mathbb{R}^N} W(x) |u_n(x)|^p dx \right|, \end{aligned}$$

where $C_2, C_3, C_4, C_5 \in (0, \infty)$. By taking the limit on both sides of the inequality above, and then using [prop. 6.1](#), we conclude that

$$c_{\text{per}} = \inf_{u \in \mathcal{N}_{\text{per}}} \mathcal{I}_{\text{per}}(u) \leq \lim \mathcal{I}_{\text{per}}(t_n u_n) = \lim \mathcal{I}_{\text{asympt}}(u_n) = c_{\text{asympt}}, \quad (6.15)$$

where we used [prop. 3.1](#), the fact that $t_n u_n \in \mathcal{N}_{\text{per}}$ and that (u_n) is a $(\text{PS})_{c_{\text{asympt}}}$ sequence for $\mathcal{I}_{\text{asympt}}$. However, [eq. \(6.15\)](#) contradicts [prop. 6.2](#); thus $u_n \rightarrow 0$ in $\mathbb{X}$. $\square$

Proof of theorem 1.2. As discussed previously in the beginning of section 5, there exists (u_n) a $(PS)_{c_{\text{asympt}}}$ sequence for the functional $\mathcal{I}_{\text{asympt}}$. Because of prop. 5.5 and the fact that $\mathbb{X}$ is reflexive, we know that there is some $u \in \mathbb{X}$ such that $u_n \rightharpoonup u$ in $\mathbb{X}$. As prop. 5.9 guarantees that this u is a critical point of $\mathcal{I}_{\text{asympt}}$, it only remains to show that $u \neq 0$. Suppose by contradiction that this was not the case; then lemma 6.3 implies that, up to subsequence, $u_n \rightarrow 0$ in $\mathbb{X}$. On one hand, this can be used with the continuity of $\mathcal{I}$ to deduce $\mathcal{I}(u_n) \rightarrow 0$. On the other hand, since (u_n) is $(PS)_{c_{\text{asympt}}}$ for functional $\mathcal{I}_{\text{asympt}}$, we have $\mathcal{I}_{\text{asympt}}(u_n) \rightarrow c_{\text{asympt}}$. The uniqueness of limit then implies $c_{\text{asympt}} = 0$, which cannot occur because $\mathcal{I}_{\text{asympt}}$ satisfies the mountain pass geometry. $\square$

7 Infinite multiplicity

The goal of this section is to prove theorem 1.4 and then corollary 1.5. Thus, let $\mathbb{B}, \mathcal{J}, \mathcal{G}, h$ be as in the statement of theorem 1.4 and assume that the conditions therein hold. For any $u \in \mathbb{B}$ and $g \in \mathcal{G}$, we will use the following shorthand notation: $g(u) := h(g, u)$. For any $u, v \in \mathbb{B}$, we say that $v \sim u$ when there exists $g \in \mathcal{G}$ such that $v = g(u)$. Clearly, $\sim$ is an equivalence relation. Now define the sets $\mathcal{K} := \{u \in \mathbb{B} \mid \mathcal{J}(u) = 0\}$ and $\tilde{\mathcal{K}} := \mathcal{K} / \sim$, where this last one denotes the quotient space of $\mathcal{K}$ by $\sim$. We will distinguish between an element $w \in \mathbb{B}$ and its equivalence class by placing a $\sim$ symbol above the variable name, e.g., $\tilde{w} \in \tilde{\mathcal{K}}$. By $(\mathcal{J}_1)$, $\mathcal{J}$ is odd; this implies that $\mathcal{K} = -\mathcal{K}$ and so there exists $\mathcal{F} \subset \mathcal{K}$ such that $\mathcal{F} = -\mathcal{F}$, $\tilde{\mathcal{K}} = \mathcal{F} / \sim$, and $u, v \in \mathcal{F}$ only if $u \sim v$.

Consider $\kappa := \inf\{\|u - v\|_{\mathbb{B}} \in \mathbb{R} \mid u, v \in \mathcal{K} \text{ with } u \neq v\}$. The first result we prove in this section is that κ is nonzero under a specific condition.

Proposition 7.1. *If $\tilde{\mathcal{K}}$ is finite, then $\kappa > 0$.*

Proof. By the definition of κ and properties of the infimum, there exists $(u_n), (v_n) \subset \mathcal{K}$ such that $u_n \neq v_n$ always and $\|u_n - v_n\|_{\mathbb{B}} \rightarrow \kappa$. The hypothesis that $\tilde{\mathcal{K}}$ is finite means that $\tilde{\mathcal{K}}^2$ also is finite. Since $(\tilde{u}_n, \tilde{v}_n) \in \tilde{\mathcal{K}}^2$, we can use the pigeonhole principle to find $(\tilde{u}, \tilde{v}) \in \tilde{\mathcal{K}}^2$ and indexes $(n_k) \in \mathbb{N}$ such that $(\tilde{u}_{n_k}, \tilde{v}_{n_k}) = (\tilde{u}, \tilde{v})$ for all $k \in \mathbb{N}$. Note that $g(0) = 0$ for every $g \in \mathcal{G}$ by $(\mathcal{G}_1)$. Thus, the class $\tilde{0}$ only has one element in it: the zero function. Therefore, if $\tilde{u} = \tilde{0} = \tilde{v}$, then $\tilde{u}_{n_k} = \tilde{0} = \tilde{v}_{n_k}$ and $u_{n_k} = 0 = v_{n_k}$ for all $k \in \mathbb{N}$; this would contradict the fact that $u_n \neq v_n$ for all $n \in \mathbb{N}$. Thus, we can assume without loss of generality that $\tilde{u} \neq \tilde{0}$ and $u \neq 0$.

By the definitions of $\mathcal{F}$ and $\sim$, we can then find $u, v \in \mathcal{F}$ and $(g_{1,k}), (g_{2,k}) \in \mathcal{G}$ such that $u \in \tilde{u}$, $v \in \tilde{v}$, $u_{n_k} = g_{1,k}(u)$ and $v_{n_k} = g_{2,k}(v)$ for every $k \in \mathbb{N}$. Now let $g_k := g_{1,k}^{-1}g_{2,k}$ and note that how $(\mathcal{G}_1)$ and $(\mathcal{G}_2)$ imply, for all $k \in \mathbb{N}$, that

$$\|u_{n_k} - v_{n_k}\|_{\mathbb{B}} = \|g_{1,k}(u) - g_{2,k}(v)\|_{\mathbb{B}} = \|g_{1,k}^{-1}(g_{1,k}(u)) - g_{1,k}^{-1}(g_{2,k}(v))\|_{\mathbb{B}} = \|u - g_k(v)\|_{\mathbb{B}}.$$

Thus, for any $k \in \mathbb{N}$,

$$g_k(v) = u \implies g_{1,k}^{-1}(g_{2,k}(v)) = u \implies g_{2,k}(v) = g_{1,k}(u) \implies v_{n_k} = u_{n_k}.$$

Therefore, $g_k(v) \neq u$ for all $k \in \mathbb{N}$. Since $u \neq 0$, $(\mathcal{G}_3)$ allows us to conclude that

$$\kappa = \lim \|u_{n_k} - v_{n_k}\|_{\mathbb{B}} = \lim \|u - g_k(v)\|_{\mathbb{B}} \geq \inf_{k \in \mathbb{N}} \|u - g_k(v)\|_{\mathbb{B}} \geq \inf_{g \in \mathcal{G}, g(v) \neq u} \|u - g(v)\|_{\mathbb{B}} > 0. \quad \square$$

The next result, which is our compactness condition, was introduced (as far as the authors know), in [2].

Proposition 7.2. *Let $(u_n), (v_n)$ be (PS) sequences for $\mathcal{J}$. If $\tilde{\mathcal{K}}$ is finite, then either $\|u_n - v_n\|_{\mathbb{B}} \rightarrow 0$ or $\limsup \|u_n - v_n\|_{\mathbb{B}} \geq \kappa$.*

Proof. Suppose $\|u_n - v_n\|_{\mathbb{B}} \not\rightarrow 0$. By $(\mathcal{G}_4)$, there are $u, v \in \mathbb{B}$ with $u \neq v$, $(g_n) \in \mathcal{G}$, and indexes $(n_k) \subset \mathbb{N}$ such that $g_{n_k}(u_{n_k}) \rightharpoonup u$, $g_{n_k}(v_{n_k}) \rightharpoonup v$, and $(g_{n_k}(u_{n_k})), (g_{n_k}(v_{n_k}))$ are (PS) sequences. Thus, $g_{n_k}(u_{n_k}) - g_{n_k}(v_{n_k}) \rightharpoonup u - v$ and, by $(\mathcal{J}_3)$, we have $\mathcal{J}(u) = 0$ and $\mathcal{J}(v) = 0$, i.e., $u, v \in \mathcal{K}$. Then, as

$u \neq v$, the properties of the limit superior, the weakly lower semicontinuity of norms, $(\mathcal{J}_2)$, $(\mathcal{G}_1)$ and [prop. 7.1](#) give

$$\begin{aligned} \limsup \|u_n - v_n\|_{\mathbb{B}} &\geq \limsup \|u_{n_k} - v_{n_k}\|_{\mathbb{B}} = \limsup \|g_{n_k}(u_{n_k}) - g_{n_k}(v_{n_k})\|_{\mathbb{B}} \\ &\geq \liminf \|g_{n_k}(u_{n_k}) - g_{n_k}(v_{n_k})\|_{\mathbb{B}} \geq \liminf \|u_{n_k} \circ \tau_{z_{n_k}} - v_{n_k} \circ \tau_{z_{n_k}}\|_{\mathbb{B}} \geq \|u - v\|_{\mathbb{B}} \geq \kappa. \quad \square \end{aligned}$$

For $a, b \in \mathbb{R}$ such that $a \leq b$, we introduce the usual sets $\mathcal{J}^a := \mathcal{J}^{-1}((-\infty, a])$, $\mathcal{J}_a := \mathcal{J}^{-1}([a, \infty))$ and $\mathcal{J}_b^a = \mathcal{J}^a \cap \mathcal{J}_b$. Our goal for most of the remainder of this section is to construct a deformation mapping that allows us to use pseudoindeindex theory to contradict the supposition that $\tilde{\mathcal{K}}$ is finite. This mapping will be the flow of a “gradient-like” field for $\mathcal{J}$. In the construction of this field, we use the function

$$\nu^d: \mathcal{J}^d \rightarrow [0, \infty); \quad \nu^d(u) := \inf_{v \in \mathcal{J}^d} (\|\mathcal{J}(v)\|_{\mathbb{B}'} + \|u - v\|_{\mathbb{B}}).$$

Because of this, we state and prove some of its properties next.

Proposition 7.3. *Let $d \in (0, \infty)$ and $(u_n) \subset \mathcal{J}^d$. Then:*

- (ν_1) $\forall u \in \mathcal{J}^d, \nu^d(u) \leq \|\mathcal{J}(u)\|_{\mathbb{B}'};$
- (ν_2) $\nu^d(u) = 0 \iff \mathcal{J}(u) = 0;$
- (ν_3) $\nu^d(u_n) \rightarrow 0 \implies \exists (v_n) \subset \mathcal{J}^d: (\|\mathcal{J}(v_n)\|_{\mathbb{B}'} \rightarrow 0 \text{ and } \|v_n - u_n\|_{\mathbb{B}} \rightarrow 0);$
- (ν_4) ν^d is Lipschitz continuous;
- (ν_5) ν^d is an even function.

Proof. (ν_1) Trivial. (ν_2) Since $\mathcal{J}$ is continuous, it suffices to note that

$$\begin{aligned} \nu^d(u) = 0 &\iff \inf_{v \in \mathcal{J}^d} (\|\mathcal{J}(v)\|_{\mathbb{B}'} + \|v - u\|_{\mathbb{B}}) = 0 \iff \exists (v_n) \subset \mathcal{J}^d: (\|\mathcal{J}(v_n)\|_{\mathbb{B}'} + \|v_n - u\|_{\mathbb{B}}) \rightarrow 0 \\ &\iff \exists (v_n) \subset \mathcal{J}^d: \|\mathcal{J}(v_n)\|_{\mathbb{B}'} = 0 \text{ and } v_n \rightarrow u \text{ in } \mathbb{B}. \end{aligned}$$

(ν_3) By the definition of infimum, we can always find $(\bar{v}_{n,m}) \in \mathcal{J}^d$ such that

$$\lim_{m \rightarrow \infty} \|\mathcal{J}(\bar{v}_{n,m})\|_{\mathbb{B}'} + \|u_n - \bar{v}_{n,m}\|_{\mathbb{B}} = \nu^d(u_n)$$

for every $n \in \mathbb{N}$, meaning that there exists $m_n \in \mathbb{N}$ for which

$$|\nu^d(u_n) - \|\mathcal{J}'(\bar{v}_{n,m_n})\|_{\mathbb{B}'} + \|u_n - \bar{v}_{n,m_n}\|_{\mathbb{B}}| < \frac{1}{n}.$$

Setting $(v_n) := (\bar{v}_{n,m_n})$ and using the hypothesis that $\nu^d(u_n) \rightarrow 0$ gives us (ν_3) .

(ν_4) Let $u_1, u_2, v \in \mathcal{J}^d$. By the triangle inequality, we have

$$\nu^d(u_1) \leq \|\mathcal{J}(v)\|_{\mathbb{B}'} + \|u_1 - v\|_{\mathbb{B}} \leq \|\mathcal{J}(v)\|_{\mathbb{B}'} + \|u_2 - v\|_{\mathbb{B}} + \|u_1 - u_2\|_{\mathbb{B}}.$$

Taking the infimum over $v \in \mathcal{J}^d$ on both sides gives us $\nu^d(u_1) \leq \nu^d(u_2) + \|u_1 - u_2\|_{\mathbb{B}}$, i.e., Thus, $\nu^d(u_1) - \nu^d(u_2) \leq \|u_1 - u_2\|_{\mathbb{B}}$. Swapping u_1 and u_2 in this previous argument yields (ν_4) .

(ν_5) Let $u \in \mathcal{J}^d$. By $(\mathcal{J}_1)$, $-u \in \mathcal{J}^d$ and $\mathcal{J}$ is odd. Then

$$\begin{aligned} \nu^d(-u) &= \inf_{v \in \mathcal{J}^d} (\|\mathcal{J}(v)\|_{\mathbb{B}'} + \|-u - v\|_{\mathbb{B}}) = \inf_{v \in \mathcal{J}^d} (\|-\mathcal{J}(v)\|_{\mathbb{B}'} + \|-u - (-v)\|_{\mathbb{B}}) \\ &= \inf_{v \in \mathcal{J}^d} (\|\mathcal{J}(-v)\|_{\mathbb{B}'} + \|v - u\|_{\mathbb{B}}) = \inf_{v \in \mathcal{J}^d} (\|\mathcal{J}(v)\|_{\mathbb{B}'} + \|u - v\|_{\mathbb{B}}) = \nu^d(u). \quad \square \end{aligned}$$

Our next step is to employ ν^d in the construction of the aforementioned “gradient-like” field for $\mathcal{J}$; this field — which we will refer to as Υ — differs from the traditional pseudogradient by requiring that $\mathcal{J}(u)(\Upsilon^d(u)) > \frac{1}{2}\nu^d(u)$, instead of the usual inequality $\mathcal{J}(u)(\Upsilon^d(u)) > \|\mathcal{J}(u)\|_{\mathbb{B}'}$. Nonetheless, its construction is similar to the usual case: it uses the hypothesis that $\mathbb{B}$ is separable to construct a partition of unity of the set that is dense in $\mathbb{B}$ and then defines Υ in terms of this partition of unity.

Proposition 7.4. *Let $d \in (0, \infty)$. Then there exists $\Upsilon^d: \mathcal{J}^d - \mathcal{K} \rightarrow \mathbb{B}$ satisfying, for every $u \in \mathcal{J}^d - \mathcal{K}$:*

- (Υ_1) *There exists $U \subset \mathbb{B}$ open such that $\Upsilon^d|_{U \cap \mathcal{J}^d - \mathcal{K}}$ is Lipschitz continuous;*

$$(\Upsilon_2) \quad \|\Upsilon^d(u)\|_{\mathbb{B}} \leq 1; \quad (\Upsilon_3) \quad \mathcal{J}(u)(\Upsilon^d(u)) > \frac{1}{2}\nu^d(u); \quad (\Upsilon_4) \quad \Upsilon^d(-u) = -\Upsilon^d(u).$$

Proof. Let $u \in \mathcal{J}^d - \mathcal{K}$ and note that $-u \in \mathcal{J}^d - \mathcal{K}$ because of $(\mathcal{J}_1)$. By (ν_1) and the definition of the dual norm, there exists $v_u \in \mathbb{B}$ with $\|v_u\|_{\mathbb{B}} = 1$ for which $\mathcal{J}(u)(v_u) > \frac{1}{2}\nu^d(u)$. Since $\mathcal{J} \in \mathcal{C}^1(\mathbb{B}, \mathbb{R})$, the mapping $u \mapsto \mathcal{J}(u)v_u$ is continuous and by (ν_4) , so is ν^d . Thus, we can find $U_u \subset \mathbb{B}$ open with $u \in U_u$ and such that $\mathcal{J}(w)v_u > \frac{1}{2}\nu^d(w)$ for every $w \in U_u$. Note that $U_u \cap (\mathcal{J}^d - \mathcal{K})$ is open in the subspace topology of $\mathcal{J}^d - \mathcal{K}$.

We now use the hypothesis that $\mathbb{B}$ is separable to find $(u_n) \subset \mathbb{B}$ which is dense therein. Then the family $\mathcal{U} = \{U_n := U_{u_n} \cap (\mathcal{J}^d - \mathcal{K}) \subset \mathcal{J}^d - \mathcal{K} \mid n \in \mathbb{N}\}$ is an open covering of $\mathcal{J}^d - \mathcal{K}$. Since $\mathcal{J}^d - \mathcal{K}$ is metric and hence paracompact, there exists a locally finite refinement of $\mathcal{U}$, that is, an open cover of $\mathcal{J}^d - \mathcal{K}$ given by a family $\mathcal{V} = \{V_n \subset \mathcal{J}^d - \mathcal{K} \mid n \in \mathbb{N}\}$, such that for each $n \in \mathbb{N}$, we can find $n' \in \mathbb{N}$ for which $V_n \subset U_{n'}$ holds, and for each $u \in \mathcal{J}^d - \mathcal{K}$, there is an open (in the subspace topology) set $U \subset \mathcal{J}^d - \mathcal{K}$ with $u \in U$ and satisfying $\#\{n \in \mathbb{N} \mid V_n \cap U \neq \emptyset\} < \infty$.

Let $n \in \mathbb{N}$ and n' be as in the paragraph above, and note that if $u \in V_n$ and $v_n := v_{u_{n'}}$, then $\mathcal{J}(u)v_n > \frac{1}{2}\nu^d(u)$ because $V_n \subset U_{n'}$. Now consider

$$\alpha_n : \mathcal{J}^d - \mathcal{K} \rightarrow [0, \infty); \quad \alpha_n(u) := \text{dist}(u, V_n^c) \left(\sum_{i=1}^{\infty} \text{dist}(u, V_i^c) \right)^{-1},$$

which is well-defined because there exists an open neighborhood of $u \in \mathcal{J}^d - \mathcal{K}$ in which all but finite terms of the sum vanish, and because the fact that $\mathcal{V}$ covers $\mathcal{J}^d - \mathcal{K}$ guarantees that the denominator never vanishes. Since the distance from a point to a set is Lipschitz continuous, finite sums of Lipschitz functions remain Lipschitz continuous, and the ratio of two Lipschitz functions is locally Lipschitz continuous (as long as the denominator does not vanish), we have that each α_n is locally Lipschitz continuous (here the fact that each point has a neighborhood in which only finitely many V_n intersect is, again, crucial). We then define

$$\Upsilon^d : \mathcal{J}^d - \mathcal{K} \rightarrow \mathbb{B}; \quad \Upsilon^d(u) := \frac{1}{2} \left(\left(\sum_{i=1}^{\infty} \alpha_i(u) v_i \right) - \sum_{i=1}^{\infty} \alpha_i(-u) v_i \right).$$

Let $u \in \mathcal{J}^d - \mathcal{K}$. Since $\mathcal{V}$ is a locally finite cover of $\mathcal{J}^d - \mathcal{K}$, there exists $U^+, U^- \subset \mathcal{J}^d - \mathcal{K}$ open in the subspace topology with $u \in U^+$, $-u \in U^-$, $k \in \mathbb{N}$ and $n_1, \dots, n_k \in \mathbb{N}$ such that $(U^+ \cup U^-) \cap V_n = \emptyset$ for all $n \in \mathbb{N} - \{n_1, \dots, n_k\}$. Now let $w \in U^+ \cup U^-$ and $m \in \mathbb{N} - \{n_1, \dots, n_k\}$. Then $w \notin V_m$ and hence $w \in V_m^c$, which, by the definition of α_m , implies that $\alpha_m(w) = 0$. In particular, $\Upsilon^d(w) = \frac{1}{2} \sum_{i=1}^k (\alpha_{n_i}(w) - \alpha_{n_i}(-w)) v_{n_i}$, and

$$\sum_{i=1}^k \alpha_{n_i}(w) = \sum_{i=1}^k \frac{\text{dist}(w, V_{n_i}^c)}{\sum_{j=1}^{\infty} \text{dist}(w, V_j^c)} = \sum_{i=1}^k \frac{\text{dist}(w, V_{n_i}^c)}{\sum_{j=1}^k \text{dist}(w, V_{n_j}^c)} = \frac{\sum_{i=1}^k \text{dist}(w, V_{n_i}^c)}{\sum_{j=1}^k \text{dist}(w, V_{n_j}^c)} = 1. \quad (7.1)$$

(Υ_1) Since each α_{n_i} is locally Lipschitz, there exists $U_{n_1}^{\pm}, \dots, U_{n_k}^{\pm} \subset \mathcal{J}^d - \mathcal{K}$ open with $\pm u \in U_{n_i}^{\pm}$ and such that $\alpha_{n_i}|_{U_{n_i}^{\pm}}$ is Lipschitz for each $i \in [1, k] \cap \mathbb{N}$. Let $K_{n_i}^{\pm} \in (0, \infty)$ be the Lipschitz constant for $\alpha_{n_i}|_{U_{n_i}^{\pm}}$ and consider

$$U := U^+ \cap \left(\bigcap_{i=1}^k U_{n_i}^+ \right) \cap (-U^-) \cap \left(\bigcap_{i=1}^k -U_{n_i}^- \right)$$

We immediately see that U is open (finite intersection of open sets) and non-empty (contains u). For each $i \in [1, k] \cap \mathbb{N}$, note that $U_{n_i}^+, -U_{n_i}^- \subset U$, and that the latter of this inclusions implies that $U_{n_i}^- \subset -U$.

Thus, each α_{n_i} is Lipschitz when restricted to any of U or $-U$. For any $w_1, w_2 \in U$, we now conclude that

$$\begin{aligned} \|\Upsilon^d(w_1) - \Upsilon^d(w_2)\|_{\mathbb{B}} &= \frac{1}{2} \left\| \sum_{i=1}^k \left(\alpha_{n_i}(w_1) - \alpha_{n_i}(-w_1) - \alpha_{n_i}(w_2) + \alpha_{n_i}(-w_2) \right) v_{n_i} \right\|_{\mathbb{B}} \\ &\leq \frac{1}{2} \sum_{i=1}^k \left(|\alpha_{n_i}(w_1) - \alpha_{n_i}(w_2)| + |\alpha_{n_i}(-w_1) - \alpha_{n_i}(-w_2)| \right) \\ &\leq \frac{1}{2} \sum_{i=1}^k K_{n_i}^+ \|w_1 - w_2\|_{\mathbb{B}} + K_{n_i}^- \|-w_1 - (-w_2)\|_{\mathbb{B}} \leq \frac{k \max_{1 \leq i \leq k} K_{n_i}^{\pm}}{2} \|w_1 - w_2\|_{\mathbb{B}}. \end{aligned}$$

(Υ_2) Since $w \in U^+ \cup U^-$, we can use eq. (7.1) and the definition of Υ^d to get

$$\begin{aligned} \|\Upsilon^d(w)\|_{\mathbb{B}} &= \frac{1}{2} \left\| \sum_{i=1}^k (\alpha_{n_i}(w) - \alpha_{n_i}(-w)) v_{n_i} \right\|_{\mathbb{B}} \leq \frac{1}{2} \sum_{i=1}^k (|\alpha_{n_i}(w)| + |\alpha_{n_i}(-w)|) \\ &= \frac{1}{2} \left(\left(\sum_{i=1}^k |\alpha_{n_i}(w)| \right) + \left(\sum_{i=1}^k |\alpha_{n_i}(-w)| \right) \right) = \frac{1}{2} (1 + 1) = 1. \end{aligned}$$

(Υ_3) We initially had chosen $n_1, \dots, n_k$ such that they were the only integers for which $(U^+ \cup U^-) \cap V_n$ could be non-empty. Now we need to refine these choices to U^+ and U^- individually. Let $k^+, k^- \in \mathbb{N}$ and $n_1^+, \dots, n_{k^+}^+, n_1^-, \dots, n_{k^-}^- \in \mathbb{N}$ such that $\pm u \in V_n$ if, and only if, $n \in \{n_1^{\pm}, \dots, n_{k^{\pm}}^{\pm}\}$. Refining each summation to the specific indexes where their summand does not vanish, and then using that $\mathcal{J}$ is odd (by ($\mathcal{J}_1$)) and that $\mathcal{J}(u)$ is linear, we have

$$\begin{aligned} \mathcal{J}(u)(\Upsilon^d(u)) &= \mathcal{J}(u) \left(\frac{1}{2} \sum_{i=1}^k (\alpha_{n_i}(u) v_{n_i} - \alpha_{n_i}(-u) v_{n_i}) \right) = \left(\sum_{i=1}^k \frac{\alpha_{n_i}(u)}{2} \mathcal{J}(u)(v_{n_i}) \right) \\ &\quad - \sum_{i=1}^k \frac{\alpha_{n_i}(-u)}{2} \mathcal{J}(u)(v_{n_i}) = \left(\sum_{i=1}^{k^+} \frac{\alpha_{n_i^+}(u)}{2} \mathcal{J}(u)(v_{n_i^+}) \right) + \sum_{i=1}^{k^-} \frac{\alpha_{n_i^-}(-u)}{2} \mathcal{J}(-u)(v_{n_i^-}). \end{aligned}$$

Since $\pm u \in V_{n_i^{\pm}}$, the construction of $\mathcal{V}$ implies $\mathcal{J}(\pm u)(v_{n_i^{\pm}}) > \frac{1}{2} \nu^d(\pm u)$. Thus, eq. (7.1) and (ν_5) give

$$\mathcal{J}(u)(\Upsilon^d(u)) > \left(\sum_{i=1}^{k^+} \frac{\alpha_{n_i^+}(u)}{4} \nu^d(u) \right) + \sum_{i=1}^{k^-} \frac{\alpha_{n_i^-}(-u)}{4} \nu^d(-u) = \frac{1}{2} \nu^d(u).$$

(Υ_4) Just note that at each point, Υ^d is the odd part of the finite sums of $\alpha_i(\cdot) v_i$. $\square$

For each $d \in \mathbb{R}$, consider the set of critical points at level d , which we define as $\mathcal{K}_d := \mathcal{J}^{-1}(\{d\}) \cap \mathcal{K}$. When $\tilde{\mathcal{K}}$ is finite, ($\mathcal{J}_2$) implies that $\#(\mathcal{J}(\mathcal{K})) < \infty$, and so for each $d \in \mathcal{J}(\mathcal{K})$, there would exist $\varepsilon_d \in (0, \infty)$ for which $\mathcal{J}_{d-2\varepsilon_d}^{d+2\varepsilon_d} \cap \mathcal{K} = \mathcal{K}_d$. Let $\varepsilon_0 := \frac{1}{2} \min_{d \in \mathcal{J}(\mathcal{K})} \varepsilon_d > 0$. If $d_0 \in (0, \infty)$ and $\mathcal{J}_{d_0-2\varepsilon_0}^{d_0+2\varepsilon_0} \cap \mathcal{K} \neq \emptyset$, then for any critical value $d \in [d_0 - 2\varepsilon_0, d_0 + 2\varepsilon_0] = \mathcal{J}(u)$, we have

$$d_0 + 2\varepsilon_0 \leq d_0 + \varepsilon_d = d + d_0 - d + \varepsilon_d \leq d + |d_0 - d| + \varepsilon_d \leq d + 2\varepsilon_0 + \varepsilon_d \leq d + 2\varepsilon_d.$$

Similarly, we can also get $d_0 - 2\varepsilon_0 > d - 2\varepsilon_d$. Thus, $\mathcal{J}_{d_0-2\varepsilon_0}^{d_0+2\varepsilon_0} \subset \mathcal{J}_{d-2\varepsilon_d}^{d+2\varepsilon_d}$ and $\mathcal{J}_{d_0-2\varepsilon_0}^{d_0+2\varepsilon_0} \cap \mathcal{K} \subset \mathcal{J}_{d-2\varepsilon_d}^{d+2\varepsilon_d} \cap \mathcal{K} = \mathcal{K}_d$, i.e., there is at most one critical value in any interval of the form $(d_0 - 2\varepsilon_0, d_0 + 2\varepsilon_0)$.

For each $U \subset \mathbb{B}$ and $\delta \in (0, \infty)$, we define the δ -neighborhood of U as $\mathcal{B}(U, \delta) := \{u \in \mathbb{B} \mid \text{dist}(u, U) < \delta\}$. We are now ready to state and prove the strict positivity of $\nu^{d+2\varepsilon_0}$ over some special set.

Proposition 7.5. *Let $d \in (0, \infty)$, $\delta \in (0, \frac{\kappa}{3})$, and suppose that $\tilde{\mathcal{K}}$ is finite. Then*

$$\zeta^{d, \delta} := \inf \left\{ \frac{1}{2} \nu^{d+2\varepsilon_0}(u) \in \mathbb{R} \mid u \in \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} \cap \left(\overline{\mathcal{B}(\mathcal{K}, 3\delta)} - \mathcal{B}(\mathcal{K}, 2\delta) \right) \right\} > 0.$$

Proof. First, note that $(\mathcal{E}_1)$ implies that $Dg(u) = g$ always; this, along with the chain rule, $(\mathcal{J}_2)$ and $(\mathcal{E}_2)$ allows us to conclude that, for all $u \in \mathbb{B}$ and $g \in \mathcal{G}$,

$$\|\mathcal{J}(g(u))\|_{\mathbb{B}'} = \|\mathcal{J}(u)\|_{\mathbb{B}'}. \quad (7.2)$$

Now suppose by contradiction that $\zeta^{d,\delta} = 0$. We can then find a sequence $(\bar{w}_n) \subset \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} \cap \left(\overline{\mathcal{B}(\mathcal{K}, 3\delta)} - \mathcal{B}(\mathcal{K}, 2\delta) \right)$ with $\nu^{d+2\varepsilon_0}(\bar{w}_n) \rightarrow 0$. Note that [prop. 7.1](#) implies that if $u, v \in \mathcal{F}$ and $g_1, g_2 \in \mathcal{G}$, then either $g_1(u) = g_2(v)$ or $\mathcal{B}(g_1(u), r) \cap \mathcal{B}(g_2(v), r) = \emptyset$ for all $r \in (0, \kappa)$. This, along with the fact that $3\delta < \kappa$ and the supposition that $\tilde{\mathcal{K}}$ is finite, gives us

$$\begin{aligned} \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} \cap \left(\overline{\mathcal{B}(\mathcal{K}, 3\delta)} - \mathcal{B}(\mathcal{K}, 2\delta) \right) &= \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} \cap \left(\left(\bigcup_{u \in \mathcal{F}} \bigcup_{g \in \mathcal{G}} \overline{\mathcal{B}(g(u), 3\delta)} \right) - \bigcup_{u \in \mathcal{F}} \bigcup_{g \in \mathcal{G}} \mathcal{B}(g(u), 2\delta) \right) \\ &= \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} \cap \left(\bigcup_{u \in \mathcal{F}} \bigcup_{g \in \mathcal{G}} \left(\overline{\mathcal{B}(g(u), 3\delta)} - \mathcal{B}(g(u), 2\delta) \right) \right). \end{aligned}$$

Therefore, we have sequences $(v_n) \in \mathcal{F}$ and $(g_n) \in \mathcal{G}$ such that $\bar{w}_n \in \overline{\mathcal{B}(g_n(v_n), 3\delta)} - \mathcal{B}(g_n(v_n), 2\delta)$. We can now use the pigeonhole principle to find $v_0 \in \mathcal{F}$ and a subsequence such that $(v_n) = (v_0)$. Therefore, up to subsequence, we have $\bar{w}_n \in \overline{\mathcal{B}(g_n(v_0), 3\delta)} - \mathcal{B}(g_n(v_0), 2\delta)$; then by $(\mathcal{E}_2)$, $w_n := g_n^{-1}(\bar{w}_n) \in \overline{\mathcal{B}(v_0, 3\delta)} - \mathcal{B}(v_0, 2\delta)$. Moreover, $(\mathcal{E}_2)$ and [eq. \(7.2\)](#) imply that $\nu^{d+2\varepsilon_0}(w_n) \rightarrow 0$.

By (ν_3) , there exists $(u_n) \subset \mathcal{J}_{d-\varepsilon_0}^{d+2\varepsilon_0}$ for which $\|u_n - w_n\|_{\mathbb{B}} \rightarrow 0$ and $\|\mathcal{J}(u_n)\|_{\mathbb{B}'} \rightarrow 0$. Since $(u_n) \subset \mathcal{J}_{d-\varepsilon_0}^{d+2\varepsilon_0}$, we have $\mathcal{J}(u_n) \leq d + 2\varepsilon_0$, and so (u_n) is a (PS) sequence. As $\|u_n - w_n\|_{\mathbb{B}} \rightarrow 0$, there is $n_0 \in \mathbb{N}$ such that $\|u_n - w_n\|_{\mathbb{B}} < \frac{\delta}{2}$ for all $n \geq n_0$. For all such n , the triangle and reverse triangle inequalities give us

$$0 < 2\delta - \frac{\delta}{2} \leq \|w_n - v_0\|_{\mathbb{B}} - \|u_n - w_n\|_{\mathbb{B}} \leq \|u_n - v_0\|_{\mathbb{B}} \leq \|u_n - w_n\|_{\mathbb{B}} + \|w_n - v_0\|_{\mathbb{B}} \leq \|u_n - w_n\|_{\mathbb{B}} + 3\delta. \quad (7.3)$$

It is easy to see from [eq. \(7.3\)](#) that $\|u_n - v_0\|_{\mathbb{B}} \not\rightarrow 0$. On the other hand, taking the limit superior of [eq. \(7.3\)](#) and using $\|u_n - w_n\|_{\mathbb{B}} \rightarrow 0$ yields $\limsup \|u_n - v_0\|_{\mathbb{B}} \leq 3\delta < \kappa$. Thus, [eq. \(7.3\)](#) and [prop. 7.1](#) (with the trivial (PS) sequence (v_0)) always contradict each other, which concludes the proof. $\square$

Consider $B_{d,\delta} := \mathcal{J}_{d-2\varepsilon_0}^{d+2\varepsilon_0} - \mathcal{B}(\mathcal{K}, \delta)$, $A_{d,\delta} := \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0} - \mathcal{B}(\mathcal{K}, 2\delta)$ and

$$\psi^{d,\delta} : \mathbb{B} \rightarrow [0, \infty); \quad \psi^{d,\delta}(u) = \text{dist}\left(u, B_{d,\delta}^c\right) \left(\text{dist}\left(u, B_{d,\delta}^c\right) + \text{dist}\left(u, A_{d,\delta}\right) \right)^{-1}.$$

Note that $\psi^{d,\delta}$ is locally Lipschitz, $\psi^{d,\delta}|_{A_{d,\delta}} \equiv 1$, $\psi^{d,\delta}|_{\overline{B_{d,\delta}}} \equiv 0$, $\psi^{d,\delta}(\text{int}(B_{d,\delta})) \subset (0, 1)$. Now let

$$\mathcal{W}_{d,\delta} : \mathbb{B} \rightarrow \mathbb{B}; \quad \mathcal{W}_{d,\delta}(u) := \begin{cases} -\psi^{d,\delta}(u) \Upsilon^{d+2\varepsilon_0}(u), & u \in B_{d,\delta}; \\ 0, & u \in B_{d,\delta}^c. \end{cases}$$

Properties (Υ_1) and (Υ_4) also make it easy to see that $\mathcal{W}_{d,\delta}$ is locally Lipschitz and odd. We now prove the existence and some properties of the gradient flow of $\mathcal{W}_{d,\delta}$; this will be our deformation mapping.

Proposition 7.6. *Suppose that $\tilde{\mathcal{K}}$ is finite and let $d \in (0, \infty)$, $\delta \in (0, \frac{\kappa}{3})$. Then there exists $\eta^{d,\delta} \in \mathcal{C}((-\infty, \infty) \times \mathbb{B}, \mathbb{B})$ satisfying, for all $t \in (-\infty, \infty)$ and $u \in \mathbb{B}$, the Initial Value Problem (IVP)*

$$\begin{cases} \frac{d}{dt} \eta^{d,\delta}(t, u) = \mathcal{W}_{d,\delta}(\eta^{d,\delta}(t, u)); \\ \eta^{d,\delta}(0, u) = u. \end{cases} \quad (7.4)$$

Moreover, for all $t_0 \in \mathbb{R}$ and $u_0 \in \mathbb{B}$, we have:

$$(\eta_1) \text{ If } u_0 \in \text{int}(B_{d,\delta}), \text{ then } \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t_0, u_0)) < 0. \text{ Otherwise, } \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t_0, u_0)) = 0.$$

$$(\eta_2) \quad \|\eta^{d,\delta}(t_1, u_0) - \eta^{d,\delta}(t_0, u_0)\|_{\mathbb{B}} \leq |t_1 - t_0| \text{ for each } t_1 \in \mathbb{R};$$

$$(\eta_3) \quad u \mapsto \eta^{d,\delta}(t_0, u) \text{ is a homeomorphism of } \mathbb{B} \text{ onto itself;} \quad (\eta_4) \quad \eta^{d,\delta}(t_0, -u_0) = -\eta^{d,\delta}(t_0, u_0);$$

Proof. Since $\mathcal{W}_{d,\delta}$ is autonomous, it could be viewed as a function from $\mathbb{R} \times \mathbb{B}$. Then proposition 6.14, notation 6.15, and theorem 6.21 of [9] guarantee the existence of $a, b: \mathbb{B} \rightarrow [-\infty, \infty]$ and $\eta^{d,\delta} \in \mathcal{C}(D, \mathbb{B})$ — where $D := \bigcup_{u \in \mathbb{B}} \left((a(u), b(u)) \times \{u\} \right)$ — such that $a(u) < 0 < b(u)$ for all $u \in \mathbb{B}$, and $\eta^{d,\delta}$ satisfies IVP (7.4) in all its domain. In order to show that $D = \mathbb{R} \times \mathbb{B}$, we need first to show the following intermediate results.

Let $u_0 \in \mathbb{B}$, $t_0, t_1 \in (a(u_0), b(u_0))$ and assume, without loss of generality, that $t_0 \leq t_1$. Since $\eta^{d,\delta}$ satisfies IVP (7.4), the fundamental theorem of calculus gives us

$$\|\eta^{d,\delta}(t_1, u_0) - \eta^{d,\delta}(t_0, u_0)\|_{\mathbb{B}} = \left\| \int_{t_0}^{t_1} \frac{d}{dt} \eta^{d,\delta}(s, u_0) ds \right\|_{\mathbb{B}} \leq \left| \int_{t_0}^{t_1} \|\mathcal{W}_{d,\delta}(\eta^{d,\delta}(s, u_0))\|_{\mathbb{B}} ds \right|. \quad (7.5)$$

For each $s \in (t_0, t_1)$, either $\eta^{d,\delta}(s, u_0) \in B_{d,\delta}$ or $\eta^{d,\delta}(s, u_0) \in B_{d,\delta}^c$. Thus, there exists $I \subset (t_0, t_1)$ such that $\eta^{d,\delta}(I, u_0) \subset B_{d,\delta}$ and $\eta^{d,\delta}((t_0, t_1) - I, u_0) \subset B_{d,\delta}^c$. The definition of $\mathcal{W}_{d,\delta}$ then allows us to write

$$\begin{aligned} \|\eta^{d,\delta}(t_1, u_0) - \eta^{d,\delta}(t_0, u_0)\|_{\mathbb{B}} &\leq \left| \int_{t_0}^{t_1} \|\mathcal{W}_{d,\delta}(\eta^{d,\delta}(s, u_0))\|_{\mathbb{B}} ds \right| \\ &= \left| \int_I \psi^{d,\delta}(\eta^{d,\delta}(s, u_0)) \|\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(s, u_0))\|_{\mathbb{B}} ds \right| \leq \left| \int_I \Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(s, u_0))\|_{\mathbb{B}} ds \right| \leq \left| \int_I 1 ds \right| \leq |t_1 - t_0|, \end{aligned} \quad (7.6)$$

where we used (Υ_2) and that $0 \leq \psi^{d,\delta}(v) \leq 1$ for all $v \in \mathbb{B}$.

On one hand, when $\eta^{d,\delta}(t_0, u_0) \in B_{d,\delta}^c$, the definition of $\mathcal{W}_{d,\delta}$ trivially gives us $\|\mathcal{W}_{d,\delta} \eta^{d,\delta}(t_0, u_0)\|_{\mathbb{B}} = 0 \leq 1$. On the other hand, if $\eta^{d,\delta}(t_0, u_0) \in B_{d,\delta}$, then ineq. (7.6) and the definition of $\mathcal{W}_{d,\delta}$ imply that

$$\begin{aligned} \|\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t_0, u_0))\|_{\mathbb{B}} &= \left| \psi^{d,\delta}(\eta^{d,\delta}(t_0, u_0)) \|\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t_0, u_0))\|_{\mathbb{B}} \right| \leq \|\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t_0, u_0))\|_{\mathbb{B}} \\ &= \left\| \frac{d\eta^{d,\delta}(\cdot, u_0)}{dt}(t_0) \right\|_{\mathbb{B}} = \left\| \lim_{t \rightarrow t_0} \frac{\eta^{d,\delta}(t, u_0) - \eta^{d,\delta}(t_0, u_0)}{t - t_0} \right\|_{\mathbb{B}} = \lim_{t \rightarrow t_0} \left\| \frac{\eta^{d,\delta}(t, u_0) - \eta^{d,\delta}(t_0, u_0)}{t - t_0} \right\|_{\mathbb{B}} \\ &= \lim_{t \rightarrow t_0} \frac{\|\eta^{d,\delta}(t, u_0) - \eta^{d,\delta}(t_0, u_0)\|_{\mathbb{B}}}{|t - t_0|} \leq \lim_{t \rightarrow t_0} \frac{|t - t_0|}{|t - t_0|} \leq 1. \end{aligned} \quad (7.7)$$

If $b(u_0) < \infty$, then proposition 6.16 of [9] implies that either $\limsup_{t \uparrow b(u_0)} \|\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t, u_0))\|_{\mathbb{B}} = \infty$ or $\lim_{t \uparrow b(u_0)} \eta^{d,\delta}(t, u_0)$ exists and is not in $\mathbb{B}$. The former is ruled out by ineq. (7.7) and the latter is impossible because $\mathbb{B}$ is complete; thus, $b(u) = \infty$. Showing $a(u_0) = -\infty$ is analogous. Now take arbitrary $t_0, t_1 \in \mathbb{R}$:

(η_1) By IVP (7.4),

$$\left. \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t, u_0)) \right|_{t=t_0} = \mathcal{J}(\eta^{d,\delta}(t_0, u_0)) \left(\left. \frac{d}{dt} \eta^{d,\delta}(t, u_0) \right|_{t=t_0} \right) = \mathcal{J}(\eta^{d,\delta}(t_0, u_0))(\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t_0, u_0))).$$

If $\eta^{d,\delta}(t_0, u_0) \in \text{int}(B_{d,\delta})$, then $\psi^{d,\delta}(\eta^{d,\delta}(t_0, u_0)) > 0$. Thus, we have

$$\begin{aligned} \left. \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t, u_0)) \right|_{t=t_0} &= \mathcal{J}(\eta^{d,\delta}(t_0, u_0))(\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t_0, u_0))) \\ &= -\psi^{d,\delta}(\eta^{d,\delta}(t_0, u_0)) \mathcal{J}(\eta^{d,\delta}(t_0, u_0))(\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t_0, u_0))) < 0 \end{aligned} \quad (7.8)$$

because (Υ_3) implies that $\mathcal{J}(\eta^{d,\delta}(t_0, u_0))(\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t_0, u_0))) > \frac{1}{2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t_0, u_0)) \geq 0$. Otherwise, $\mathcal{J}(\eta^{d,\delta}(t_0, u_0))(\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t_0, u_0))) = \mathcal{J}(\eta^{d,\delta}(t_0, u_0))(0) = 0$.

(η_2) The same calculations used in ineq. (7.6) still work with $t_0, t_1 \in \mathbb{R}$.

(η_3) Item 2 of theorem 6.23 of [9] implies that $\eta^{d,\delta}(t_0, \eta^{d,\delta}(-t_0, u)) = \text{Id}|_{\mathbb{B}}$ and $\eta^{d,\delta}(-t_0, \eta^{d,\delta}(t_0, u)) = \text{Id}|_{\mathbb{B}}$ for all $u \in \mathbb{B}$. Therefore, $\eta^{d,\delta}(t_0, \cdot)$ is invertible with inverse $\eta^{d,\delta}(-t_0, \cdot)$. Since $\eta^{d,\delta}$ is also continuous, the mapping is a homeomorphism.

(η_4) By the definitions of $\eta^{d,\delta}$, the oddness of $\mathcal{W}_{d,\delta}$ and the fundamental theorem of calculus:

$$\eta^{d,\delta}(t_0, -u) = \eta^{d,\delta}(0, -u) + \int_0^{t_0} \frac{d}{dt} \eta^{d,\delta}(t, -u) \Big|_{t=s} ds = -u + \int_0^{t_0} \mathcal{W}_{d,\delta}(\eta^{d,\delta}(s, -u)) ds = -\eta^{d,\delta}(t_0, u). \quad \square$$

Henceforth, $\eta^{d,\delta}$ refers to its homonymous with domain $\mathbb{R} \times \mathbb{B}$ in the above proposition. We now show an intermediate result, which can roughly be stated as: if the path taken by deforming some u has its image strictly above $d - 2\varepsilon_0$ and is never near $\mathcal{K}$, then this path has a defined limit at infinity. This conclusion will be used later as a technical tool to show a crucial property of the deformation map $\eta^{d,\delta}$.

Proposition 7.7. *Suppose $\tilde{\mathcal{K}}$ is finite and let $d \in (0, \infty)$, $\delta \in (0, \frac{\kappa}{3})$, and $u \in \mathcal{J}^{d+\varepsilon_0}$. If $\lim_{t \rightarrow \infty} \mathcal{J}(\eta^{d,\delta}(t, u)) > d - \varepsilon_0$ and $\eta^{d,\delta}([0, \infty), u) \cap \mathcal{B}(\mathcal{K}, 2\delta) = \emptyset$, then $\lim_{t \rightarrow \infty} \eta^{d,\delta}(t, u)$ exists.*

Proof. The hypotheses of the proposition together with (η_1) imply $\eta^{d,\delta}([0, \infty), u) \subset \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0}$, which then allows us to conclude that $\eta^{d,\delta}([0, \infty), u) \subset A_{d,\delta}$. Now suppose by contradiction that there exists $\bar{\varepsilon}_0 \in (0, \infty)$ such that for all $t \in [0, \infty)$, we can find some $s_t \in (t, \infty)$ for which $\|\eta^{d,\delta}(t, u) - \eta^{d,\delta}(s_t, u)\|_{\mathbb{B}} \geq \bar{\varepsilon}_0 \geq \bar{\varepsilon} := \frac{1}{2} \min\{\bar{\varepsilon}_0, \frac{\kappa}{2}\}$. Note that $f_t: [t, s_t] \rightarrow [0, \infty)$; $f_t(s) := \|\eta^{d,\delta}(t, u) - \eta^{d,\delta}(s, u)\|_{\mathbb{B}}$ is always continuous, $f_t(t) = 0$, and $f_t(s_t) \geq \bar{\varepsilon}$. The intermediate value theorem then allows us to find $t' \in [s, s_t]$ such that $f_{s_t}(t') = \bar{\varepsilon}$.

We introduce the recursively defined sequence

$$t_1 := 0; \quad t_n := t'_{n-1}, \quad \forall n \in \mathbb{N} - \{1\}.$$

By construction, $\|\eta^{d,\delta}(t_n, u) - \eta^{d,\delta}(t_{n+1}, u)\|_{\mathbb{B}} = \bar{\varepsilon}$ always and $t_n \rightarrow \infty$. In order to avoid repeating arguments in the future, we consider, inside this proof only, $[b, a] := [a, b]$ (resp. $(b, a) := (a, b)$) for $a, b \in \mathbb{R}$ with $b > a$. Let $n \in \mathbb{N}$ be arbitrary and $i \in \{0, 1\}$; since $\eta^{d,\delta}(\cdot, u)$ is continuous, so is the mapping $g_{i,n}: [t_n, t_{n+1}] \rightarrow [0, \infty)$; $g_{i,n}(t) := \|\eta^{d,\delta}(t_{n+i}, u) - \eta^{d,\delta}(t, u)\|_{\mathbb{B}}$. Note that $g_{i,n}(t_{n+i}) = 0$ and $g_{i,n}(t_{n+1-i}) \geq \bar{\varepsilon}$. The intermediate value theorem then says that $S_{i,n} := g_{i,n}^{-1}\left(\left\{\frac{\bar{\varepsilon}}{3}\right\}\right) \cap [t_n, t_{n+1}] \neq \emptyset$. As $[t_n, t_{n+1}]$ is compact and $g_{i,n}^{-1}\left(\left\{\frac{\bar{\varepsilon}}{3}\right\}\right)$ is closed, $S_{i,n}$ is also compact. Thus, there exists $t_n^0 := \min S_{0,n}$ and $t_n^1 := \max S_{1,n}$. If $t \in [t_n^i, t_{n+i}^i]$, then $g_{i,n}(t) < \frac{\bar{\varepsilon}}{3}$ by construction.

We now define $\kappa_n^i := \inf_{t \in [t_{n+i}^i, t_n^i]} \frac{1}{2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u))$. Since $[t_{n+i}^i, t_n^i]$ is compact, (ν_4) implies that there exists $t_n^{2+i} \in [t_{n+i}^i, t_n^i]$ such that $\kappa_n^i = \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t_n^{2+i}, u))$. By hypothesis, $\eta^{d,\delta}([0, \infty), u) \cap \mathcal{B}(\mathcal{K}, 2\delta) = \emptyset$, which leads us to $\eta^{d,\delta}(t_n^{2+i}, u) \notin \mathcal{K}$. Thus, by (ν_2), $\kappa_n^i > 0$. We introduce the shorthands

$$a_n^i := \begin{cases} t_n, & i=0; \\ t_n^1, & i=1, \end{cases} \quad b_n^i := \begin{cases} t_n^0, & i=0; \\ t_{n+1}, & i=1, \end{cases}$$

and observe that $a_n^i \leq b_n^i$ always. Then, using the fundamental theorem of calculus, the definitions of $\eta^{d,\delta}$ and $\mathcal{W}_{d,\delta}$, (Υ_3) and the fact that $\eta^{d,\delta}([0, \infty), u) \subset A_{d,\delta}$, we have

$$\begin{aligned} \mathcal{J}(\eta^{d,\delta}(a_n^i, u)) - \mathcal{J}(\eta^{d,\delta}(b_n^i, u)) &= \int_{b_n^i}^{a_n^i} \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t, u)) dt \\ &= \int_{b_n^i}^{a_n^i} \mathcal{J}(\eta^{d,\delta}(t, u)) (\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t, u))) dt = \int_{a_n^i}^{b_n^i} \psi^{d,\delta}(\eta^{d,\delta}(t, u)) \mathcal{J}(\eta^{d,\delta}(t, u)) (\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u))) dt \\ &> \int_{a_n^i}^{b_n^i} \frac{1}{2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u)) dt \geq \kappa_n^i \int_{a_n^i}^{b_n^i} 1 dt = \kappa_n^i (b_n^i - a_n^i), \quad (\eta_2) \\ \frac{\bar{\varepsilon}}{3} &= \|\eta^{d,\delta}(a_n^i, u) - \eta^{d,\delta}(b_n^i, u)\|_{\mathbb{B}} \leq |a_n^i - b_n^i| < \frac{1}{\kappa_n^i} \left(\mathcal{J}(\eta^{d,\delta}(a_n^i, u)) - \mathcal{J}(\eta^{d,\delta}(b_n^i, u)) \right). \end{aligned} \quad (7.9)$$

As $\lim_{t \rightarrow \infty} \mathcal{J}(\eta^{d,\delta}(t, u)) > d - 2\varepsilon > \varepsilon_0$, (η_1) implies $\lim_{t \rightarrow \infty} \mathcal{J}(\eta^{d,\delta}(t, u))$ exists. Since $a_n^i, b_n^i \in [t_n, t_{n+1}]$ and $t_n \rightarrow \infty$ force $a_n^i, b_n^i \rightarrow \infty$, we have $\mathcal{J}(\eta^{d,\delta}(a_n^i, u)) - \mathcal{J}(\eta^{d,\delta}(b_n^i, u)) \rightarrow 0$. Thus, the only way for the ineq. (7.9) to hold, for all $n \in \mathbb{N}$, is if $\kappa_n^i \rightarrow 0$. This means that $\frac{1}{2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t_n^{2+i}, u)) = \kappa_n^i \rightarrow 0$,

and by (ν_3) , there exists $(v_n^i) \subset \mathcal{J}^{d+2\varepsilon_0}$ such that $\|\mathcal{J}(v_n^i)\|_{\mathbb{B}'} \rightarrow 0$ and $\|v_n^i - \eta^{d,\delta}(t_n^{2+i}, u)\|_{\mathbb{B}} \rightarrow 0$. Since $(v_n^i) \subset \mathcal{J}^{d+2\varepsilon_0}$, we have that (v_n^0) and (v_n^1) are (PS) sequences. Then, [prop. 7.2](#) implies either $\|v_n^1 - v_n^0\|_{\mathbb{B}} \rightarrow 0$ or $\limsup \|v_n^1 - v_n^0\|_{\mathbb{B}} \geq \kappa$. However, since $t_n^{2+i} \in [t_{n+i}, t_n^i]$, the triangular inequality gives us

$$\begin{aligned} \|\eta^{d,\delta}(t_{n+1}, u) - \eta^{d,\delta}(t_n, u)\|_{\mathbb{B}} &\leq \|\eta^{d,\delta}(t_{n+1}, u) - \eta^{d,\delta}(t_n^3, u)\|_{\mathbb{B}} + \|\eta^{d,\delta}(t_n^3, u) - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \\ &\quad + \|\eta^{d,\delta}(t_n^2, u) - \eta^{d,\delta}(t_n, u)\|_{\mathbb{B}} \leq \frac{2}{3}\bar{\varepsilon} + \|\eta^{d,\delta}(t_n^3, u) - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \\ &\leq \frac{2}{3}\bar{\varepsilon} + \|\eta^{d,\delta}(t_n^3, u) - v_n^1\|_{\mathbb{B}} + \|v_n^1 - v_n^0\|_{\mathbb{B}} + \|v_n^0 - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}}, \quad \therefore \\ \|v_n^1 - v_n^0\|_{\mathbb{B}} &\geq \|\eta^{d,\delta}(t_{n+1}, u) - \eta^{d,\delta}(t_n, u)\|_{\mathbb{B}} - \frac{2}{3}\bar{\varepsilon} - \|\eta^{d,\delta}(t_n^3, u) - v_n^1\|_{\mathbb{B}} - \|v_n^0 - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \\ &= \frac{\bar{\varepsilon}}{3} - \|\eta^{d,\delta}(t_n^3, u) - v_n^1\|_{\mathbb{B}} - \|v_n^0 - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}}. \end{aligned}$$

Because $n \in \mathbb{N}$ is arbitrary and $\|v_n^i - \eta^{d,\delta}(t_n^{2+i}, u)\|_{\mathbb{B}} \rightarrow 0$, it cannot be the case that $\|v_n^1 - v_n^0\|_{\mathbb{B}} \rightarrow 0$. We can then contradict our initial supposition by using the subadditivity of the limit superior, as follows:

$$\begin{aligned} \limsup \|v_n^1 - v_n^0\|_{\mathbb{B}} &\leq \limsup \|v_n^1 - \eta^{d,\delta}(t_n^3, u)\|_{\mathbb{B}} + \limsup \|\eta^{d,\delta}(t_n^3, u) - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \\ &\quad + \limsup \|\eta^{d,\delta}(t_n^2, u) - v_n^0\|_{\mathbb{B}} = \limsup \|\eta^{d,\delta}(t_n^3, u) - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \\ &\leq \limsup \|\eta^{d,\delta}(t_n^3, u) - \eta^{d,\delta}(t_{n+1}, u)\|_{\mathbb{B}} + \limsup \|\eta^{d,\delta}(t_{n+1}, u) - \eta^{d,\delta}(t_n, u)\|_{\mathbb{B}} \\ &\quad + \limsup \|\eta^{d,\delta}(t_n, u) - \eta^{d,\delta}(t_n^2, u)\|_{\mathbb{B}} \leq \frac{\bar{\varepsilon}}{3} + \bar{\varepsilon} + \frac{\bar{\varepsilon}}{3} < 2\bar{\varepsilon} < 2\frac{\kappa}{2} < \kappa. \end{aligned}$$

Thus, for each $\bar{\varepsilon} \in (0, \infty)$, there exists $t_{\bar{\varepsilon}} \in [0, \infty)$ for which $\|\eta^{d,\delta}(t, u) - \eta^{d,\delta}(t_{\bar{\varepsilon}}, u)\|_{\mathbb{B}} < \frac{\bar{\varepsilon}}{2}$ whenever $t \in (t_{\bar{\varepsilon}}, \infty)$. This means that for any $(t_n) \subset [0, \infty)$ with $t_n \rightarrow \infty$, we can find $n_0 \in \mathbb{N}$ such that $t_n > t_{\bar{\varepsilon}}$ for all $n \in [n_0, \infty) \cap \mathbb{N}$. Then, for all $n, m \in [n_0, \infty) \cap \mathbb{N}$,

$$\|\eta^{d,\delta}(t_n, u) - \eta^{d,\delta}(t_m, u)\|_{\mathbb{B}} \leq \|\eta^{d,\delta}(t_n, u) - \eta^{d,\delta}(t_{\bar{\varepsilon}}, u)\|_{\mathbb{B}} + \|\eta^{d,\delta}(t_{\bar{\varepsilon}}, u) - \eta^{d,\delta}(t_m, u)\|_{\mathbb{B}} \leq \bar{\varepsilon},$$

meaning that $(\eta^{d,\delta}(t_n, u))$ is a Cauchy sequence. The completeness of $\mathbb{B}$ then completes the proof. $\square$

The following lemma is the culmination of all work done in this section so far; it will eventually imply that $\eta^{d,\delta}$ can take compact sets to a level below $d - \varepsilon$ for some $\varepsilon \in (0, \infty)$. In fact, this is almost the statement of the corollary that will follow it.

Lemma 7.8. *Suppose that $\tilde{\mathcal{K}}$ is finite and let $d \in (0, \infty)$, $\delta \in (0, \frac{\kappa}{3})$. Then there exists $\varepsilon \in (0, \varepsilon_0)$ and a continuous function $s: \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \rightarrow [0, \infty)$ such that $\mathcal{J}(\eta^{d,\delta}(s(u), u)) \leq d - \frac{3}{2}\varepsilon$ for all $u \in \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta)$.*

Proof. Let $\varepsilon := \frac{1}{2} \min \left\{ \frac{\varepsilon_0}{2}, \frac{\delta \kappa^{d,\delta}}{3} \right\}$, $u \in \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta)$, and $C_0 := \eta^{d,\delta}([0, -\infty), u) \cap \mathcal{B}(\mathcal{K}, 2\delta)$. First, suppose that $C_0 \neq \emptyset$, i.e., there exists $t_0 \in [0, \infty)$ such that $\eta^{d,\delta}(t_0, u) \in \mathcal{B}(\mathcal{K}, 2\delta) \subset \mathcal{B}(\mathcal{K}, 3\delta)$.

Since $\eta^{d,\delta}(0, u) = u \in \mathcal{B}(\mathcal{K}, 3\delta)^c$, we can use [prop. A.6](#) to find $t_3 \in (0, t_0)$ such that $\eta^{d,\delta}(t_3, u) \in \partial \mathcal{B}(\mathcal{K}, 3\delta)$. Thus, the set $C_3 := \{t \in [0, t_0] \mid \eta^{d,\delta}(t, u) \in \partial \mathcal{B}(\mathcal{K}, 3\delta)\}$ is non-empty and bounded. We can then define $t'_3 = \sup C_3$ and $C_2 := \{t \in [t_3, t_0] \mid \eta^{d,\delta}(t, u) \in \partial \mathcal{B}(\mathcal{K}, 2\delta)\}$. As $\eta^{d,\delta}(t_0, u) \in \mathcal{B}(\mathcal{K}, 2\delta)$ and $\eta^{d,\delta}(t_3, u) \in \mathcal{B}(\mathcal{K}, 2\delta)^c$, we can use [prop. A.6](#) once more to find $t_2 \in [t_3, t_0]$ such that $\eta^{d,\delta}(t_2, u) \in \partial \mathcal{B}(\mathcal{K}, 2\delta)$. Thus, $C_2 \neq \emptyset$, and we can define $t'_2 := \inf C_2$ because C_2 is clearly bounded.

It is easy to check that $t_2 \in C_2$ and $t_3 \in C_3$, which then leads to $\|\eta^{d,\delta}(t_2, u) - \eta^{d,\delta}(t_3, u)\|_{\mathbb{B}} \geq \text{dist}(\partial \mathcal{B}(\mathcal{K}, 3\delta), \partial \mathcal{B}(\mathcal{K}, 2\delta)) = \delta$. On the other hand, (η_2) implies that $|t_2 - t_3| \geq \|\eta^{d,\delta}(t_2, u) - \eta^{d,\delta}(t_3, u)\|_{\mathbb{B}} \geq \delta$.

Suppose by contradiction that there exists some $s_0 \in [t_3, t_2]$ such that $\eta^{d,\delta}(s_0, u) \notin \overline{\mathcal{B}(\mathcal{K}, 3\delta) - \mathcal{B}(\mathcal{K}, 2\delta)}$. Because $\eta^{d,\delta}(t_0, u) \in \mathcal{B}(\mathcal{K}, 2\delta)$, we could use [prop. A.6](#) to find $s_3 \in [s_0, t_2]$ such that $\eta^{d,\delta}(s_3, u) \in \partial \mathcal{B}(\mathcal{K}, 3\delta)$. By the definition of t_3 , we would have $s_3 = t_3$, which would therefore imply that $s_0 = t_3$. This would contradict $\eta^{d,\delta}(s_0, u) \notin \overline{\mathcal{B}(\mathcal{K}, 3\delta) - \mathcal{B}(\mathcal{K}, 2\delta)}$ and $\eta^{d,\delta}(t_3, u) \in \partial \mathcal{B}(\mathcal{K}, 3\delta)$. Similarly, if $\eta^{d,\delta}(s_0, u) \in \mathcal{B}(\mathcal{K}, 2\delta)$, then we could use [prop. A.6](#) to find $s_2 \in [t_3, s_0]$ such that $\eta^{d,\delta}(s_2, u) \in \partial \mathcal{B}(\mathcal{K}, 2\delta)$. By the definition of t_2 , we

would have $s_2 = t_2$, which would therefore imply that $s_0 = t_2$; this would contradict the sets in which $\eta^{d,\delta}(s_0, u)$ and $\eta^{d,\delta}(t_2, u)$ lie. Thus, $\eta^{d,\delta}([t_3, t_2], u) \subset \overline{\mathcal{B}(\mathcal{K}, 3\delta) - \mathcal{B}(\mathcal{K}, 2\delta)}$.

Suppose now by contradiction that $\mathcal{J}(\eta^{d,\delta}(t_2, u)) > d - 2\varepsilon$. Then (η_1) , the definition of ε , and the fact that $u \in \mathcal{J}^{d+\varepsilon}$ imply that $\eta^{d,\delta}([t_3, t_2], u) \subset \mathcal{J}_{d-2\varepsilon}^{d+\varepsilon} \subset \mathcal{J}_{d-\varepsilon_0}^{d+\varepsilon_0}$. These two inclusions further imply that $\eta^{d,\delta}([t_3, t_2], u) \subset A_{d,\delta}$. Therefore, $\text{dist}(\eta^{d,\delta}([t_3, t_2], u), A_{d,\delta}) = 0$, i.e., $\psi^{d,\delta}(\eta^{d,\delta}([t_3, t_2], u)) = \{1\}$. The fundamental theorem of calculus, the definitions of $\eta^{d,\delta}$ and $\mathcal{W}_{d,\delta}$, (Υ_3) and [prop. 7.5](#) gives us

$$\begin{aligned} \mathcal{J}(\eta^{d,\delta}(t_2, u)) - \mathcal{J}(\eta^{d,\delta}(t_3, u)) &= \int_{t_3}^{t_2} \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(s, u)) ds = \int_{t_3}^{t_2} \mathcal{J}'(\eta^{d,\delta}(s, u)) \left(\frac{d\eta^{d,\delta}}{dt}(s, u) \right) ds \\ &= \int_{t_3}^{t_2} \mathcal{J}'(\eta^{d,\delta}(s, u)) (-\psi^{d,\delta}(\eta^{d,\delta}(s, u)) \Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(s, u))) ds = - \int_{t_3}^{t_2} \mathcal{J}'(\eta^{d,\delta}(s, u)) (\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(s, u))) ds \\ &< - \int_{t_3}^{t_2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(s, u)) ds \leq - \int_{t_3}^{t_2} \zeta^{d,\delta} ds = -(t_3 - t_2) \zeta^{d,\delta} \leq -\delta \zeta^{d,\delta}. \end{aligned}$$

Thus, we can use that $-\delta \zeta^{d,\delta} < -3\varepsilon$ (because $\varepsilon < \delta \zeta^{d,\delta} / 3$) to get

$$\mathcal{J}(\eta^{d,\delta}(t_2, u)) \leq \mathcal{J}(\eta^{d,\delta}(t_3, u)) - \delta \zeta^{d,\delta} \leq d + \varepsilon - \delta \zeta^{d,\delta} < d + \varepsilon - 3\varepsilon = d - 2\varepsilon,$$

which contradicts $\mathcal{J}(\eta^{d,\delta}(t_2, u)) > d - 2\varepsilon$. Therefore, $\mathcal{J}(\eta^{d,\delta}(t_2, u)) < d - 2\varepsilon$.

Now we consider the case $C_0 = \emptyset$ and suppose by contradiction that $\lim_{t \rightarrow \infty} \mathcal{J}(\eta^{d,\delta}(t, u)) > d - 2\varepsilon > d - \varepsilon_0$. Since $u \in \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \subset \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 2\delta)$ and $\eta^{d,\delta}([0, \infty), u) \cap \mathcal{B}(\mathcal{K}, 2\delta) = \emptyset$, [prop. 7.7](#) implies that $w := \lim_{t \rightarrow \infty} \eta^{d,\delta}(t, u)$ exists. It is not hard to see that the existence of w implies that $K := \eta^{d,\delta}([0, \infty), u) \cup \{w\}$ is compact. By (ν_4) , there exists $u_0 \in K$ such that $h := \inf_{u \in K} \frac{1}{2} \nu^{d+2\varepsilon}(u) = \frac{1}{2} \nu^{d+2\varepsilon}(u_0)$. Because $C_0 = \emptyset$, we have $\text{dist}(u_0, \mathcal{K}) \geq \text{dist}(K, \mathcal{K}) \geq \delta$, which means that $u_0 \notin \mathcal{K}$. Therefore, by (ν_2) , $h = \frac{1}{2} \nu^{d+2\varepsilon}(u_0) > 0$.

By the continuity of $\mathcal{J}$, there exists $\bar{\delta}_2 \in (0, \infty)$ such that $|\mathcal{J}(v)| > d - 2\varepsilon > d - \varepsilon_0$ for all $v \in \mathcal{B}(w, \bar{\delta})$. Then the definition of w gives us $T_0 \in (0, \infty)$ for which $\eta^{d,\delta}(t, u) \in \mathcal{B}(w, \bar{\delta}_2)$ whenever $t \in [T_0, \infty)$.

Let $s \in [T_0, \infty)$ arbitrary. Then, by the definitions of $\eta^{d,\delta}$ and $\mathcal{W}_{d,\delta}$, the fact that $\eta^{d,\delta}([0, \infty), u) \cup \{w\} \subset A_{d,\delta}$ (recall that $\mathcal{J}(w) > d - 2\varepsilon > d - \varepsilon_0$), and the fundamental theorem of calculus, we have

$$\begin{aligned} (\mathcal{J}(\eta^{d,\delta}(T_0, u)) - \mathcal{J}(\eta^{d,\delta}(s, u))) &= \int_s^{T_0} \frac{d}{dt} \mathcal{J}(\eta^{d,\delta}(t, u)) dt = \int_s^{T_0} \mathcal{J}'(\eta^{d,\delta}(t, u)) \left(\frac{d\eta^{d,\delta}}{dt}(t, u) \right) dt \\ &= \int_s^{T_0} \mathcal{J}'(\eta^{d,\delta}(t, u)) (\mathcal{W}_{d,\delta}(\eta^{d,\delta}(t, u))) dt = \int_s^{T_0} \mathcal{J}'(\eta^{d,\delta}(t, u)) (-\psi^{d,\delta}(\eta^{d,\delta}(t, u)) \Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u))) dt \\ &= \int_{T_0}^s \psi^{d,\delta}(\eta^{d,\delta}(t, u)) \mathcal{J}'(\eta^{d,\delta}(t, u)) (\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u))) dt \\ &= \int_{T_0}^s \mathcal{J}'(\eta^{d,\delta}(t, u)) (\Upsilon^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u))) dt = \int_{T_0}^s \frac{1}{2} \nu^{d+2\varepsilon_0}(\eta^{d,\delta}(t, u)) dt \geq h \int_{T_0}^s 1 dt = h(s - T_0), \end{aligned}$$

from which, aided by the continuity of $\mathcal{J}$ and the fact that $h > 0$, we get

$$\infty = \lim_{s \rightarrow \infty} h(s - T_0) \leq \lim_{s \rightarrow \infty} (\mathcal{J}(\eta^{d,\delta}(T_0, u)) - \mathcal{J}(\eta^{d,\delta}(s, u))) = \mathcal{J}(\eta^{d,\delta}(T_0, u)) - \mathcal{J}(w).$$

From this contradiction, we conclude that $\lim_{t \rightarrow \infty} \mathcal{J}(\eta^{d,\delta}(t, u)) \leq d - 2\varepsilon$, i.e., there exists some $t_{u,\varepsilon} \in [0, \infty)$ such that $\mathcal{J}(\eta^{d,\delta}(t_{u,\varepsilon}, u)) \leq d - 2\varepsilon$.

Now note that the set $\mathcal{J}^{d-\frac{3}{2}\varepsilon} \cap \eta^{d,\delta}([0, t_{u,\varepsilon}], u)$ is compact, and we have just showed that it is also non-empty. Thus, the function

$$s : \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \rightarrow [0, \infty); \quad s(u) := \min \left\{ s \in [0, \infty) \mid \mathcal{J}(\eta^{d,\delta}(s, u)) \leq d - \frac{3}{2}\varepsilon \right\}$$

is well-defined, and for each u in its domain, $\mathcal{J}(\eta^{d,\delta}(s(u), u)) \leq d - \frac{3}{2}\varepsilon \leq d - \varepsilon$. It remains to show the continuity of s ; this will be done by proving that it is both upper and lower semicontinuous.

We start by the lower semicontinuity. Let $\bar{\delta} \in (0, \infty)$ be arbitrary. By the definition of $\mathfrak{s}$, we have $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) - \delta, u)) > d - \frac{3}{2}\varepsilon$. As $\mathcal{J}$ and $\eta^{d,\delta}(t, \cdot)$ are continuous, there exists $r \in (0, \infty)$ such that $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) - \delta, v)) > d - \frac{3}{2}\varepsilon$ for any $v \in \mathcal{B}(u, r)$. Thus, for any such v , we have $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) - \delta, v)) > d - \frac{3}{2}\varepsilon \geq \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(v), v))$. By (η_1) , it holds that $\mathfrak{s}(v) \geq \mathfrak{s}(u) - \delta$ for all $v \in \mathcal{B}(u, r)$. Therefore,

$$\begin{aligned} \liminf_{v \rightarrow u} \mathfrak{s}(v) &= \lim_{\varepsilon \rightarrow 0} \left(\inf \{ \mathfrak{s}(v) \in [0, \infty) \mid v \in \mathcal{B}(u, \varepsilon) \cap (\mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, \delta)) \} \right) \\ &\geq \lim_{\varepsilon \rightarrow 0} \left(\inf \{ \mathfrak{s}(v) \in [0, \infty) \mid v \in \mathcal{B}(u, \bar{r}) \cap (\mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, \delta)) \} \right) \geq \mathfrak{s}(u) - \bar{\delta}. \end{aligned}$$

Since $\bar{\delta}$ is arbitrary, we get $\liminf_{v \rightarrow u} \mathfrak{s}(v) \geq \mathfrak{s}(u)$; this is the lower semicontinuity of $\mathfrak{s}$ at u .

Now suppose by contradiction that $\mathfrak{s}$ is not upper semicontinuous at u , i.e., $\limsup_{v \rightarrow u} \mathfrak{s}(v) > \mathfrak{s}(u)$. Since the inequality is strict, we can find $\bar{\delta} \in (0, \infty)$ such that $\limsup_{v \rightarrow u} \mathfrak{s}(v) > \mathfrak{s}(u) + \delta$. Let $U := \{(v_n) \subset \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \mid v_n \rightarrow u \text{ in } \mathbb{B}\}$. By properties of them limit superior of real functions, we have $\sup_{(v_n) \subset U} \limsup \mathfrak{s}(v_n) = \limsup_{v \rightarrow u} \mathfrak{s}(v) > \mathfrak{s}(u) + \delta$. Therefore, we can find a sequence $(u_n) \in U$ such that $\limsup \mathfrak{s}(u_n) > \mathfrak{s}(u) + \delta$. By properties of the limit superior of sequences, we can find a subsequence such that $\mathfrak{s}(u_n) > \mathfrak{s}(u) + \delta$. Furthermore, the definition of $\mathfrak{s}$ gives us that $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) \in [0, d - \frac{3}{2}\varepsilon]$. Since $[0, d - \frac{3}{2}\varepsilon]$ is bounded, there exists $d' \in [0, d - \frac{3}{2}\varepsilon]$ such that $\limsup \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) = d'$.

Take $v \in \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta)$ such that $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(v), v)) \neq d - \frac{3}{2}\varepsilon$, and note that the definition of $\mathfrak{s}$ implies $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(v), v)) \leq d - \frac{3}{2}\varepsilon$. As an intermediate step, we now show that $\mathfrak{s}(v) = 0$. Suppose by contradiction that $\mathfrak{s}(v) \neq 0$; this allows us to find some $s \in [0, \mathfrak{s}(v))$. By (η_1) , we have $\mathcal{J}(\eta^{d,\delta}(s, v)) \geq \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(v), v))$. The continuity of $\mathcal{J}$ and $\eta^{d,\delta}(\cdot, v)$ then permit us to use the intermediate value theorem to find $s' \in [s, \mathfrak{s}(v))$ for which $\mathcal{J}(\eta^{d,\delta}(s', v)) \leq d - \frac{3}{2}\varepsilon$. But since $s' < \mathfrak{s}(v)$, this violates the definition of $\mathfrak{s}$ and, therefore, the assumption that $\mathfrak{s}(v) \neq 0$ was wrong.

Now $d' = \limsup \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) < d - \frac{3}{2}\varepsilon$ implies that $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) \geq d - \frac{3}{2}\varepsilon$ for only finitely many $n \in \mathbb{N}$. This and the preceding paragraph then allows us to conclude that there exists $n_0 \in \mathbb{N}$ such that $\mathfrak{s}(u_n) = 0$ for all $n \in [n_0, \infty) \cap \mathbb{N}$. Thus, for all n sufficiently large, we have $\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) = \mathcal{J}(\eta^{d,\delta}(0, u_n))$. Taking the limit and using the continuity of the composition of $\mathcal{J}$ with $\eta^{d,\delta}(0, \cdot)$ (see [prop. 7.6](#)) gives us

$$\mathcal{J}(\eta^{d,\delta}(0, u)) = \lim \mathcal{J}(\eta^{d,\delta}(0, u_n)) = \lim \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) = d' < d - \frac{3}{2}\varepsilon.$$

However, the definition of $\mathfrak{s}$ forces $\mathfrak{s}(u) = 0$, which would cause the contradiction

$$0 = \lim \mathfrak{s}(u_n) = \limsup \mathfrak{s}(u_n) > \mathfrak{s}(u) + \delta = 0 + \delta = \delta,$$

where we emphasize the fact that no further subsequence was taken, as the fact that the limit superior of a subsequence can be smaller than that of the original sequence would ruin the argument.

We may now consider $d' = d - \frac{3}{2}\varepsilon$. By the continuity of $\mathcal{J}$ and of $\eta^{d,\delta}(\mathfrak{s}(u) + \delta, \cdot)$ and by (η_1) , we have

$$\mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) + \delta, u)) = \limsup \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) + \delta, u_n)) \geq \limsup \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u_n), u_n)) = d' = d - \frac{3}{2}\varepsilon.$$

On one hand, if $u \in \text{int}(B_{d,\delta})$, then (η_1) allows us to arrive at the contradiction

$$d - \frac{3}{2}\varepsilon \leq \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u) + \delta, u)) < \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u), u)) \leq d - \frac{3}{2}\varepsilon.$$

On the other hand, if $u \notin \text{int}(B_{d,\delta})$, then $u \in \mathcal{J}^{d-2\varepsilon_0}$. Thus, (η_1) implies that $\mathcal{J}(\eta^{d,\delta}(t, u)) = c_0 \leq d - 2\varepsilon_0 < d - \varepsilon$ for all $t \in [0, \infty)$, i.e., it is constant. The definition of $\mathfrak{s}$ then implies $\mathfrak{s}(u) = 0$. Now by continuity, there exists $r \in (0, \infty)$ such that $\mathcal{J}(\eta^{d,\delta}(t, v)) < c_0 + \varepsilon_0 \leq d - \varepsilon_0 < d - \varepsilon$, for all $v \in \mathcal{B}(u, r)$ and $t \in (\mathfrak{s}(u) - r, \mathfrak{s}(u) + r) = (-r, r)$. Since $u_n \rightarrow u$ in $\mathbb{B}$, there is some $n_0 \in \mathbb{N}$ for which $u_n \in \mathcal{B}(u, r)$ whenever $n \in [n_0, \infty) \cap \mathbb{N}$. Letting $n \in [n_0, \infty) \cap \mathbb{N}$, we have $\mathcal{J}(\eta^{d,\delta}(0, u_n)) = \mathcal{J}(\eta^{d,\delta}(\mathfrak{s}(u), u_n)) < d - \varepsilon$. This allows us to conclude that $\mathfrak{s}(u_n) = 0$, i.e., $\lim \mathfrak{s}(u_n) = 0 = \mathfrak{s}(u)$. $\square$

Corollary 7.9. *Suppose that $\tilde{\mathcal{K}}$ is finite and let $d \in (0, \infty)$, $\delta \in (0, \frac{\kappa}{3})$. Then there exists $\varepsilon \in (0, \varepsilon_0)$ such that for all $K \subset \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta)$ compact we can find $t_K \in (0, \infty)$ for which $\eta^{d,\delta}(t_K, K) \subset \mathcal{J}^{d-\varepsilon}$.*

Let $\Sigma := \{A \subset \mathbb{B} \mid A = -A, \text{ and } A \text{ is compact}\},$

$\mathcal{H} := \{h: \mathbb{B} \rightarrow \mathbb{B} \mid \mathcal{J}(h(u)) \leq \mathcal{J}(u) \ \forall u \in \mathbb{B}, \text{ and } h \text{ is an odd homeomorphism onto } h(\mathbb{B})\},$
and denote the Krasnoselskii's genus of some set $A \in \Sigma$ by $\text{gen}(A)$. For $\mathbf{r}$ as in (MP₁), we define the following pseudoindex:

$$i^*: \Sigma \rightarrow \mathbb{N} \cup \{0\}; \quad i^*(A) := \min_{h \in \mathcal{H}} \text{gen}(h(A) \cap \partial \mathcal{B}(0, \mathbf{r})).$$

Some properties of i^* are in order:

Proposition 7.10. *Let $A, B \in \Sigma$, $h \in \mathcal{H}$ and take any $k \in \mathbb{N}$. Then the following holds:*

(i₁^{*}) *If $A \subset B$, then $i^*(A) \leq i^*(B)$;*

(i₂^{*}) *If $C \subset \mathbb{B}$ satisfies $\overline{C} = -\overline{C}$ and $A \cup \overline{C} \in \Sigma$, then $i^*(A \cup \overline{C}) \leq i^*(A) + \text{gen}(\overline{C})$;*

(i₃^{*}) *$h(A) \in \Sigma$ and $i^*(A) \leq i^*(h(A))$;*

(i₄^{*}) *There exists $A_k \subset \mathbb{B}$ such that $i^*(A_k) \geq k$.*

Proof. Let $\phi \in \mathcal{H}$ be arbitrary.

(i₁^{*}) Because $A \subset B$, item 2 of proposition 5.4 of [24] gives $\text{gen}(\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r})) \leq \text{gen}(\phi(B) \cap \partial \mathcal{B}(0, \mathbf{r}))$. Since $\phi \in \mathcal{H}$ is arbitrary, taking the minimum of both sides when $\phi \in \mathcal{H}$ gives

$$i^*(A) = \min_{\phi \in \mathcal{H}} \text{gen}(\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r})) \leq \min_{\phi \in \mathcal{H}} \text{gen}(\phi(B) \cap \partial \mathcal{B}(0, \mathbf{r})) = i^*(B).$$

(i₂^{*}) By items 2 and 4 of proposition 5.4 of [24], we have

$$\begin{aligned} \text{gen}(\phi(A \cup \overline{C}) \cap \partial \mathcal{B}(0, \mathbf{r})) &= \text{gen}((\phi(A) \cup \phi(\overline{C})) \cap \partial \mathcal{B}(0, \mathbf{r})) \\ &= \text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r})) \cup (\phi(\overline{C}) \cap \partial \mathcal{B}(0, \mathbf{r}))) \leq \text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r}))) + \text{gen}((\phi(\overline{C}) \cap \partial \mathcal{B}(0, \mathbf{r}))) \\ &\leq \text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r}))) + \text{gen}(\phi(\overline{C})) \leq \text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r}))) + \text{gen}(\overline{C}). \end{aligned}$$

Since $\phi \in \mathcal{H}$ is arbitrary, taking the minimum of both sides when $\phi \in \mathcal{H}$ gives

$$\begin{aligned} i^*(A \cup \overline{C}) &= \min_{\phi \in \mathcal{H}} \text{gen}(\phi(A \cup \overline{C}) \cap \partial \mathcal{B}(0, \mathbf{r})) \leq \min_{\phi \in \mathcal{H}} (\text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r}))) + \text{gen}(\overline{C})) \\ &= \text{gen}(\overline{C}) + \min_{\phi \in \mathcal{H}} \text{gen}((\phi(A) \cap \partial \mathcal{B}(0, \mathbf{r}))) = i^*(A) + \text{gen}(\overline{C}). \end{aligned}$$

(i₃^{*}) It is easy to see that $h(A) \in \Sigma$ and so $\phi \circ h \in \mathcal{H}$. Therefore, $S := \{\phi \circ h \mid \phi \in \mathcal{H}\} \subset \mathcal{H}$, and then $i^*(A) = \min_{\varphi \in \mathcal{H}} \text{gen}(\varphi(A) \cap \partial \mathcal{B}(0, \mathbf{r})) \leq \min_{\tilde{h} \in S} \text{gen}(\tilde{h}(A) \cap \partial \mathcal{B}(0, \mathbf{r})) = \min_{\varphi \in \mathcal{H}} \text{gen}(\varphi(h(A)) \cap \partial \mathcal{B}(0, \mathbf{r})) = i^*(h(A)).$

(i₄^{*}) Let $\mathbb{Y} \subset \mathbb{B}$ a k -dimensional linear subspace. Condition (J₄) gives us the existence of some $R = R(\mathbb{Y}) \in (0, \infty)$ such that $\mathcal{J}(u) < \mathcal{J}(0)$ for all $u \in \mathcal{B}(0, R)^c \cap \mathbb{Y}$.

Now consider $A := \overline{\mathcal{B}(0, R)} \cap \mathbb{Y}$; it is easy to see that $A \in \Sigma$. Suppose by contradiction that $i^*(A) < k$. By the definition of i^* , there exists $\tilde{h} \in \mathcal{H}$ such that $\text{gen}(\tilde{h}(A) \cap \partial \mathcal{B}(0, \mathbf{r})) < k$. The definition of the Krasnoselskii's genus then implies the existence of a continuous, odd map $g: \tilde{h}(A) \cap \partial \mathcal{B}(0, \mathbf{r}) \rightarrow \mathbb{R}^{k-1} - \{0\}$. Let $B := \tilde{h}^{-1}(\mathcal{B}(0, \mathbf{r})) \cap \mathbb{Y}$ and $u \in B$. As $\tilde{h} \in \mathcal{H}$ and

$$\tilde{h}(u) \in \tilde{h}(B) = \tilde{h}(\tilde{h}^{-1}(\mathcal{B}(0, \mathbf{r})) \cap \mathbb{Y}) \subset \tilde{h}(\tilde{h}^{-1}(\mathcal{B}(0, \mathbf{r}))) \cap \tilde{h}(\mathbb{Y}) \subset \mathcal{B}(0, \mathbf{r}) \cap \tilde{h}(\mathbb{Y}) \subset \mathcal{B}(0, \mathbf{r}),$$

(MP₁) allows us to write $\mathcal{J}(0) \leq \alpha \leq \mathcal{J}(\tilde{h}(u)) \leq \mathcal{J}(u)$. Since $u \in \mathbb{Y}$ also, our choice of R then precludes $u \in \mathcal{B}(0, R)^c$, i.e., $u \in \mathbb{Y} \cap \overline{\mathcal{B}(0, R)} = A$. This means that $B \subset \mathbb{Y} \cap \overline{\mathcal{B}(0, R)}$ and, therefore, is a bounded set in $\mathbb{B}$. Moreover, since B is open in $\mathbb{Y}$ and $0 \in B$, we conclude that B is a neighborhood of 0 in $\mathbb{Y}$. Finally, we have $B = -B$ because $\tilde{h}$ is odd.

Take $u \in \partial B$ and suppose by contradiction that $\tilde{h}(u) \in \mathcal{B}(0, R)$. Then there exists $U \subset \mathcal{B}(0, R)$ an open neighborhood of $\tilde{h}(u)$ in $\mathbb{B}$. Since $\tilde{h}$ is a homeomorphism, $\tilde{h}^{-1}(U) \cap \mathbb{Y}$ is an open neighborhood of u in B ; this contradicts the fact that $u \in \partial B$. Thus, $\tilde{h}(u) \in \partial \mathcal{B}(0, R)$ whenever $u \in \partial B$, which means that the composition $g \circ (\tilde{h}|_{\partial B})$ is well-defined. It also is a continuous, odd mapping between the

boundary of a bounded, symmetric open neighborhood of 0 (the set B) in a k -dimensional vector space over R (the subspace $\mathbb{Y}$) and $\mathbb{R}^{k-1} - \{0\}$. This contradicts the Borsuk-Ulam theorem, from which we conclude that $i^*(A) \geq k$. $\square$

Before we finally prove [theorem 1.4](#), one more result is needed; it establishes an important qualitatively property of the intersection of $\mathcal{J}_{d-\varepsilon}^{d+\varepsilon}$ and $\mathcal{B}(\mathcal{K}, 3\delta)$.

Proposition 7.11. *Suppose $\tilde{\mathcal{K}}$ is finite and let $d \in (0, \infty)$. If d is a critical value of $\mathcal{J}$, then there exists $\delta \in (0, \infty)$ such that for any $\varepsilon \in (0, \varepsilon_0)$, we have for all $\delta' \in (0, \delta)$, that $\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta') = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}_d, 3\delta)$. If d is not a critical value of $\mathcal{J}$, then there exists $\delta, \varepsilon \in (0, \infty)$ for which $\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta) = \emptyset$.*

Proof. Suppose d is a critical value of $\mathcal{J}$, but the first statement of the proposition is false. Then for every $N \in \mathbb{N}$, we can find $\varepsilon_n \in (0, \varepsilon_0)$ and $\delta'_n \in (0, \frac{1}{n})$ such that $\mathcal{J}_{d-\varepsilon_n}^{d+\varepsilon_n} \cap \mathcal{B}(\mathcal{K}, 3\delta'_n) \not\subset \mathcal{J}_{d-\varepsilon_n}^{d+\varepsilon_n} \cap \mathcal{B}(\mathcal{K}_d, 3\delta'_n)$. Thus, there is some $(u_n) \subset \mathbb{B}$ for which $u_n \in \mathcal{J}_{d-\varepsilon_n}^{d+\varepsilon_n} \cap \mathcal{B}(\mathcal{K}, 3\delta'_n)$ and $u_n \notin \mathcal{B}(\mathcal{K}_d, 3\delta'_n)$ for all $n \in \mathbb{N}$. Therefore, there also exists $(v_n) \subset \mathcal{K}$ that always satisfies $\|u_n - v_n\|_{\mathbb{B}} \leq \delta'_n < \frac{1}{n}$. Let $d_n := \mathcal{J}(v_n)$. [Condition \(\$\mathcal{J}_2\$ \)](#) and the finiteness of $\tilde{\mathcal{K}}$ imply that there are only finitely many critical values. The pigeonhole principle then allows us to find $\bar{d} \in (0, \infty)$ and a subsequence such that $\mathcal{J}(v_n) = \bar{d}$ for every $n \in \mathbb{N}$. By continuity, we have $0 = \lim |\mathcal{J}(u_n) - \mathcal{J}(v_n)| = \lim |\mathcal{J}(u_n) - \bar{d}|$, implying that $d \in (d - \varepsilon_n, d + \varepsilon_n) \subset (d - 2\varepsilon_0, d + 2\varepsilon_0)$. Since there is at most one critical value in $(d - 2\varepsilon_0, d + 2\varepsilon_0)$ and we supposed that d is a critical value, we must conclude that $\bar{d} = d$ and that each v_n is a critical point at level d . However, this and $\|u_n - v_n\|_{\mathbb{B}} < \frac{1}{n}$ would imply that $u_n \in \mathcal{B}(\mathcal{K}_d, \frac{1}{n})$, contradicting our choice of (u_n) .

Now suppose the second part of the proposition does not hold whilst d is not a critical value of $\mathcal{J}$; this means we can find $(u_n) \subset \mathcal{J}_{d-\frac{1}{n}}^{d+\frac{1}{n}} \cap \mathcal{B}(\mathcal{K}, \frac{1}{n})$ and $(v_n) \subset \mathcal{K}$ with $\|u_n - v_n\|_{\mathbb{B}} \leq \frac{1}{n}$. Let $d_n := \mathcal{J}(v_n)$. A similar use of the pigeonhole principle allows us to find $\bar{d} \in (0, \infty)$ and a subsequence such that $\mathcal{J}(v_n) = \bar{d}$ for every $n \in \mathbb{N}$. By continuity, we have $0 = \lim |\mathcal{J}(u_n) - \mathcal{J}(v_n)| = \lim |\mathcal{J}(u_n) - \bar{d}| = |d - \bar{d}|$, implying that $\bar{d} = d$. Thus, each v_n was a critical point at level d , contradicting the supposition that d was not a critical value of $\mathcal{J}$. $\square$

Proof of [theorem 1.4](#). Suppose by contradiction that $\tilde{\mathcal{K}}$ is finite, let $k \in \mathbb{N}$ be arbitrary, and define

$$d_k := \inf \left\{ \sup_{u \in A} \mathcal{J}(u) \mid A \in \Sigma, \text{ and } i^*(A) \geq k \right\}.$$

For brevity, we will use $d := d_k$ when appropriate. By [\(\$i_4^*\$ \)](#), d_k exists and is a finite number. Suppose by contradiction that there exists $A \in \Sigma$ with $i^*(A) \geq k$ such that $\sup_{u \in A} \mathcal{J}(u) < \alpha$. Since $\text{gen}(\emptyset) = 0$ and $i^*(A) \geq k > 0$, the definition of i^* implies that $h(A) \cap \partial \mathcal{B}(0, \mathbf{r}) \neq \emptyset$ for all $h \in \mathcal{H}$. Taking $h = \text{Id}$, we conclude that $A \cap \partial \mathcal{B}(0, \mathbf{r}) \neq \emptyset$. Thus, there exists $v \in A$ that also satisfies $v \in \partial \mathcal{B}(0, \mathbf{r})$. This means that $\mathcal{J}(v) \geq \alpha$ by [\(MP₁\)](#), which is a contradiction; therefore, $i^*(A) < k$ whenever $\sup_{u \in A} \mathcal{J}(u) < \alpha$.

Suppose, once more by contradiction, that k is such that $\mathcal{K}_{d_k} = \emptyset$. Since d is not a critical value of $\mathcal{J}$, we can use the second part of [prop. 7.11](#) to find $\delta, \varepsilon_1 \in (0, \infty)$ for which $\mathcal{J}_{d-\varepsilon_1}^{d+\varepsilon_1} \cap \mathcal{B}(\mathcal{K}, 3\delta) = \emptyset$. For this δ and d , let $\varepsilon_2 \in (0, \varepsilon_0)$ be as in [corollary 7.9](#) and then choose $\varepsilon := \min \frac{1}{2} \{\varepsilon_1, \varepsilon_2\}$. The definition of d allows us to find $A \in \Sigma$ with $i^*(A) \geq k$ and also satisfying $d \leq \sup_{u \in A} \mathcal{J}(u) < d + \varepsilon < d + \varepsilon_2$. Since $\varepsilon < \varepsilon_1$ implies that $\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta) = \emptyset$, we have

$$\begin{aligned} \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) &= \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta)^c = (\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta)^c) \cup \emptyset \\ &= (\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta)^c) \cup (\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta)) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap (\mathcal{B}(\mathcal{K}, 3\delta)^c \cup \mathcal{B}(\mathcal{K}, 3\delta)) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathbb{B} = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon}. \end{aligned}$$

[Corollary 7.9](#) allows us to find t_A such that $\eta^{d, \delta}(t_A, A) \subset \mathcal{J}^{d-\varepsilon_2}$ because

$$A \subset \mathcal{J}_d^{d+\varepsilon} \subset \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \subset \mathcal{J}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \subset \mathcal{J}^{d+\varepsilon_2} - \mathcal{B}(\mathcal{K}, 3\delta).$$

Moreover, by (η_1) , (η_3) , and (η_4) , we have $\eta^{d,\delta}(t_A, \cdot) \upharpoonright_A \in \mathcal{H}$. Then (i_3^*) gives us $k \leq i^*(A) \leq i^*(\eta^{d,\delta}(t_A, A))$. Furthermore, (η_3) , (η_4) , and $A \in \Sigma$ imply that $\eta^{d,\delta}(t_A, A) \in \Sigma$; this leads us to the following contradiction:

$$d = \inf \left\{ \sup_{u \in A} \mathcal{J}(u) \mid A \in \Sigma, \text{ and } i^*(A) \geq k \right\} \leq \sup_{u \in \eta^{d,\delta}(t_A, A)} \mathcal{J}(u) < d - \varepsilon_2.$$

So $\mathcal{K}_{d_k} \neq \emptyset$, i.e., d_k is a critical. It is easy to see that $d_k \leq d_{k+1}$, so it suffices now to show that $d_k \neq d_{k+1}$. We will proceed by contradiction, i.e., suppose $d_k = d_{k+1} =: d$. Note that $\mathcal{K}_d$ must be at most a countable set because $\tilde{\mathcal{K}}$ is finite, and because the elements of $\mathcal{G}$ form a countable set. We must then have $\text{gen}(\mathcal{K}_d) = 1$ due to the facts that the two-dimensional sphere is non-enumerable and that $\mathcal{K}_d \neq \emptyset$. According to theorem 9.11 of [12], there exists $\delta_1 \in (0, \infty)$ such that $\text{gen}(\overline{\mathcal{B}(\mathcal{K}_d, 3\delta_1)}) = \text{gen}(\mathcal{K}_d) = 1$.

Let $\delta_2 \in (0, \infty)$ be as in the first part of [prop. 7.11](#), $\delta := \frac{1}{2} \min\{\delta_1, \delta_2, \frac{\kappa}{3}\}$, and for this choice of δ , take $\varepsilon \in (0, \varepsilon_0)$ as in [corollary 7.9](#). The definition of d_{k+1} allows us to find $A \in \Sigma$ with $i^*(A) \geq k+1$ such that $d \leq \sup_{u \in A} \mathcal{J}(u) < d + \varepsilon$. Because $\varepsilon \in (0, \varepsilon_0)$ and $\delta \in (0, \delta_2)$, we have

$$\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - (\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}, 3\delta)) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - (\mathcal{J}_{d-\varepsilon}^{d+\varepsilon} \cap \mathcal{B}(\mathcal{K}_d, 3\delta)) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}_d, 3\delta)$$

by [prop. 7.11](#). Since the difference of a compact set and an open set is compact, the set

$$K := A - \mathcal{B}(\mathcal{K}_d, 3\delta) \subset \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}_d, 3\delta) \subset \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}_d, 3\delta) = \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta) \subset \mathcal{J}_{d-\varepsilon}^{d+\varepsilon} - \mathcal{B}(\mathcal{K}, 3\delta)$$

is compact. By [corollary 7.9](#), there exists $t_K \in (0, \infty)$ such that $\eta^{d,\delta}(t_K, K) \subset \mathcal{J}^{d-\varepsilon}$. Therefore,

$$k+1 \leq i^*(A) \leq i^*\left((A - \mathcal{B}(\mathcal{K}_d, 3\delta)) \cup \overline{\mathcal{B}(\mathcal{K}_d, 3\delta)}\right) \leq i^*(A - \mathcal{B}(\mathcal{K}_d, 3\delta)) + \text{gen}(\overline{\mathcal{B}(\mathcal{K}_d, 3\delta)})$$

$$= i^*(K) + \text{gen}(\overline{\mathcal{B}(\mathcal{K}_d, 3\delta)}) \leq i^*(\eta^{d,\delta}(t_K, K)) + \text{gen}(\overline{\mathcal{B}(\mathcal{K}_d, 3\delta)})$$

$$= i^*(\eta^{d,\delta}(t_K, K)) + \text{gen}(\mathcal{K}_d) = i^*(\eta^{d,\delta}(t_K, K)) + 1,$$

where we used (i_1^*) , (i_2^*) , and (i_3^*) with $h = \eta^{d,\delta}(t_K, \cdot) \in \mathcal{H}$; this choice of h is possible because of (η_1) , (η_3) , and (η_4) . This means that $i^*(\eta^{d,\delta}(t_K, K)) \geq k$ by definition. Once more, as $K \in \Sigma$, (η_3) , and (η_4) imply that $\eta^{d,\delta}(t_K, K) \in \Sigma$, which yields the following contradiction:

$$d = d_k = \inf \left\{ \sup_{u \in A} \mathcal{J}(u) \mid A \in \Sigma, \text{ and } i^*(A) \geq k \right\} \leq \sup_{u \in \eta^{d,\delta}(t_K, K)} \mathcal{J}(u) < d - \varepsilon. \quad \square$$

Now the proof of [corollary 1.5](#) is reduced to finding a group $\mathcal{G}$ and a group action h , and showing that $\mathbb{X}$, $\mathcal{I}$ (when $\beta=0$, which we will assume from now on), $\mathcal{G}$, and h satisfy the hypotheses of [theorem 1.4](#).

The following facts can either be found in the literature, have already been shown in this work, or are easy to be deduced: $\mathbb{X}$ is a separable, reflexive Banach space over $\mathbb{R}$, $\mathcal{I} \in \mathcal{C}^1(\mathbb{X}, \mathbb{R})$, $(\mathbb{Z}^N, +)$ is a group whose elements form a countable set, and

$$h: \mathbb{Z}^N \times \mathbb{X} \rightarrow \mathbb{X}; \quad h(z, u) := u \circ \tau_z \quad (7.10)$$

is a group action. Moreover, $(\mathcal{J}_1)$, $(\mathcal{J}_2)$, $(\mathcal{E}_1)$ and $(\mathcal{E}_2)$ can also be included in the previous list. Finally, $(\mathcal{J}_3)$ has been proved in [prop. 5.9](#), while (MP_1) has been shown in [lemma 3.4](#). It thus only remains to show $(\mathcal{J}_4)$, $(\mathcal{E}_3)$, and $(\mathcal{E}_4)$.

Proposition 7.12. *The functional $\mathcal{I}$ satisfies $(\mathcal{J}_4)$.*

Proof. Let $\mathbb{Y} \subset \mathbb{X}$ be a finite-dimensional linear subspace. Since its dimension is finite, we have that $\|\cdot\|_{\mathbb{X}}, \|\cdot\|_{\theta}$ and $\|\cdot\|_{p_s^*}$ are all equivalent there, i.e., there are $c_{\theta}(\mathbb{Y}), c_{p_s^*}(\mathbb{Y}) \in (0, \infty)$ such that

$$c_{\theta}(\mathbb{Y})\|u\|_{\mathbb{X}} \leq \|u\|_{\theta} \leq C_{\theta}\|u\|_{\mathbb{X}} \quad \text{and} \quad c_{p_s^*}(\mathbb{Y})\|u\|_{\mathbb{X}} \leq \|u\|_{p_s^*} \leq C_{p_s^*}\|u\|_{\mathbb{X}} \quad (7.11)$$

for all $u \in \mathbb{Y}$, where $C_{\theta}, C_{p_s^*} \in (0, \infty)$ are the embedding constants of $\mathbb{X} \hookrightarrow L^{\theta}(\mathbb{R}^N), L^{p_s^*}(\mathbb{R}^N)$.

Now $p < \theta < p_s^*$ and [prop. 2.5](#) imply that the function

$$f: (0, \infty) \rightarrow \mathbb{R}; \quad f(x) = \frac{x^p}{p} + \frac{C_{\theta}(\mathbb{Y})^{\theta} x^{\theta}}{\theta^2} + C_2 x^{\theta} - \ln(x) c_{\theta}(\mathbb{Y})^{\theta} x^{\theta} - \beta c_{p_s^*}(\mathbb{Y})^{p_s^*} \frac{x^{p_s^*}}{p_s^*}$$

satisfies $\lim_{x \rightarrow \infty} f(x) = -\infty$. Thus, there exists $R = R(c_\theta(\mathbb{Y}), C_\theta, c_{p_s^*}(\mathbb{Y})) \in (0, \infty)$ such that $f(x) < \mathcal{I}(0)$ for all $x \in (R, \infty)$. For every $u \in \mathcal{B}(0, R)^c \cap \mathbb{Y}$, [inequations \(7.11\)](#) and [prop. 2.5](#) imply that

$$\begin{aligned} \mathcal{I}(u) &= \frac{\|u\|_{\mathbb{X}}^p}{p} + \frac{\|u\|_{\theta}^\theta}{\theta^2} - \frac{1}{\theta} \mathcal{L}(u) - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} \leq \frac{\|u\|_{\mathbb{X}}^p}{p} + \frac{\|u\|_{\theta}^\theta}{\theta^2} + C_2 \|u\|_{\mathbb{X}}^\theta - \ln(\|u\|_{\mathbb{X}}) \|u\|_{\theta}^\theta - \beta \frac{\|u\|_{p_s^*}^{p_s^*}}{p_s^*} \\ &\leq \frac{\|u\|_{\mathbb{X}}^p}{p} + \frac{C_\theta^\theta \|u\|_{\mathbb{X}}^\theta}{\theta^2} + C_2 \|u\|_{\mathbb{X}}^\theta - \ln(\|u\|_{\mathbb{X}}) c_\theta^\theta \|u\|_{\mathbb{X}}^\theta - \beta c_{p_s^*}^{p_s^*} \frac{\|u\|_{\mathbb{X}}^{p_s^*}}{p_s^*} = f(\|u\|_{\mathbb{X}}) < \mathcal{I}(0) = 0. \end{aligned} \quad \square$$

Proposition 7.13. *The functional $\mathcal{I}$, the group $(\mathbb{Z}^N, +)$, and the group action h satisfy [\(\mathcal{G}_3\)](#).*

Proof. Let $u \in \mathbb{X} - \{0\}$ and $v \in \mathbb{X}$. By the properties of the infimum, there exists $(z_n) \in \mathbb{Z}^N$ with $h(z_n, v) = v \circ \tau_{z_n} \neq u$ such that

$$\|u - v \circ \tau_{z_n}\|_{\mathbb{X}} = \|u - h(z_n, v)\|_{\mathbb{X}} \rightarrow \inf_{z \in \mathbb{Z}^N, h(z, v) \neq u} \|u - h(z, v)\|_{\mathbb{X}}.$$

There are now two possibilities: either $|z_n| \rightarrow \infty$ or not. If so, then it is not hard to show that $v \circ \tau_{z_k} \rightarrow 0$ in $\mathbb{X}$ and then that $u - v \circ \tau_{z_k} \rightarrow u$ in $\mathbb{X}$ also. Using [corollary 2.4](#), Fatou's Lemma, and the fact that $u \neq 0$, we can get, up to subsequence, that

$$\lim \|u - v \circ \tau_{z_k}\|_{\mathbb{X}} \geq \liminf \|u - v \circ \tau_{z_k}\|_{\mathbb{X}} \geq \|u - \liminf v \circ \tau_{z_k}\|_{\mathbb{X}} = \|u\|_{\mathbb{X}} > 0.$$

On the other hand, if $|z_n| \not\rightarrow \infty$, then it remains bounded. Since any bounded subset of $\mathbb{Z}^N$ is finite, we can find $z \in \mathbb{Z}^N$ and subindexes $(n_k) \subset \mathbb{N}$ such that $z_{n_k} = z$ always. Because $u \neq v \circ \tau_{n_{n_k}} = v \circ \tau_z$, we conclude that $\lim \|u - v \circ \tau_{z_{n_k}}\|_{\mathbb{X}} = \lim \|u - v \circ \tau_z\|_{\mathbb{X}} > 0$. The fact that the limit of any subsequence of a converging sequence coincides with the limit of the original sequence completes the proof. $\square$

Before we prove that [\(\mathcal{G}_4\)](#) also holds, we need a certain inequality involving $\langle \cdot, \cdot \rangle_{\mathbb{X}}$.

Proposition 7.14. *There exists $C \in (0, \infty)$ such that for all $u, v \in \mathbb{X}$, we have*

$$\|u - v\|_{\mathbb{X}}^p \leq C(\langle u, u - v \rangle_{\mathbb{X}}^p - \langle v, u - v \rangle_{\mathbb{X}}^p).$$

Proof. Take arbitrary $u, v \in \mathbb{X}$, $x, y \in \mathbb{R}^N$. By [prop. A.5](#) with $a := u(x) - u(y)$ and $b := v(x) - v(y)$, we get

$$\begin{aligned} |u(x) - u(y) - (v(x) - v(y))|^p &\leq C \left(|u(x) - u(y)|^{p-2} (u(x) - u(y)) \right. \\ &\quad \left. - |v(x) - v(y)|^{p-2} (v(x) - v(y)) \right) (u(x) - u(y) - (v(x) - v(y))) \\ &= C \left(|u(x) - u(y)|^{p-2} (u(x) - u(y)) (u(x) - u(y) - (v(x) - v(y))) \right. \\ &\quad \left. - |v(x) - v(y)|^{p-2} (v(x) - v(y)) (u(x) - u(y) - (v(x) - v(y))) \right), \end{aligned}$$

where we draw the reader's attention to the fact that the constant C does not depend on u or v . Multiplying both sides by $\mathcal{K}(x, y)$ and integrating on $\mathbb{R}^{2N}$ gives us $[u - v]^p \leq C([u, u - v]^p - [v, u - v]^p)$. A similar calculation shows that

$$\int_{\mathbb{R}^N} V(x) |u(x) - v(x)|^p dx \leq C \int_{\mathbb{R}^N} V(x) (|u(x)|^{p-2} u(x) - |v(x)|^{p-2} v(x)) (u(x) - v(x)) dx. \quad \square$$

Proposition 7.15. *The functional $\mathcal{I}$, the group $(\mathbb{Z}^N, +)$, and the group action h satisfy [\(\mathcal{G}_4\)](#).*

Proof. Let $(u_n), (v_n)$ be (PS) sequences for $\mathcal{I}$ such that $\|u_n - v_n\|_{\mathbb{X}} \not\rightarrow 0$. We will first show that

$$\limsup_{y \in \mathbb{R}^N} \|u_n - v_n\|_{L^p(\mathcal{B}(y, 1))} \neq 0. \quad (7.12)$$

Suppose by contradiction that it was not the case, and let $n \in \mathbb{N}$. Note that [prop. 7.14](#) gives us

$$\begin{aligned}
\|u_n - v_n\|_{\mathbb{X}}^p &\leq C(\langle u_n, u_n - v_n \rangle_{\mathbb{X}}^p - \langle v_n, u_n - v_n \rangle_{\mathbb{X}}^p) \\
&= C(\mathcal{I}'(u_n)(u_n - v_n) + \mathcal{I}'(v_n)(v_n - u_n) + \mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n)) \\
&\leq C(\|\mathcal{I}'(u_n)\|_{\mathbb{X}'} \|u_n - v_n\|_{\mathbb{X}} + \|\mathcal{I}'(v_n)\|_{\mathbb{X}'} \|v_n - u_n\|_{\mathbb{X}} + \mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n)) \\
&= C\left((\|\mathcal{I}'(u_n)\|_{\mathbb{X}'} + \|\mathcal{I}'(v_n)\|_{\mathbb{X}'})\|u_n - v_n\|_{\mathbb{X}} + \mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n)\right) \\
&\quad \left(\varepsilon_n \|u_n - v_n\|_{\mathbb{X}} + \mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n)\right), \quad (7.13)
\end{aligned}$$

where $\varepsilon_n := \|\mathcal{I}'(u_n)\|_{\mathbb{X}'} + \|\mathcal{I}'(v_n)\|_{\mathbb{X}'} \rightarrow 0$ because $(u_n), (v_n)$ are (PS) sequences. For the logarithmic term, we have

$$\begin{aligned}
\mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n) &= \int_{\mathbb{R}^N} |u_n(x)|^{\theta-2} u_n(x) (u_n(x) - v_n(x)) \ln(|u_n(x)|) dx \\
&\quad - \int_{\mathbb{R}^N} |v_n(x)|^{\theta-2} v_n(x) (u_n(x) - v_n(x)) \ln(|v_n(x)|) dx \\
&\leq \int_{\mathbb{R}^N} (|u_n(x)|^{\theta-1} |\ln(|u_n(x)|)| + |v_n(x)|^{\theta-1} |\ln(|v_n(x)|)|) |u_n(x) - v_n(x)| dx \\
&\leq C_2 \int_{\mathbb{R}^N} (|u_n(x)|^{\theta-1+\delta} + |u_n(x)|^{\theta-1-\delta} + |v_n(x)|^{\theta-1+\delta} + |v_n(x)|^{\theta-1-\delta}) |u_n(x) - v_n(x)| dx,
\end{aligned}$$

where $\delta := \frac{1}{2} \min\left\{\frac{p_s^* - \theta}{p_s^*}, \frac{(\theta - p)}{p}\right\}$, and we used [corollary A.2](#) with $a = \delta$ and $b = -\delta$. Then

$$\begin{aligned}
&\mathcal{L}'(u_n)(u_n - v_n) - \mathcal{L}'(v_n)(u_n - v_n) \\
&\leq C_2 \int_{\mathbb{R}^N} (|u_n(x)|^{\theta-1+\delta} + |u_n(x)|^{\theta-1-\delta} + |v_n(x)|^{\theta-1+\delta} + |v_n(x)|^{\theta-1-\delta}) |u_n(x) - v_n(x)| dx \\
&\leq C_2 \left(\left(\|u_n\|_{\theta}^{\frac{\theta-1+\delta}{\theta}} + \|v_n\|_{\theta}^{\frac{\theta-1+\delta}{\theta}} \right) \|u_n - v_n\|_{\frac{\theta}{1-\delta}}^{\frac{\theta}{1-\delta}} + \left(\|u_n\|_{\theta}^{\frac{\theta-1-\delta}{\theta}} + \|v_n\|_{\theta}^{\frac{\theta-1-\delta}{\theta}} \right) \|u_n - v_n\|_{\frac{\theta}{1+\delta}}^{\frac{\theta}{1+\delta}} \right) \quad (7.14)
\end{aligned}$$

as a consequence of [prop. A.4](#) and of Hölder's inequality (with $p' = \frac{\theta}{\theta-1+\delta}, q' = \frac{\theta}{1+\delta}$). Now note that

$$p < \theta < \frac{\theta}{1-\delta} < \frac{\theta}{1 - \frac{p_s^* - \theta}{p_s^*}} = \frac{\theta p_s^*}{p_s^* - p_s^* + \theta} = p_s^* \quad \text{and} \quad p_s^* > \theta > \frac{\theta}{1+\delta} > \frac{\theta}{1 + \frac{\theta-p}{p}} = \frac{p\theta}{p + \theta - p} = p.$$

This, the supposition that [inequality \(7.12\)](#) is false, and [prop. 5.6](#) gives us $\|u_n - v_n\|_{\frac{\theta}{1-\delta}} \rightarrow 0$, $\|u_n - v_n\|_{\frac{\theta}{1+\delta}} \rightarrow 0$. Since (u_n) and (v_n) are (PS) sequences, it follows from [prop. 5.5](#) and [corollary 2.2](#) that

$$\sup_{n \in \mathbb{N}} \left(\|u_n\|_{\theta}^{\frac{\theta-1+\delta}{\theta}} + \|v_n\|_{\theta}^{\frac{\theta-1+\delta}{\theta}} \right) < C_3 \quad \text{and} \quad \sup_{n \in \mathbb{N}} \left(\|u_n\|_{\theta}^{\frac{\theta-1-\delta}{\theta}} + \|v_n\|_{\theta}^{\frac{\theta-1-\delta}{\theta}} \right) < C_3$$

for some $C_3 \in (0, \infty)$. Combining this with ineqs (7.13) and (7.14) gives us

$$\|u_n - v_n\|_{\mathbb{X}}^p \leq C \left(\varepsilon_n \|u_n - v_n\|_{\mathbb{X}} + C_2 C_3 \left(\|u_n - v_n\|_{\frac{\theta}{1-\delta}}^{\frac{\theta}{1-\delta}} + \|u_n - v_n\|_{\frac{\theta}{1+\delta}}^{\frac{\theta}{1+\delta}} \right) \right).$$

Since $0 \leq \|u_n - v_n\|_{\mathbb{X}} \not\rightarrow 0$, we have that $\limsup \|u_n - v_n\|_{\mathbb{X}} > 0$. Taking the limit superior on both sides of the inequation above, we find then that the right-hand side must remain strictly positive along some subsequence. However, since ε_n and the second term in the right-hand side go to 0, the only way it could remain positive is if $\limsup \|u_n - v_n\|_{\mathbb{X}} = \infty$ along that subsequence; this would make the left-hand side eventually surpass the right-hand side, which begets a contradiction. Thus, $\|u_n - v_n\|_{\mathbb{X}} \rightarrow 0$.

As [inequality \(7.12\)](#) holds, there exists $\varepsilon \in (0, \infty)$ such that $\limsup_{y \in \mathbb{R}^N} \|u_n - v_n\|_{L^p(\mathcal{B}(y, 1))} > \varepsilon > 0$. Using the same argument as in the end of the proof of [prop. 5.7](#), we may assume that there exists $(z_n) \subset \mathbb{Z}^N$ and $R \in (0, \infty)$ such that

$$\|u_n - v_n\|_{L^p(\mathcal{B}(z_n, R))} > \frac{\varepsilon}{2} > 0. \quad (7.15)$$

Then $(h(z_n, u_n)) = (u_n \circ \tau_{z_n})$ and $(h(z_n, v_n)) = (v_n \circ \tau_{z_n})$ are still (PS) sequences by [prop. 5.4](#). By [prop. 5.5](#), this implies that they are bounded in $\mathbb{X}$ and also that $\|T'(u_n \circ \tau_{z_n})\|_{\mathbb{X}'}, \|T'(v_n \circ \tau_{z_n})\|_{\mathbb{X}'} \rightarrow 0$.

Since $\mathbb{X}$ is reflexive, we can find $u, v \in \mathbb{X}$ and a strictly increasing sequence of indexes $(n_k) \subset \mathbb{N}$ such that $u_{n_k} \circ \tau_{z_{n_k}} \rightharpoonup u$ and $v_{n_k} \circ \tau_{z_{n_k}} \rightharpoonup v$ in $\mathbb{X}$; thus $u_{n_k} \circ \tau_{z_{n_k}} - v_{n_k} \circ \tau_{z_{n_k}} \rightharpoonup u - v$ in $\mathbb{X}$. Furthermore, since $(u_{n_k} \circ \tau_{z_{n_k}} - v_{n_k} \circ \tau_{z_{n_k}})$ is also bounded in $\mathbb{X}$ and [ineq. \(7.15\)](#) still holds, [prop. 5.3](#) implies that $u - v \neq 0$, i.e., $u \neq v$. Considering $(g_n) := (z_n)$, the proof is complete. $\square$

A Auxiliary results

Proposition A.1. *Let $x, r \in (0, \infty)$. Then $\ln(x) < \frac{x^r}{r}$.*

Proof. Consider the function $f: (0, \infty) \rightarrow \mathbb{R}; f(y) = y - \ln(y)$. It is easy to see that $x_0 = 1$ is its global minimum. Since $f(x_0) = 1 > 0$, it follows that $\ln(y) < y$ for all $y \in (0, \infty)$. Making $y = x^r$ in this last inequality, and then using properties of the logarithm yields the result. $\square$

Corollary A.2. *Let $a, b \in (0, \infty)$. Then there exists $C = C(a, b) \in (0, \infty)$ such that for all $x \in (0, \infty)$,*

$$|\ln(x)| < C(x^a + x^{-b}).$$

Proof. Let $x \in (0, \infty)$ and $y = \frac{1}{x} \in (0, \infty)$. By [prop. A.1](#), we have that $\ln(y) < \frac{y^b}{b}$ for all $y \in (0, \infty)$; that is

$$-\ln(x) = \ln(x^{-1}) = \ln\left(\frac{1}{x}\right) = \ln(y) < \frac{y^b}{b} = \frac{1}{b} \frac{1}{x^b} = \frac{x^{-b}}{b}. \quad (\text{A.1})$$

Now by the properties of the logarithm and of the absolute value, we can get that $|\ln(x)| = -\mathbb{1}_{(0,1]}(x)\ln(x) + \mathbb{1}_{(1,\infty)}(x)\ln(x)$. Using [prop. A.1](#) again, we also infer $\ln(x) < \frac{x^a}{a}$. Combining these previous two expressions with [ineq. \(A.1\)](#), we conclude that

$$\begin{aligned} |\ln(x)| &= -\mathbb{1}_{(0,1]}(x)\ln(x) + \mathbb{1}_{(1,\infty)}(x)\ln(x) < \mathbb{1}_{(0,1]}(x) \frac{x^{-b}}{b} + \mathbb{1}_{(1,\infty)}(x) \frac{x^a}{a} \\ &\leq \frac{x^{-b}}{b} + \frac{x^a}{a} \leq \max\left\{\frac{1}{a}, \frac{1}{b}\right\} (x^a + x^{-b}). \end{aligned} \quad \square$$

Proposition A.3. *Let $x \in (-1, 1) - \{0\}$. Then*

$$\ln(1+x) = x \int_0^1 \frac{1}{1+xt} dt.$$

Proof. Let $g_x: [0, 1] \rightarrow \mathbb{R}; g_x(s) = 1 + xs$. It is easy to see that g_x with its codomain is restricted to its range is a diffeomorphism and that $g'_x(s) = x$. Using the theorem of change of variables with g_x , we then get

$$\begin{aligned} x \int_0^1 \frac{1}{1+xt} dt &= \int_0^1 \frac{1}{1+xt} x dt = \int_0^1 \frac{1}{g_x(t)} g'_x(t) dt = \int_{g_x(0)}^{g_x(1)} \frac{1}{s} ds = \int_1^{1+x} \frac{1}{s} ds \\ &= \ln(|1+x|) - \ln(|1|) = \ln(1+x). \end{aligned} \quad \square$$

Proposition A.4. *Let $a, b, q \in (0, \infty)$. Then $(a+b)^q \leq 2^q(a^q + b^q)$.*

Proof.

$$(a+b)^q \leq (\max\{a, b\} + \max\{a, b\})^q = (2\max\{a, b\})^q = 2^q \max\{a, b\}^q = 2^q \max\{a^q, b^q\} \leq 2^q(a^q + b^q). \quad \square$$

Proposition A.5. *Let $q \in (2, \infty)$. Then there exists $C \in (0, \infty)$ such that $|a-b|^q \leq C(|a|^{q-2}a - |b|^{q-2}b)(a-b)$ for all $a, b \in \mathbb{R}$.*

Proof. Without loss of generality, we may assume that $a > b$. Let $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(t) := |t|^{q-2}t$; it is clear that $f'(t) = (q-1)|t|^{q-2}$ and that it is continuous. The fundamental theorem of calculus then gives us

$$f(a) - f(b) = \int_b^a f'(t) dt = (q-1) \int_b^a |t|^{q-2} dt. \quad (\text{A.2})$$

Setting $h := a - b > 0$, we can perform the change of variables $t = b + hs$ to get

$$\int_b^a |t|^{q-2} dt = \int_0^1 h|b + hs|^{q-2} ds = h^{q-1} \int_0^1 \left| \frac{b}{h} + s \right|^{q-2} ds. \quad (\text{A.3})$$

Consider now the function $g: \mathbb{R} \rightarrow \mathbb{R}$; $g(a) = \int_0^1 |a + s|^{q-2} ds$. On one hand, if $a \geq 0$, then $g(a) \geq \int_0^1 s^{q-2} ds = \frac{1}{q-1} > 0$. On the other hand, if $a \leq -1$, then

$$g(a) \geq \int_0^{\frac{1}{2}} |a + s|^{q-2} ds = \int_0^{\frac{1}{2}} (-a - s)^{q-2} ds > \int_0^{\frac{1}{2}} (1 - \frac{1}{2})^{q-2} ds = (1 - \frac{1}{2})^{q-2} > 0.$$

Finally, if $a \in (-1, 0)$, then either $-a \in (\frac{1}{2}, 1)$ or $-a \in (0, \frac{1}{2})$. In the first case, we have

$$g(a) \geq \int_0^{\frac{1}{4}} |a + s|^{q-2} ds = \int_0^{\frac{1}{4}} (-a - s)^{q-2} ds \geq \int_0^{\frac{1}{4}} (\frac{1}{2} - \frac{1}{4})^{q-2} ds = \frac{1}{4^{q-2}}$$

and, in the latter,

$$g(a) \geq \int_{\frac{3}{4}}^1 |a + s|^{q-2} ds = \int_{\frac{3}{4}}^1 (a + s)^{q-2} ds \geq \int_{\frac{3}{4}}^1 (-\frac{1}{2} + \frac{3}{4})^{q-2} ds = \frac{1}{4^{q-2}}.$$

Thus, $C := \frac{1}{(q-1)\inf_{a \in \mathbb{R}} g(a)} > 0$. We can then combine this with eqs. (A.2) and (A.3) to get $\frac{1}{C}(a-b)^{q-1} \leq |a|^{q-2}a - |b|^{q-2}b$. To conclude the proof, we multiply both sides by $(a-b) > 0$, yielding $|a-b|^q = (a-b)^{q-1}(a-b) \leq (|a|^{q-2}a - |b|^{q-2}b)(a-b) = (|b|^{q-2}b - |a|^{q-2}a)(b-a)$. $\square$

Proposition A.6. *Let $\mathbb{Y}$ be a normed vector space, $I \subset \mathbb{R}$ be connected, $\gamma: I \rightarrow \mathbb{Y}$ be continuous and $U \subset \mathbb{Y}$ be open. If there exists $t_1, t_2 \in I$ such that $\gamma(t_1) \in U^c$ and $\gamma(t_2) \in U$, then there also exists $t_0 \in I' := [\min\{t_1, t_2\}, \max\{t_1, t_2\}]$ such that $\gamma(t_0) \in \partial U$.*

Proof. If $\gamma(t_1) \in \partial U$, we are done. Otherwise, $\gamma(t_1) \in \overline{U}^c$ since U is open. Note that $I' \subset I$ by connectedness. Suppose now by contradiction that $\gamma(I') \cap \partial U = \emptyset$, and consider the sets $A := \gamma(I') \cap U$ and $B := \gamma(I') \cap \overline{U}^c$. It is clear that $A \neq \emptyset$ (because $\gamma(t_2) \in A$), $B \neq \emptyset$ (because $\gamma(t_1) \in B$), and $A \cap B = \emptyset$ (because U is open). Furthermore,

$$A \cup B = \gamma(I') \cap (U \cup \overline{U}^c) = \gamma(I') \cap (U \cup \overline{U}^c \cup \emptyset) = \gamma(I') \cap (U \cup \overline{U}^c \cup \partial U) = \gamma(I') \cap \mathbb{Y} = \gamma(I').$$

The definition of subspace topology and the fact that U is open, allows us to deduce that A is open in the subspace topology of $\gamma(I')$. Similarly, since $\overline{U}^c$ is open, we also conclude that B is open in that same subspace topology. Thus, the pair (A, B) disconnect $\gamma(I')$. However, since γ is continuous and I' is connected, $\gamma(I')$ must also be connected, which is a contradiction. Thus, $\gamma(I') \cap \partial U \neq \emptyset$. $\square$

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used the freely available versions of Chat GPT and Claude for literature review, early exploration of arguments, and for writing suggestions. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

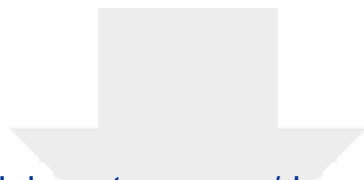
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